



A framework for user centred privacy and security in the cloud

Requirements Specification V1

Type (distribution level)	Public
Contractual date of Delivery	30-06-2015
Actual date of delivery	30-07-2015
Deliverable number	D2.2
Deliverable name	Requirements Specification V1
Version	V1.0
Number of pages	109
WP/Task related to the deliverable	Task 2.2 – User Requirements
WP/Task responsible	AKKA
Author(s)	AKKA (Monique Damiot, Frédéric Brouillé, Thierry Chevallier), Montimage (Antonio M Ortiz), Trust-IT (Stephanie Parker, Roberto Cascella), OFFIS (Stefan Janacek)
Partner(s) Contributing	All
Document ID	CLARUS-D2.2-Requirements Specification V1-V0.5

Abstract *This document defines the use cases of the CLARUS solution and the requirements at the functional and non-functional levels*

Disclaimer

CLARUS (G.A. 644024) is a Research and Innovation Actions project funded by the EU Framework Programme for Research and Innovation Horizon 2020. This document contains information on CLARUS core activities, findings and outcomes. Any reference to content in this document should clearly indicate the authors, source, organisation and publication date. The content of this publication is the sole responsibility of the CLARUS consortium and cannot be considered to reflect the views of the European Commission.

Revision History

Version	Date	Author	Description
0.1	18/06/2015	AKKA	AKKA internal initial iteration
0.2	17/07/2015	AKKA	First complete version of deliverable. Montimage contribution on monitoring.
0.3	22/07/2015	Trust-IT	First internal review and editing of the deliverable
0.4	24/07/2015	AKKA	Non-functional requirements completed, functional requirements adding and document conclusion
0.5	28/07/2015	AKKA	New release following TRUST-IT and OFFIS reviews. All sections revised.
1.0	30/07/2015	AKKA	Executive Summary added : deliverable version.

Executive Summary

The main objective of CLARUS is to enhance trust in cloud computing services by **developing a secure framework for the storage and processing of data outsourced to the cloud** that allows end users to monitor, audit and apply control over data, with some **security and privacy features**, without impairing the functionality and cost-saving benefits of cloud services.

The CLARUS solution proposed, relies on the choice of two main **business application cases** on which the solution could be applied in two sectors, namely: geopublication and e-Health. The term goal would be to deliver a more generic solution, extensible to other business domains where privacy and confidentiality of data are high concerns.

The choice of these two **business application cases** (geopublication and e-Health) has led to **several demonstration cases** in D2.1 document (e.g., data storage, publication, computation for statistical purposes, etc.) on which the CLARUS solution implemented could be further tested and validated.

For **design and implementation** of the CLARUS solution we apply a rigorous and relevant methodology in order to ensure that all quality measures for the development of the software solution are applied.

This document deals with the collection work and the specification of requirements. These requirements are the **input** keys for the design work and the complete architecture definition. This is why the stated requirements should cover both functional features (based on both the technical and social needs), as non-functional aspects (we can mention e.g., quality of service, security, etc.), in order to cover all the facets of the CLARUS solution proposed.

The **methodology** applied consists in collecting the requirements through a UML based modeling approach.

This approach builds on **use cases** determination, derived from the D2.1 demonstration cases under a more generic angle, not necessarily correlated to the domain of application of e-Health or geopublication. These use cases take different **perspectives from actors** who interact with the system to be implemented. The specification of use cases lead to several **scenarios** of possible use of CLARUS.

These **scenarios** allow predefining what the CLARUS solution should be, and the expected outcomes:

- o A technical solution regarded as a proxy running on trusted local premises, transparently for the end user, but may also be part of the CLARUS adopting CSP.
- o A configurable tool by a dedicated actor, embodied by the IT security manager of the organization considered,
- o A necessary integration in the IT system of an organization, including access control management capabilities and security policy management capabilities,
- o A technical tool which encompasses some monitoring functionalities, able to face potential intrusion attempt in an 'Attack Tolerant System' mode, which delivers appropriated counter-measures, depending on the intrusion type encountered.
- o An "enhanced-security solution", able to support multi usage scenarios of outsourcing data management to the cloud. In this prospect, the way how CLARUS could protect the datasets according to their usage, where the security manager will specify the usage(s) and the outcomes' accuracy for a given dataspace are outlined.

This first framework of CLARUS solution is proposed at this stage of the project, however, without any presumption on security techniques to be provisioned. This will be contemplated in WP3.

In order to strengthen the work of requirements collection, the work performed in this deliverable also provides for each requirement submitted:

- o A solution for requirements traceability by the means of the definition and application of a naming scheme (this will be a must utility for the coming steps of implementation, testing and validation).
- o A 'priorities' classification, from which a level of criticality assigned to the requirement helps to define the mandatory capabilities of the solution.

Finally these scenarios must completely fit with the **business demonstration cases** previously described in D2.1. This point of control is described by the appendix document also joined with this deliverable.

Table of Contents

1	INTRODUCTION.....	13
1.1	SCOPE AND PURPOSE	13
1.2	METHODOLOGY	14
1.3	DOCUMENT STRUCTURE	14
1.4	KEY POINTS EMERGING FROM THE ANALYSIS.....	15
1.5	THE SPECIFICATIONS IN RELATION TO THE PROJECT WORKFLOW	15
1.6	APPLICABLE AND REFERENCE DOCUMENTS.....	16
1.7	NOTATIONS, ABBREVIATIONS AND ACRONYMS	16
2	ACTORS.....	18
2.1	OVERVIEW	18
2.2	ACTORS INVOLVED IN INFRASTRUCTURE MANAGEMENT	18
2.3	END USERS.....	18
3	USE CASES.....	19
3.1	OVERVIEW	19
3.1.1	<i>Definitions</i>	20
3.1.2	<i>Naming Scheme</i>	20
3.1.3	<i>Classification codes</i>	20
3.2	CLARUS PROXY INSTALLATION.....	21
3.2.1	<i>Overview</i>	21
3.2.1.1	System	21
3.2.1.2	UML diagram	21
3.2.2	<i>Use cases</i>	22
3.2.2.1	Install CLARUS proxy (UC-ADM-01)	22
3.2.2.2	Upgrade CLARUS proxy (UC-ADM-02)	22
3.2.2.3	Configure CLARUS proxy (UC-ADM-03)	23
3.2.2.4	Configure user repository (UC-ADM-04)	23
3.2.2.5	Configure user authentication (UC-ADM-05)	24
3.2.2.6	Configure access to the CSP(s) (UC-ADM-06)	24

3.2.2.7	Configure failover mode (UC-ADM-07)	25
3.2.2.8	Implement CLARUS module (UC-IMPL-01)	26
3.2.2.9	Add CLARUS module (UC-ADM-08)	26
3.3	CLARUS ADOPTING CSP INSTALLATION.....	27
3.3.1	Overview	27
3.3.1.1	System	27
3.3.1.2	UML diagram	27
3.3.2	Use cases.....	27
3.3.2.1	Install CLARUS on cloud platform (UC-ADM-09).....	27
3.3.2.2	Upgrade CLARUS on cloud platform (UC-ADM-10)	28
3.3.2.3	Adapt cloud service (UC-IMPL-02).....	28
3.4	ACCESS CONTROL MANAGEMENT	29
3.4.1	Overview	29
3.4.1.1	System	29
3.4.1.2	UML diagram	29
3.4.2	Use cases.....	30
3.4.2.1	Manage CLARUS proxies (UC-SEC-01).....	30
3.4.2.2	Register CLARUS users (UC-SEC-02).....	31
3.4.2.3	Import a user registered with another CLARUS proxy (UC-SEC-03).....	32
3.4.2.4	Deactivate CLARUS users (UC-SEC-04).....	33
3.4.2.5	Activate CLARUS users (UC-SEC-05).....	34
3.4.2.6	Unregister a CLARUS user (UC-SEC-06).....	35
3.4.2.7	Edit a user profile (UC-SEC-07)	36
3.4.2.8	View authorisations of users on an outsourced dataset (UC-SEC-08)	37
3.4.2.9	Edit authorisations of users on an outsourced dataset (UC-SEC-09).....	38
3.5	SECURITY POLICY MANAGEMENT	40
3.5.1	Overview	40
3.5.1.1	System	40
3.5.1.2	UML diagram	40
3.5.2	Use cases.....	41
3.5.2.1	Create dataspace (UC-SEC-10).....	41

3.5.2.2	Select protocol(s) (UC-SEC-11)	41
3.5.2.3	Configure protocol endpoint(s) (UC-SEC-12)	42
3.5.2.4	Select attribute(s) to protect (UC-SEC-13)	42
3.5.2.5	Select usage(s) of the outsourced dataset(s) (UC-SEC-14)	44
3.5.2.6	Define outcomes' accuracy (UC-SEC-15)	44
3.5.2.7	Configure appropriate security technique(s) (UC-SEC-16)	45
3.6	STANDARD MONITORING	46
3.6.1	Overview	46
3.6.1.1	System	46
3.6.1.2	UML diagram	46
3.6.2	Use cases	46
3.6.2.1	Monitor information / Retrieve metrics (UC-MON-01)	46
3.6.2.2	Detect incidents (UC-MON-02)	47
3.6.2.3	Manage back up states (UC-MON-03)	48
3.6.2.4	Check monitoring tools (UC-MON-04)	48
3.6.2.5	Check the correctness of results (UC-MON-05)	49
3.6.2.6	Correlate information from other CLARUS proxies (UC-MON-06)	49
3.7	ATTACK MONITORING	50
3.7.1	Overview	50
3.7.1.1	System	50
3.7.1.2	UML diagram	50
3.7.2	Use cases	51
3.7.2.1	Implement an Attack Tolerant System (UC-MON-07)	51
3.7.2.2	Manage Intrusion Prevention System (UC-MON-08)	52
3.7.2.3	Manage Intrusion Detection System (UC-MON-09)	52
3.8	USER DATA STORAGE	53
3.8.1	Overview	53
3.8.1.1	System	53
3.8.1.2	UML diagram	53
3.8.2	Use cases	55
3.8.2.1	Store by id through proxy (UC-EUP-01)	55

3.8.2.2	Retrieve by id through proxy (UC-EUP-02)	56
3.8.2.3	Delete by id through proxy (UC-EUP-03)	57
3.9	SEARCH IN THE CLOUD	57
3.9.1	Overview	57
3.9.1.1	System	57
3.9.1.2	UML diagrams	58
3.9.2	Use cases	59
3.9.2.1	Search through proxy (UC-EUP-04)	59
3.9.2.2	Search directly in the cloud (UC-EUP-05)	60
3.10	UPDATE IN THE CLOUD	60
3.10.1	Overview	60
3.10.1.1	System	61
3.10.1.2	UML diagrams	61
3.10.2	Use cases	62
3.10.2.1	Update by attributes through proxy (UC-EUP-06)	62
3.11	COMPUTATION IN THE CLOUD	62
3.11.1	Overview	62
3.11.1.1	System	62
3.11.1.2	UML diagrams	63
3.11.2	Use cases	64
3.11.2.1	Computation through proxy (UC-EUP-07)	64
3.11.2.2	Computation directly in the cloud (UC-EUP-08)	65
4	REQUIREMENTS	66
4.1	OVERVIEW	66
4.1.1	Naming Scheme	66
4.1.2	Priorities	66
4.1.3	Description	67
4.1.4	Categories code	67
4.2	FUNCTIONAL REQUIREMENTS	68
4.2.1	Business Rules	68
4.2.2	Features	70

4.2.2.1	Secure data storage	70
4.2.2.2	Secure data search	72
4.2.2.3	Secure data update	73
4.2.2.4	Secure data computation	74
4.2.2.5	Manage access control	75
4.2.2.6	Manage security policy	84
4.2.2.7	Manage monitoring	87
4.2.2.8	Manage installation/configuration	88
4.2.3	<i>User Interface</i>	89
4.2.3.1	General	89
4.2.3.2	User management dashboard	90
4.2.3.3	Security policy dashboard	93
4.2.3.4	Monitoring and attack management interface	95
4.2.4	<i>Social needs</i>	96
4.3	NON FUNCTIONAL REQUIREMENTS	97
4.3.1	<i>Delivery</i>	97
4.3.2	<i>Implementation</i>	99
4.3.3	<i>Quality of Service</i>	100
4.3.4	<i>Operational</i>	101
4.3.5	<i>Persistence</i>	101
4.3.6	<i>Security</i>	102
4.3.7	<i>Performance</i>	103
4.3.8	<i>Scalability</i>	104
4.3.9	<i>Transport</i>	104
4.3.10	<i>Standardisation</i>	104
4.3.11	<i>Legal and ethical</i>	105
4.3.12	<i>Open Source</i>	106
5	CONCLUSION	108

Table of Figures

Figure 1 - Template Table for Use Case Definition	14
Figure 2 - Use Case ID Naming Scheme	20
Figure 3 – CLARUS Proxy Installation Use Case Diagram	21
Figure 4 - CLARUS Proxy adopting CSP Installation Use Case Diagram.....	27
Figure 5 - Access Control Management Use Case Diagram	30
Figure 6 - Security Policy Management Use Case Diagram	40
Figure 7 - Standard Monitoring Use Case Diagram.....	46
Figure 8 - Attack Monitoring Use Case Diagram.....	51
Figure 9 – Single-User Data Storage Use Case Diagram	54
Figure 10 – Muti-User Data Storage Use Case Diagram	54
Figure 11 - User Data Storage Activity Diagram	55
Figure 12 - Search Through Proxy Use Case Diagram	58
Figure 13 - Search Directly in the Cloud Use Case Diagram.....	58
Figure 14 - Search Activity Diagram	59
Figure 15 - Update by Attributes Use Case Diagram	61
Figure 16 - Update by Attributes Activity Diagram.....	61
Figure 17 - Computation Through Proxy Use Case Diagram.....	63
Figure 18 - Computation Directly in the Cloud Use Case Diagram	63
Figure 19 - Computation Activity Diagram.....	64
Figure 20 - Requirement Naming Scheme	66
Figure 21 - Classification of Requirements	68

1 Introduction

1.1 Scope and Purpose

The main objective of CLARUS is to enhance trust in cloud computing services by **developing a secure framework for the storage and processing of data outsourced to the cloud** that allows end users to monitor, audit and retain control of the stored data without impairing the functionality and cost-saving benefits of cloud services. The **CLARUS solution will provide the end user with a dedicated proxy located in a trusted domain implementing security and privacy features** towards the cloud provider. The proxy is intended for deployment within the client computer, in a server within the user's domain, in an edge device (e.g. a router), or in any other location trusted by the user. CLARUS will also provide **a set of security auditing services** enabling the user to supervise the security operations performed by the CLARUS framework as well as other trust-enhancing features.

The focus of WP2 (user and technical requirements) is on eliciting and specifying the requirements for CLARUS drawing on the application cases (eHealth and Geopublication) that are used to demonstrate the appropriateness and applicability of the methodologies, technologies and tools developed. Briefly, this entails:

- Eliciting and defining requirements for the architecture and security framework,
- Selecting and defining significant application cases for demonstrating and implementing the complete CLARUS toolkit,
- Defining key concepts concerning the security and privacy framework,
- Specifying legal guidance and establishing legal fundamentals, starting from the constraints and contexts encountered in the application domains targeted by the application cases,
- Defining a standardisation roadmap for CLARUS based on the main requirements identified and linking this to the ETSI Cloud Standards Coordination report (as well as new relevant insights emerging from the second phase of this activity).

The purpose of this document is to provide the CLARUS requirement specifications. It defines the use cases of the CLARUS solution and the requirements at the functional (i.e. the features that the solution needs to offer, e.g. business rules, features, user interface, social requirements) and the non-functional levels (e.g. quality of service, performance, scalability, legal and ethical, standardisations).

More precisely this document aims at specifying:

- Which functionalities should be supported by the CLARUS solution, in order to help cloud customers increase their control over outsourced data, give them the guarantee that their confidential or sensitive data will not be disclosed at any moment to the Cloud Service Provider (CSP), and maintain cloud functionalities and benefits.
- What are the expected outcome of these functionalities, i.e. efficiency, data utility for both cloud customers and CSPs.
- What are the needs in terms of security (i.e. the degree of security to be provided in different use cases) .

1.2 Methodology

The initial analysis of the CLARUS requirement specifications is based on the collection and analysis of user needs in terms of security and privacy in the business domains of eHealth and Geopublication (the application case defined in D2.1 [A4]). We have focused on the needs where the set of CLARUS innovative techniques are designed to remove the current business constraints to the adoption of cloud technologies. We have paid special attention to avoiding premature design assumptions, resolving conflicts and requirements redundancy while ensuring traceability.

We have adopted the following template to detail the use cases. The template contains a set of characteristics as a framework for the description of the use cases.

<Use Case Id>	<i>Use case name</i>
Actors	<i><Actors></i>
Description	<i>Description</i>
Pre-conditions	
<i>[Pre-condition 1]</i>	
...	
Main scenario	
Step 1	
...	
[Variation 1]	
Step 1	
...	
[Variation ...]	
Step 1	
...	
Post-conditions	
<i>[Post-condition 1]</i>	
...	
Notes	
<i>[Note 1]</i>	
...	

Figure 1 - Template Table for Use Case Definition

1.3 Document structure

This document is structured as follows:

- **Section 2:** describes the actors involved in the context of the CLARUS solution, either as a stakeholder in the infrastructure management or as an end-user. Actors involved in the infrastructure management are IT team/administrator, architect, developer, cloud service provider. End-users can be security manager, data provider, data consumer.
- **Section 3:** focuses on the CLARUS use cases, from the perspective of the different actors identified in the previous section. It covers scenarios derived from the business cases and more

generic use cases; the CLARUS proxy installation; CLARUS adopting CSP installation; access control management; security policy management; standard monitoring; user data storage. It also covers use cases related to search in the cloud; update in the cloud; and computation in the cloud.

- **Section 4:** describes the requirements for the CLARUS solution, both functional and non-functional. Specifically, it covers the business rules, features (e.g. secure data storage, secure data search, secure data computation, manage access control), user interface, including the user management dashboard and security policy dashboard, monitoring and attack management interface, and social needs. From the perspective of non-functional requirements, it covers quality of service, operational requirements, performance, scalability, security, persistence, transport, implementation, legal and ethical compliance, standardisation, open source initiative.
- **Section 5:** presents the conclusions of this analysis.

1.4 Key points emerging from the analysis

The overall analysis of the CLARUS use cases led to a series of key elements:

- The CLARUS solution can be considered as a proxy installed on premises, but can also need to be partly integrated in the cloud platform by a CLARUS adopting CSP.
- The CLARUS solution includes a monitoring solution and attack prevention and detection tools.
- The notion of “dataspace” – a virtual space linking the applications running on premises with the cloud services – is very important to understand how CLARUS could protect the outsourced datasets according to their usage.
- To configure a security policy, the security manager must specify the usage and outcome accuracy for a given dataspace (and the supported protocols, the appropriate endpoints of the cloud storage services, the supported types of data and the attributes to be protected).
- The CLARUS proxy helps the security manager to select the security techniques which are the most appropriate for the given usages and outcomes accuracy.
- Only authorized users can access existing outsourced datasets in a dataspace (managed by the security manager).
- From the data providers/consumers perspective, four scenarios are proposed: User data storage, Search in the cloud, Update in the cloud and Computation in the cloud. These scenarios are a combination of basic features which define a “data usage” (create, retrieve by id, update by id, search by attributes, update by attributes, compute).
- From the CSP perspective, CLARUS provides an implemented API with the functions needed to process and operate protected data.

1.5 The specifications in relation to the project workflow

The aim of this deliverable is to help in the architecture design in WP4 (architecture design), and implementation in WP5 (CLARUS implementation) according to the different scenarios described in Section 3 and finally, demonstration and validation process in WP6 (evaluation, testing and validation).

As part of work package 2 (user and technical requirements), this deliverable is part of a suite of related documents produced in the early phase of the project.

D2.1 Definition of application cases defines the application cases that inform all further developments with testing and validation under WP6.

D2.4 Legal and ethical requirements specifies concrete legal requirements for the CLARUS system, describing the means for surveillance and guidance.

D2.5 Standardisation requirements reports the necessary requirements, results, and implementation roadmap, effort and analysis, considering the EU and global landscape.

An updated version of this document (D2.2), based on intermediate results and analysis of the requirements herein identified, is also planned in August 2016 :

D2.3 Requirements Specification V2 (August 2016) refines the requirement specifications for the final version of the CLARUS platform, taking into account the intermediate results of the project in WP5 and WP6.

1.6 Applicable and Reference documents

App. /Ref.	Title
[A1]	CLARUS Grant Agreement 644024
[A2]	CLARUS 644024 Consortium Agreement v3.0, dec.2014
[A3]	CLARUS Glossary
[A4]	CLARUS D2.1 – Definition of Application Cases
[A5]	CLARUS D2.2 – Requirements Specification – Annex I
[R1]	CLARUS D2.4 – Legal and Ethical Requirements
[R2]	CLARUS D2.5 – Standardisation Requirements
[R3]	Directive 95/46/EC of the European Parliament and of the Council of 24 October 1995 on the protection of individuals with regard to the processing of personal data and on the free movement of such data (Data Protection Directive), O.J. L 281, http://eurlex.europa.eu/LexUriServ/LexUriServ.do?uri=CELEX:31995L0046:en:HTML

1.7 Notations, Abbreviations and Acronyms

<i>ADM</i>	Use case classification code for 'Administration'
<i>API</i>	Application Programming Interface
<i>CH</i>	Requirement priority for 'Could Have'
<i>CSP</i>	Cloud Service Provider
<i>ETSI</i>	European Telecommunication Standards Institute
<i>EU</i>	European Union
<i>EUP</i>	Use case classification code for 'End-User Perspective'
<i>GA</i>	Grant Agreement

<i>IaaS</i>	Infrastructure as a Service
<i>IMPL</i>	Use case classification code for 'Design and Implementation'
<i>IS</i>	Information System
<i>MAMD</i>	Monitoring and Attack Management Dashboard
<i>MH</i>	Requirement priority for 'Must Have'
<i>MON</i>	Use case classification code for 'Monitoring'
<i>PaaS</i>	Platform as a Service
<i>REQ</i>	Requirement naming scheme prefix for 'Requirement'
<i>SaaS</i>	Software as a Service
<i>SEC</i>	Use case classification code for 'Security Management'
<i>SH</i>	Requirement priority for 'Should Have'
<i>UML</i>	Unified Modeling Language
<i>URI</i>	Uniform Resource Identifier
<i>WL</i>	Requirement priority for 'would like'
<i>WP</i>	Work Package

2 Actors

2.1 Overview

The actors are described here in the context of the CLARUS solution. It is possible that some of these actors may be involved in other activities within their organizations (for instance the IT team, the Security Manager, etc.), but these activities are not within the scope of this Section.

The different actors involved in the CLARUS solution can be divided into two main groups:

- Stakeholders in the infrastructure management,
- End-users.

2.2 Actors involved in Infrastructure Management

The **IT team / Administrator** is responsible for installing CLARUS in the Information System and configuring it with respect to the infrastructure.

The **Architect** is responsible for the design of the solution to be implemented by the development team.

The **Developer**, also referred to as “IT Development team”, is responsible for the development of additional modules, extending CLARUS functionalities in order to support protocols, data types, functions, etc. that are not natively supported by the solution.

The **Cloud Service Provider** is the company/organisation providing an IaaS, PaaS and/or SaaS type of service that is used by the Data Providers and Data Consumers, when storing and accessing data in/from the cloud through the CLARUS proxy. In the CLARUS context, the cloud service providers are considered “*Honest but curious*”, meaning that they correctly execute the operations but they might look at the information they store.

2.3 End Users

The **Security Manager** is in charge of managing access control of CLARUS users and defining security policies regarding data usages, data access protocols, data types and outcome accuracy. This is the user that probably interacts the most with CLARUS through a user interface.

The **Data Provider**, also referred to as “Data owner”, when the notion of data ownership is at stake, wants to securely store data in the cloud through the CLARUS proxy, with the guarantee that the data will not be disclosed to the cloud service provider. Data providers may also want to update their data or the data of other data providers (given that they have the proper authorizations) through CLARUS.

The **Data Consumer** is the potential user of the data stored in the cloud services through the CLARUS solution. Data consumers may need to access and retrieve these data, or to perform an attribute-based search, depending on whether they want to consult a specific record or to do research studies. Data consumers want to have the guarantee that their query or the result of their query will not be disclosed to the cloud service provider or any third-party.

3 Use Cases

3.1 Overview

In this section we focus on the CLARUS use cases. We define the **main scenarios** of the possible use of the CLARUS solution. We use UML use case diagrams to help define CLARUS software functional requirements in relation to the scenarios.

Depending on the considered scenario, CLARUS can be regarded as:

- A proxy running on fully trusted local premises which is transparent for the local applications and the end-users,.
- A PaaS/SaaS service deployed on CSPs adopting CLARUS to properly manage/operate encrypted data via well-defined APIs. This implementation integrating CLARUS libraries in the cloud will be subsequently referred to as “CLARUS inside”.

The following scenarios are derived mainly from the business cases [A5] but also, from more generic use cases in other fields than geo-publication or e-Health. This allows us the identification of the following **features**, that CLARUS needs to support and implement by using privacy-preserving techniques:

- Data creation,
- Data retrieval by id,
- Data update by id,
- Attribute-based search,
- Attribute-based update,
- Computation in the cloud.

Combining these features together we propose the following **main scenarios** that reflect the different use cases of the CLARUS solution, from the data provider/consumer perspective :

- Scenario 1 : User data storage,
- Scenario 2 : Search in the cloud,
- Scenario 3 : Update in the cloud,
- Scenario 4 : Computation in the cloud.

Each of these scenarios is illustrated with an UML use case diagram, representing the relationships between the main actors and the “system” embodied by the CLARUS solution to implement. The use cases describe how the system interacts with the different elements around it: users/actors, other systems, etc.

In addition scenarios showing perspectives other than the data provider/consumer are described, notably:

- CLARUS installation,
- Access control management,
- Security policy management,
- Monitoring.

3.1.1 Definitions

Primary actor requests the system to deliver a goal (usually represented on the left of the use case diagram).

Secondary actor is an actor from which the system needs assistance to achieve the primary actor's goal (usually represented on the right of the use case diagram).

Main success scenario is the typical and illustrative sequence of actions and interactions leading to a goal delivery by the system, even though it may not be the only success path.

Alternate scenario (Variation) is any other scenario or scenario fragment written as an extension to the main success scenario.

3.1.2 Naming Scheme

Each use case involves an identifier <Use Case Id> defined by the following nomenclature :

$\text{<Use Case Id> = UC-}<\text{Classification code}>-<\text{Progressive number}>$
--

Figure 2 - Use Case ID Naming Scheme

The classification codes refer to some criteria according to the following classification.

3.1.3 Classification codes

The use cases are classified according to:

- Security Management (SEC),
- End-User Perspective (EUP),
- Monitoring (MON),
- Administration (ADM),
- Design and Implementation (IMPL).

3.2 CLARUS proxy installation

3.2.1 Overview

This scenario describes the actions required for installing the CLARUS proxy.

3.2.1.1 System

In this scenario the system refers to the IT system on premises, i.e. in a server within the user's domain (trusted zone). The CLARUS proxy may also be deployed within the client computer, or in any other location trusted by the user.

The CLARUS proxy includes, among others, the monitoring solution to analyse different business and security indicators and is composed of the necessary monitoring, and attack prevention and detection tools.

3.2.1.2 UML diagram

The UML diagram below represents the corresponding use case (and their actors):

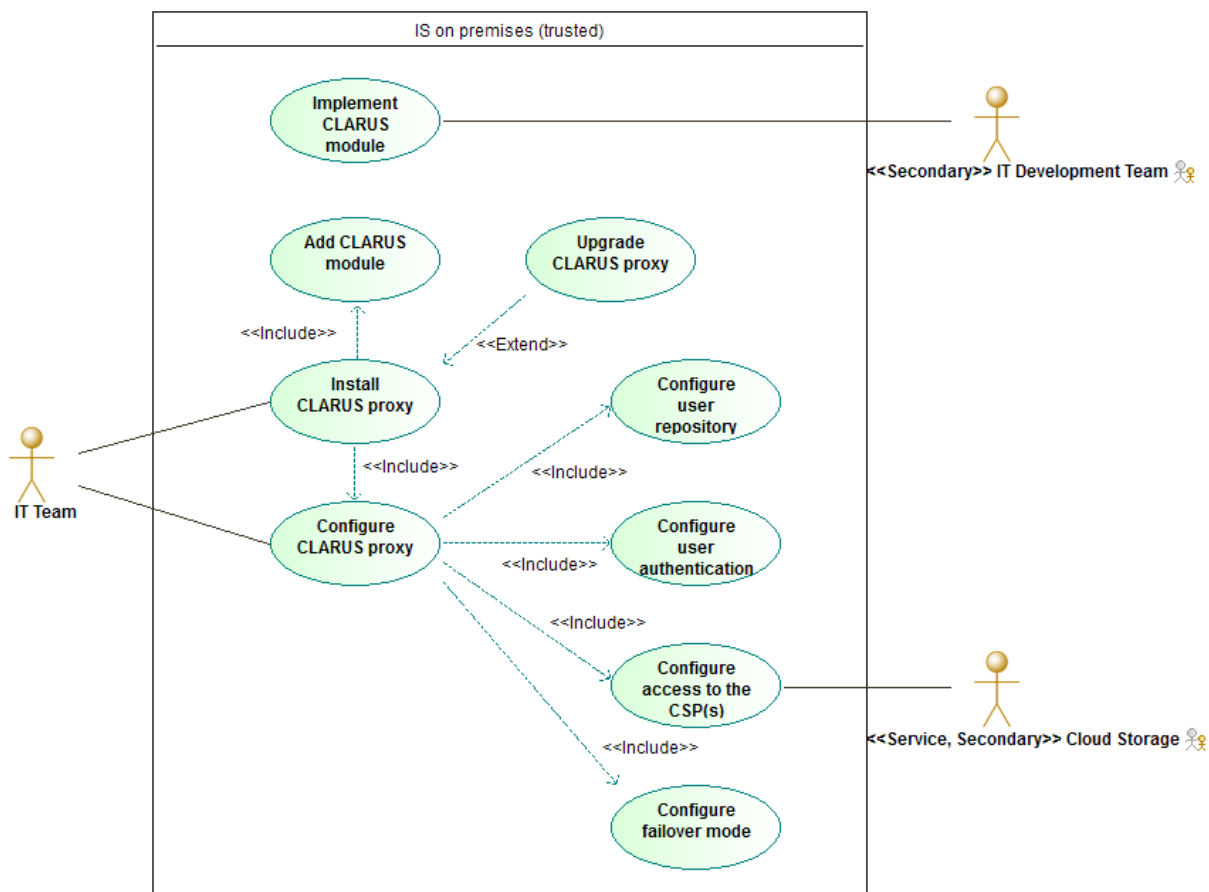


Figure 3 – CLARUS Proxy Installation Use Case Diagram

3.2.2 Use cases

3.2.2.1 Install CLARUS proxy (UC-ADM-01)

UC-ADM-01	<i>Install CLARUS proxy</i>
Actors	IT team
Description	The IT team wants to install a CLARUS proxy on a dedicated server in the local (trusted) IT system
Pre-conditions	
[Pre-condition 1]	No CLARUS proxy is installed yet
[Pre-condition 2]	A CLARUS proxy installation package is available for the target operation system
Main scenario	<i>Install</i>
Step 1	The IT team gets the installation guide of CLARUS proxy in order to follow its instructions
Step 2	The member of the IT team in charge of the installation opens an administrator session on the dedicated server
Step 3	The system controls the access rights of the IT team on the server
Step 4	The member of the IT team launches a command that automatically retrieves the package for CLARUS proxy, installs it on the server and starts it
Variation 1	<i>Add CLARUS module (see UC-ADM-08 : (3.2.2.9))</i>
Post-conditions	
[Post-condition 1]	The CLARUS proxy is installed and is running on the server

3.2.2.2 Upgrade CLARUS proxy (UC-ADM-02)

UC-ADM-02	<i>Upgrade CLARUS proxy</i>
Actors	IT team
Description	The IT team wants to upgrade a CLARUS proxy on a dedicated server in the local (trusted) IT system
Pre-conditions	
[Pre-condition 1]	An obsolete version of CLARUS proxy is installed
[Pre-condition 2]	A new 'CLARUS proxy' package (and its installation guide) is available for the target operation system
Main scenario	<i>Upgrade</i>
Step 1	The member of the IT team in charge of the upgrade opens an administrator session on the dedicated server, and follows the upgrade instructions from the installation guide of the version to install.
Step 2	The system controls the access rights of the IT team on the server
Step 3	The member of the IT team launches a command that stops the CLARUS proxy
Step 4	The member of the IT team launches a command, specifying the CLARUS version as argument, that upgrades CLARUS to the corresponding version and starts it
Post-conditions	

[Post-condition 1]	The CLARUS proxy is upgraded and is running on the server
[Post-condition 2]	Configuration data generated by CLARUS (registered users, outsourced datasets, authorizations, appropriate materials, etc.) are preserved

3.2.2.3 Configure CLARUS proxy (UC-ADM-03)

UC-ADM-03	<i>Configure CLARUS proxy</i>
Actors	IT team
Description	The IT team wants to configure a CLARUS proxy on the local (trusted) IT system
Pre-conditions	
[Pre-condition 1]	A CLARUS proxy, in its last version, has been installed in the local (trusted) IT system (see UC-ADM-01: 3.2.2.1 or UC-ADM-02: 3.2.2.2)
Main scenario	<i>Configure CLARUS</i>
Step 1	The IT team gets the configuration manual of CLARUS proxy corresponding to the installed version
Step 2	The member of the IT team in charge of the configuration opens an administrator session on the server where the CLARUS proxy is running
Step 3	The system controls the access rights of the IT team on the server
Step 4	The member of the IT team configures the user repository (see UC-ADM-04: 3.2.2.4)
Step 5	The member of the IT team configures the user authentication (see UC-ADM-05: 3.2.2.5)
Step 6	The member of the IT team configures the access to the CSP(s) (see UC-ADM-06: 3.2.2.6)
Variation 1	<i>Configure failover mode (see UC-ADM-07: 3.2.2.7)</i>
Post-conditions	
[Post-condition 1]	The CLARUS proxy is configured

3.2.2.4 Configure user repository (UC-ADM-04)

UC-ADM-04	<i>Configure user repository</i>
Actors	IT team
Description	The IT team wants to configure the CLARUS proxy on the local (trusted) information system to connect to a user repository for the management of users' access rights : - Either the IT user repository delivered by the organization (e.g. company LDAP) to which the CLARUS proxy must integrate if possible to search for users and their information (<i>main scenario</i>) - Or an embedded user repository (<i>alternate scenario, e.g., for CLARUS use outside the organization, CLARUS testing purpose</i>).
Pre-conditions	
[Pre-condition 1]	The [Pre-condition 1] and 'Step 1' to '3' described 'UC-ADM-03' are fulfilled.
[Pre-condition 2]	In case of an existing IT user repository, it must be configured to accept queries from the CLARUS proxy

Main scenario	<i>Configure to use the IT user repository</i>
Step 1	The member of the IT team sets and saves the access parameters of the IT user repository that will be used to search for users and their information
Step 2	The CLARUS proxy checks that the IT user repository is accessible
Variation 1	<i>Configure to use the embedded user repository (no IT user repository)</i>
Step 1	The member of the IT team sets and saves the parameters to use an embedded user repository
Step 2	The CLARUS proxy checks that the embedded user repository is accessible
Post-conditions	
[Post-condition 1]	The CLARUS proxy is configured to use the user repository made available (either IT user repository or an embedded user repository)

3.2.2.5 Configure user authentication (UC-ADM-05)

UC-ADM-05	<i>Configure user authentication</i>
Actors	IT team
Description	The IT team wants to configure the authentication mechanism of the CLARUS proxy running on the local (trusted) IT system
Pre-conditions	
[Pre-condition 1]	The [Pre-condition 1] and 'Step 1' to '3' described 'UC-ADM-03' are fulfilled.
[Pre-condition 2]	If a CLARUS proxy has been configured to use the embedded user repository (see UC-ADM-04: 3.2.2.4), it must be configured to use the embedded authentication mechanism (<i>alternate scenario</i>)
Main scenario	<i>Configure to use the IT authentication service</i>
Step 1	The member of the IT team sets and saves the access parameters of the IT authentication service (e.g. company SSO) that will be used to authenticate the users which access the CLARUS proxy
Step 2	The CLARUS proxy checks that the IT authentication service is accessible
Variation 1	<i>Configure to use the embedded authentication mechanism</i>
Step 1	The member of the IT team sets and saves the parameters to use the embedded authentication mechanism
Step 2	The member of the IT team launches a command that restarts the CLARUS proxy
Post-conditions	
[Post-condition 1]	The CLARUS proxy is configured to use either the IT authentication service or the embedded authentication mechanism

3.2.2.6 Configure access to the CSP(s) (UC-ADM-06)

UC-ADM-06	<i>Configure access to the CSP(s)</i>
Actors	IT team, cloud storage service (secondary)

Description	The IT team wants to configure the access to the cloud storage service of one (or more) CSP(s) towards which the end-users will be directed
Pre-conditions	
[Pre-condition 1]	The [Pre-condition 1] and 'Step 1' to '3' described 'UC-ADM-03' are fulfilled.
[Pre-condition 2]	For each CSP, the IT team has the cloud storage endpoint (URL) and credentials
Main scenario	<i>Configure access to the CSP(s)</i>
Step 1	The member of the IT team sets and saves the cloud storage endpoints (URLs) and credentials of one (or more) cloud storage service(s)
Step 2	The member of the IT team launches a command that restarts the CLARUS proxy
Step 3	The CLARUS proxy checks that each cloud storage service is accessible
Post-conditions	
[Post-condition 1]	The CLARUS proxy is configured to access the cloud storage service of one (or more) CSP(s) towards which the end-users will be directed.

3.2.2.7 Configure failover mode (UC-ADM-07)

UC-ADM-07	<i>Configure failover mode</i>
Actors	IT team
Description	The IT team wants to configure the CLARUS proxy running on the local (trusted) information system to be part of a failover cluster of CLARUS proxies
Pre-conditions	
[Pre-condition 1]	The [Pre-condition 1] and 'Step 1' described 'UC-ADM-03' are fulfilled <u>for several CLARUS proxies</u> .
[Pre-condition 2]	The CLARUS proxies has been configured to use the IT user repository
Main scenario	<i>Activate failover mode</i>
Step 1	The member of the IT team in charge of the CLARUS configuration opens an administrator session on each server where a CLARUS proxy is running
Step 2	On each server, the system controls the access rights of the IT team on the server
Step 3	On each server, the member of the IT team sets and saves the parameters so that the CLARUS proxies are part of the same failover cluster of CLARUS proxies
Step 4	On each server, the member of the IT team launches a command that restarts the CLARUS proxy
Step 5	On each server, the CLARUS proxy checks that it is part of a failover cluster of CLARUS proxies
Variation 1	<i>Deactivate failover mode</i>
Step 4	The member of the IT team sets and saves the parameters so that the CLARUS proxy is not part of any failover cluster of CLARUS proxies
Step 5	The member of the IT team launches a command that restarts the CLARUS proxy
Post-conditions	
[Post-condition 1]	If failover mode is activated, several CLARUS proxies are part of the same failover

	cluster of CLARUS proxies
[Post-condition 2]	If failover mode is activated, the CLARUS proxies that are part of the same failover cluster are accessible as a single CLARUS proxy
[Post-condition 3]	If failover mode is deactivated, the CLARUS proxy is not part of any failover cluster of CLARUS proxies

3.2.2.8 Implement CLARUS module (UC-IMPL-01)

UC-IMPL-01	<i>Implement CLARUS module</i>
Actors	IT development team
Description	The IT development team wants to implement a module in order to allow the CLARUS proxy to support features not natively supported (e.g. a protocol used to transfer data or a type of file used to store the data)
Pre-conditions	<i>None</i>
Main scenario	<i>Implement a module</i>
Step 1	The IT development team gets the development manual for the CLARUS proxy corresponding to the version of CLARUS that is/will be installed on the local (trusted) IT system
Step 2	The IT development team gets the CLARUS development API, and install it on the development platform
Step 3	The IT development team implements and tests a module following the development manual for the CLARUS proxy and using the CLARUS development API(s)
Post-conditions	
[Post-condition 1]	A new module has been implemented to allow the CLARUS proxy supporting features not natively implemented

3.2.2.9 Add CLARUS module (UC-ADM-08)

UC-ADM-08	<i>Add CLARUS module</i>
Actors	IT team
Description	The IT team wants to install one (or more) CLARUS module on the running CLARUS proxy, in order to support features not natively supported
Pre-conditions	
[Pre-condition 1]	A CLARUS proxy has been installed and configured on the IT system (e.g. a server in the users domain) by the IT team (see UC-ADM-01 to -07 :3.2.2.1)
[Pre-condition 1]	An additional CLARUS module has been developed (see UC-IMPL-01: 3.2.2.8)
Main scenario	<i>Add CLARUS module</i>
Step 1	The member of the IT team in charge of the installation opens an administrator session on the server where the CLARUS proxy is running
Step 2	The system controls the access rights of the IT team on the server
Step 3	The member of the IT team downloads the selected module on the server where the CLARUS proxy is running

Step 4	The member of the IT team updates the CLARUS proxy configuration to take the new module into account (e.g. copy in a dedicated folder)
Post-conditions	
[Post-condition 1]	The CLARUS proxy supports the new features brought by the add-on module

3.3 CLARUS adopting CSP installation

3.3.1 Overview

This scenario describes the actions required by a CSP for the integration of CLARUS in the cloud platform in order to provide the end-users with security-enhanced and trust-enhanced cloud services.

3.3.1.1 System

In this scenario the system is the information system on CSP.

3.3.1.2 UML diagram

The UML diagram below represents the corresponding use case (and main actor):

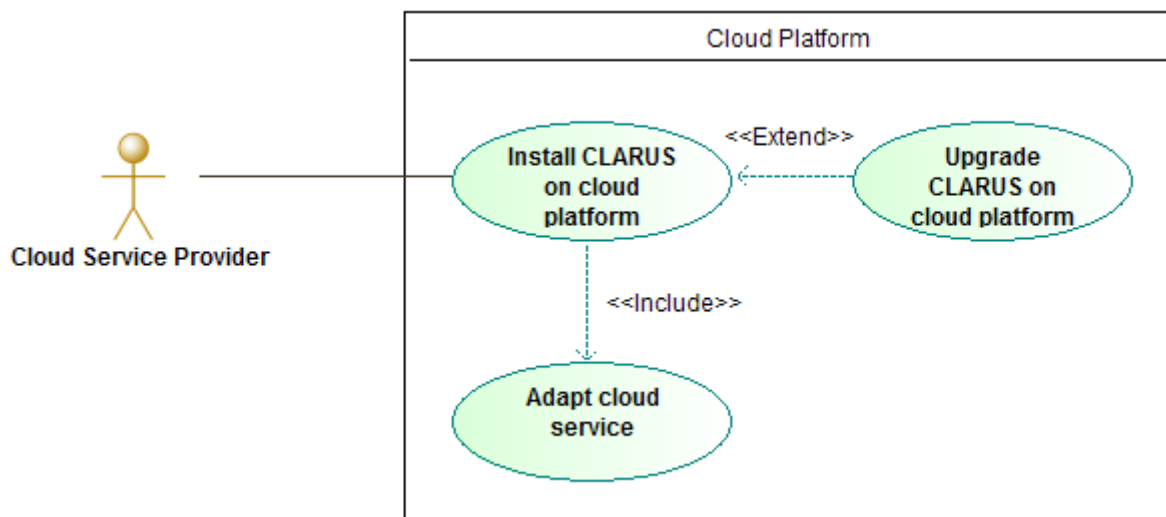


Figure 4 - CLARUS Proxy adopting CSP Installation Use Case Diagram

3.3.2 Use cases

3.3.2.1 Install CLARUS on cloud platform (UC-ADM-09)

UC-ADM-09	<i>Install CLARUS on cloud platform</i>
Actors	Cloud service provider
Description	The cloud service provider wants to integrate CLARUS in the cloud platform
Pre-conditions	
[Pre-condition 1]	No CLARUS software is installed yet

<i>[Pre-condition 2]</i>	Installation package(s) of CLARUS components is(are) available to enhance security and trust of cloud services
Main scenario	<i>Install CLARUS components</i>
Step 1	The cloud service provider gets the CSP installation guide of CLARUS in order to follow its instructions
Step 2	The cloud service provider installs one (or more) CLARUS component(s) (e.g. tool(s), service(s), library(ies))
Variation 1	<i>Adapt cloud service(s) (see UC-IMPL-02: 3.3.2.3)</i>
Post-conditions	
<i>[Post-condition 1]</i>	The cloud platform provides the end-users with security-enhanced and trust-enhanced cloud services, in a transparent way.

3.3.2.2 Upgrade CLARUS on cloud platform (UC-ADM-10)

UC-ADM-10	<i>Upgrade CLARUS on cloud platform</i>
Actors	Cloud service provider
Description	The cloud service provider wants to upgrade the CLARUS components installed on the cloud platform
Pre-conditions	
<i>[Pre-condition 1]</i>	Obsolete versions of CLARUS components are installed
<i>[Pre-condition 2]</i>	New installation package(s) of CLARUS components is(are) available
Main scenario	<i>Upgrade CLARUS components</i>
Step 1	The cloud service provider gets the CSP installation guide of CLARUS corresponding to the version to install in order to follow its upgrade instructions
Step 2	The cloud service provider upgrades one (or more) CLARUS component(s) (e.g. tool(s), service(s), library(ies))
Variation 1	<i>Adapt cloud service(s) (see UC-IMPL-02)</i>
Post-conditions	
<i>[Post-condition 1]</i>	The cloud platform provides the end-users with upgraded CLARUS security services

3.3.2.3 Adapt cloud service (UC-IMPL-02)

UC-IMPL-02	<i>Adapt cloud service</i>
Actors	Cloud service provider
Description	The cloud service provider wants to modify implementation of a cloud service in order to enhance security and trust capabilities
Pre-conditions	<i>None</i>
Main scenario	<i>Adapt cloud service</i>
Step 1	The cloud service provider gets the CLARUS development manual for CSP corresponding to the version of CLARUS library(ies) that is/will be installed on the cloud

	platform
Step 2	The cloud service provider gets the CLARUS library(ies), and installs it on the development platform
Step 3	The cloud service provider modifies the implementation of the cloud service following the CLARUS development manual for CSP and using the CLARUS library(ies)
Step 4	The cloud service provider deploys the new implementation of the cloud service with the CLARUS library(ies) on the cloud platform
Post-conditions	
<i>[Post-condition 1]</i>	A security-enhanced and trust-enhanced version of the cloud service has been deployed

3.4 Access control management

3.4.1 Overview

This scenario refers to the actions carried out for the purpose of access control management for both users of the organization and users of external organizations.

The users have a dedicated profile that includes access rights on the dataspace and appropriate materials to protect or access the outsourced datasets.

The users are also authorised to access the datasets.

3.4.1.1 System

In this scenario the system is the CLARUS proxy running on trusted local premises.

One CLARUS proxy is involved.

Additional CLARUS proxies may be involved (running on premises of the organization or of an external organization(s)) for the purpose of exchanging the user profiles (including access rights on dataspace and appropriate materials) and his/her authorizations in order to implement collaborative services.

3.4.1.2 UML diagram

The UML diagram below represents the corresponding use cases (and actors).

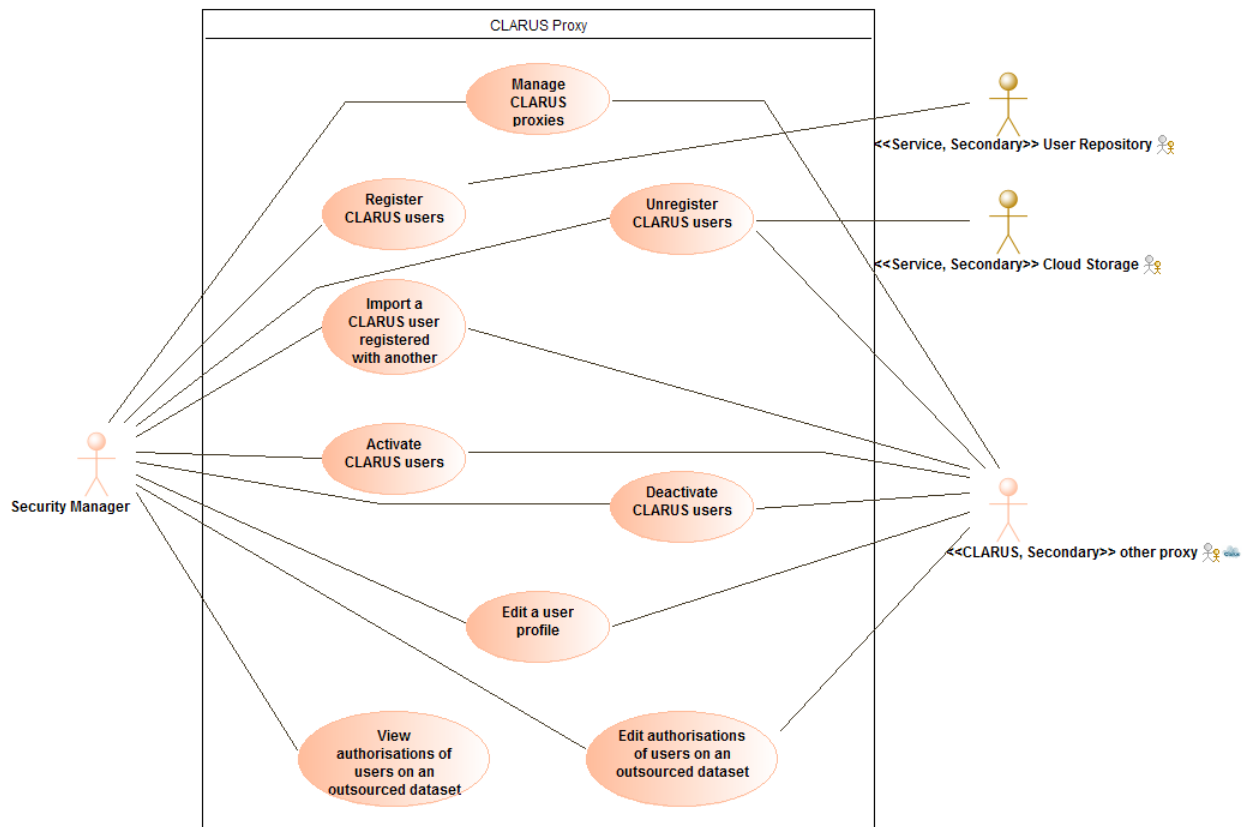


Figure 5 - Access Control Management Use Case Diagram

3.4.2 Use cases

3.4.2.1 Manage CLARUS proxies (UC-SEC-01)

UC-SEC-01	Manage CLARUS proxies
Actors	Security manager, other CLARUS proxy(ies) (secondary)
Description	The security manager wants to securely manage the list of CLARUS proxies that will interact with the system. The system must interact with other CLARUS proxies to exchange users' profile and authorizations in the following cases: - when a user of the organization needs to access outsourced dataset(s) through another CLARUS proxy running on premises of the organization (e.g. a proxy running on mobile device). In that case the other CLARUS proxy is part of the trusted domain, - when an external user of the organization need to access to outsourced dataset(s) managed by the system. In that case the other CLARUS proxy is not part of the trusted domain
Pre-conditions	
[Pre-condition 1]	The system is operational
[Pre-condition 2]	The security manager is connected to the CLARUS dashboard and has opened the 'CLARUS proxies manager'.

Main scenario	<i>Register a new CLARUS proxy</i>
Step 1	The security manager displays the registered CLARUS proxies list (with their status indicating if they are part of the trusted domain) to add a new one.
Step 2	The system prompts the security manager for the endpoint of the CLARUS proxy
Step 3	The security manager fills the endpoint of the CLARUS proxy
Step 4	The system checks the other CLARUS proxy is accessible
Step 5	The other CLARUS proxy returns basic information (e.g. organization name, CLARUS version)
Step 6	The system provides the security manager with the CLARUS proxy information and prompts to specify if the CLARUS proxy is part of the trusted domain
Step 7	The security manager indicates if the CLARUS proxy is part of the trusted domain
Step 8	The system registers the new CLARUS proxy and marks it as being (or not) part of the trusted domain
Variation 1	<i>Unregister a CLARUS proxy</i>
Step 1	The security manager selects a registered CLARUS proxy and wants to remove it
Step 2	The system checks that there is no imported users from the selected CLARUS proxy registered in the system
Step 3	The system prompts the security manager for confirmation
Step 4	The security manager confirms
Step 5	The system unregisters the selected CLARUS proxy
Post-conditions	
[Post-condition 1]	The system lists the registered CLARUS proxies with their status indicating if they are part (or not) of the trusted domain

3.4.2.2 Register CLARUS users (UC-SEC-02)

UC-SEC-02	<i>Register CLARUS users</i>
Actors	Security manager, user repository (secondary)
Description	The security manager wants to securely register users of the organization so that these users will be able to access the outsourced dataset(s) (according to their authorizations) A new registered user has a default profile: • no access rights on the existing dataspace(s), • appropriate materials for creation of outsourced dataset(s)) A new registered user has no authorizations on the outsourced dataset(s)
Pre-conditions	
[Pre-condition 1]	The system is configured to access the user repository provided by the IT infrastructure
[Pre-condition 2]	The security manager is connected to the CLARUS dashboard and has opened the 'CLARUS users manager'.
Main scenario	
Step 1	The security manager displays the registered users list to register a new one.

Step 2	The system prompts the security manager to fill user search attributes (e.g. login, mail address, name)
Step 3	The security manager partially or entirely fills the search attributes
Step 4	The system requests the user repository to search the users matching the given attributes
Step 5	The user repository returns the users matching the given attributes
Step 6	The system lists the users matching the given attributes. The system indicates if users are already registered
Step 7	The security manager selects one or several users to register
Step 8	The system registers the selected users, setting the default profile (no access rights on the existing dataspace(s) and appropriate materials required to create outsourced dataset(s) with the system)
Post-conditions	
[Post-condition 1]	The new registered users are activated, and have a default profile: - no access rights on the existing dataspace(s) - appropriate materials for creation of outsourced dataset(s) The new registered users have no authorizations on the outsourced dataset(s)

3.4.2.3 Import a user registered with another CLARUS proxy (UC-SEC-03)

UC-SEC-03	<i>Import a user registered with another CLARUS proxy</i>
Actors	Security manager, other CLARUS proxy (secondary)
Description	The security manager wants to securely register a user which is registered with another CLARUS proxy If the other CLARUS proxy is part of the trusted domain, then the profile and authorizations of the user are imported If the other CLARUS proxy is not part of trusted domain, authorizations must be empty and a default profile must be created for the imported user: - no access rights on the existing dataspace(s), - appropriate materials for creation of outsourced dataset(s) must be imported
Pre-conditions	
[Pre-condition 1]	The system is configured to access the other CLARUS proxy
[Pre-condition 2]	The import of the user is authorized by the other CLARUS proxy
[Pre-condition 3]	The security manager is connected to the CLARUS dashboard and has opened the 'CLARUS user manager'.
Main scenario	<i>The other CLARUS proxy is not part of the trusted domain</i>
Step 1	The security manager displays the registered users list to import a user
Step 2	The system lists the registered CLARUS proxies and prompts the security manager to select one and to fill user login
Step 3	The security manager selects a CLARUS proxy and fulfills the user login

Step 4	The system requests the other CLARUS proxy to identify the user
Step 5	The other CLARUS proxy validates if the system is authorized to import the user
Step 6	The system shows the user found, indicates if the user is already imported in the system and prompts for confirmation
Step 7	The security manager confirms
Step 8	The system registers the user with a default profile (no access rights on the existing dataspace(s) and importing appropriate materials for creation of outsourced dataset(s)) Moreover the imported user has no authorizations on the outsourced dataset(s)
Step 9	The system notifies the other CLARUS proxy the user has been imported by the system
Step 10	The other CLARUS proxy logs that the user is exported to the system
Variation 1	<i>The other CLARUS proxy is part of the trusted domain</i>
Step 8	The system requests the other CLARUS proxy for the user profile and his/her authorizations on the outsourced dataset(s)
Step 9	The other CLARUS proxy returns the user profile and his/her authorizations on the outsourced dataset(s)
Step 10	The system registers the user with his/her profile and his/her authorizations
Step 11	The system notifies the other CLARUS proxy that the user has been imported by the system
Step 12	The other CLARUS proxy logs that the user is exported to the system
Post-conditions	
[Post-condition 1]	The imported user is activated. The profile and authorizations depends on the other CLARUS proxy belonging (or not) to the trusted domain: If the other CLARUS proxy is part of the trusted domain: - the user profile is imported: - access rights on dataspace(s), - appropriate materials for creation of outsourced dataset(s), - user authorizations on outsourced dataset(s) are imported: - with appropriate materials for accessing to outsourced dataset(s), If the other CLARUS proxy is not part of the trusted domain: - the imported user has a default profile: - noaccess rights on dataspace(s), - appropriate materials for creation of outsourced dataset(s), - no authorizations on outsourced dataset(s)
[Post-condition 2]	The system lists the imported users with the registered users

3.4.2.4 Deactivate CLARUS users (UC-SEC-04)

UC-SEC-04	<i>Deactivate CLARUS users</i>
Actors	Security manager, other CLARUS proxy(ies) (secondary)
Description	The security manager wants to securely deactivate registered users, so that these users will be denied to access the outsourced dataset(s)
Pre-conditions	

<i>[Pre-condition 1]</i>	At least one registered user is activated
<i>[Pre-condition 2]</i>	The security manager is connected to the CLARUS dashboard and has opened the 'CLARUS user manager'.
Main scenario	<i>Deactivate users which are not imported and which are not exported</i>
Step 1	The security manager displays the registered users list (with their activation status) to deactivate one or several users.
Step 2	The system prompts the security manager for confirmation
Step 3	The security manager confirms : the system deactivates the selected users
Variation 1	<i>Deactivate users which are exported or which are imported</i>
Step 4	The system notifies the other CLARUS proxy(ies) that the users are deactivated
Step 5	The other CLARUS proxy(ies) deactivate access for the users to the outsourced dataset(s) managed by the system
Post-conditions	
<i>[Post-condition 1]</i>	The selected users are deactivated and are denied to access the outsourced dataset(s)
<i>[Post-condition 2]</i>	The system keeps the profiles, authorizations and all appropriate materials of the deactivated users
<i>[Post-condition 3]</i>	On the other CLARUS proxy(ies) , the selected users are deactivated and are denied to access the outsourced dataset(s) managed by the system
<i>[Post-condition 4]</i>	On the other CLARUS proxy(ies) , the system keeps the profiles, authorizations and all appropriate materials of the deactivated users
<i>[Post-condition 5]</i>	The system lists the registered users with their new activation status

3.4.2.5 Activate CLARUS users (UC-SEC-05)

UC-SEC-05	<i>Activate CLARUS users</i>
Actors	Security manager, other CLARUS proxy(ies) (secondary)
Description	The security manager wants to securely activate registered users, so that these users will be allowed to access the outsourced dataset(s)
Pre-conditions	
<i>[Pre-condition 1]</i>	At least one registered user is deactivated
<i>[Pre-condition 2]</i>	The security manager is connected to the CLARUS dashboard and has opened the 'CLARUS user manager'
Main scenario	<i>Activate user(s) which are not imported and which are not exported</i>
Step 1	The security manager displays the registered users list (with their activation status) to activate one or several
Step 2	The system prompts the security manager for confirmation
Step 3	The security manager confirms : the system activates the selected users
Variation 1	<i>Activate user(s) which are exported or which are imported</i>
Step 4	The system notifies the other CLARUS proxy(ies) that the users are activated
Step 5	The other CLARUS proxies activates access for the users to the outsourced dataset(s) managed by the system

Post-conditions	
[Post-condition 1]	The selected users are activated and are allowed to access the outsourced dataset(s)
[Post-condition 2]	On the other CLARUS proxy(ies) , the selected users are activated and are allowed to access the outsourced dataset(s) managed by the system
[Post-condition 3]	The system lists the registered users with their new activation status

3.4.2.6 Unregister a CLARUS user (UC-SEC-06)

UC-SEC-06	Unregister a CLARUS user
Actors	Security manager, other CLARUS proxy(ies) (secondary) , cloud storage service(s) (secondary)
Description	The security manager wants to securely unregister a user, deleting his/her profile, his/her authorizations and all appropriate materials so that the user could definitively not access the outsourced dataset(s) Outsourced dataset(s) owned by the unregistered user are also definitively deleted if no other user has authorizations on it
Pre-conditions	
[Pre-condition 1]	At least one registered user is deactivated
[Pre-condition 2]	The security manager is connected to the CLARUS dashboard and has opened the 'CLARUS user manager'.
Main scenario	Unregister a user which is not imported and which is not exported
Step 1	The security manager displays the registered users list (with their activation status) to unregister a deactivated user.
Step 2	The system prompts the security manager for confirmation
Step 3	The security manager confirms : the system unregisters the selected user, deleting his/her profile and his/her authorizations
Variation 0 bis	Remove user's outsourced dataset(s) on which no other users have authorizations
Step 4	The system warns the security manager that the outsourced dataset(s) created by the user will be definitively deleted because no other users have authorizations on it. The system prompts for acknowledgment
Step 5	The security manager validates
Step 6	The system requests the cloud storage service(s) to delete the outsourced dataset(s) created by the user
Variation 0 ter	Remove user's outsourced dataset(s) on which other users have authorizations
Step 6	The system warns the security manager that other users have authorizations on the outsourced dataset(s) created by the unregistered user and prompts for confirmation of the deletion
Step 7	The security manager confirms
Step 8	The system requests the cloud storage service(s) to delete the outsourced dataset(s) created by the user
Variation 1	Unregister an exported user

Step 6	The system notifies the other CLARUS proxy(ies) that the user is unregistered from the system
Step 7	The other CLARUS proxy(ies) unregisters the selected user, deleting his/her profile and his/her authorizations on the outsourced dataset(s)
Variation 2	<i>Unregister an imported user</i>
Step 6	The system notifies the other CLARUS proxy that the user is unregistered from the system
Step 7	The other CLARUS proxy deletes authorizations on the outsourced dataset(s)
Post-conditions	
[Post-condition 1]	The selected users is unregistered and could definitively not access the outsourced dataset(s) managed by the system
[Post-condition 2]	The profile (with appropriate materials for creation of outsourced dataset(s)) and the authorizations (with appropriate materials for accessing to outsourced dataset(s) managed by the system) of the unregistered user are deleted
[Post-condition 3]	On other CLARUS proxy(ies) where the user is exported , the profile (with appropriate materials for creation of outsourced dataset(s)) and the authorizations (with appropriate materials for accessing to outsourced dataset(s) managed by the system) of the unregistered user are deleted
[Post-condition 4]	On the other CLARUS proxy from which the user has been imported, the authorizations (with appropriate materials for accessing to outsourced dataset(s) managed by the system) are deleted
[Post-condition 5]	The outsourced dataset(s) created by the unregistered user are deleted
[Post-condition 6]	The system lists the registered users. The unregistered user is not listed anymore

3.4.2.7 Edit a user profile (UC-SEC-07)

UC-SEC-07	<i>Edit a user profile</i>
Actors	Security manager, other CLARUS proxy(ies) (secondary)
Description	The security manager wants to securely edit a user profile A user profile contains the access rights of a user on the dataspace(s) Access rights consist of read permission and write permission on a dataspace
Pre-conditions	
[Pre-condition 1]	At least one registered user
[Pre-condition 3]	The security manager is connected to the CLARUS dashboard and has opened the 'CLARUS user manager'.
Main scenario	<i>Edit profile of a user which is not exported</i>
Step 1	The security manager displays the registered users list to edit the profile
Step 2	The system shows the user profile: the list of dataspace(s) with the user access rights (read/write permissions)
Step 3	The security manager selects/unselects (read or write) permissions for each dataspace

Step 4	The system updates the user profile with the new access rights on the dataspace(s)
Variation 0 bis	<i>Edit profile of an exported user</i>
Step 5	The system notifies the other CLARUS proxy(ies) which are part of the trusted domain that the user profile has changed
Step 6	The other CLARUS proxy(ies) which are part of the trusted domain updates the user profile with the new access rights on the dataspace(s)
Variation 1	<i>Manage appropriate materials for creation of outsourced dataset(s) of a user which is not exported</i>
Step 3	The security manager wants to generate materials for creation of outsourced dataset(s)
Step 4	The system prompts the security manager for confirmation
Step 5	The system generates the appropriate materials and updates the user profile with it
Variation 1 bis	<i>Manage appropriate materials for creation of outsourced dataset(s) of an exported user</i>
Step 6	The system notifies the other CLARUS proxy(ies) that the appropriate materials for creation of outsourced dataset(s) has changed for the user
Step 7	The other CLARUS proxy(ies) updates the user profile with the new appropriate materials
Post-conditions	
[Post-condition 1]	The user profile (access rights on the dataspace(s) and the eventual new appropriate materials for creation of outsourced dataset(s)) are modified
[Post-condition 2]	On other CLARUS proxies where the user is exported , the user profile (access rights on the dataspace(s) and the eventual new appropriate materials for creation of outsourced dataset(s)) are modified

3.4.2.8 View authorisations of users on an outsourced dataset (UC-SEC-08)

UC-SEC-08	<i>View authorisations of users on an outsourced dataset</i>
Actors	Security manager
Description	The security manager wants to securely view the authorizations (read, update, delete) and access level of the users on an existing outsourced dataset
Pre-conditions	
[Pre-condition 1]	
[Pre-condition 2]	At least one outsourced dataset has been created within a dataspace through the system
[Pre-condition 2]	The security manager is connected to the CLARUS dashboard and has opened the 'CLARUS dataspace manager'
Main scenario	
Step 1	The security manager displays the dataspace(s) list to select a dataspace
Step 2	The system lists the outsourced dataset(s) stored within the selected dataspace
Step 3	The security manager selects an outsourced dataset to view the authorizations
Step 4	The system shows the owner of the outsourced dataset and lists the users which have

	authorizations on the outsourced dataset
Step 5	The security manager selects a user (among the authorized ones)
Step 6	The system shows: - the authorizations the selected user has on the outsourced dataset: read, update, delete - the access level the selected user has on the outsourced dataset (e.g. a degree of coarsening/generalization or a percentage of the accuracy)
Post-conditions	
[Post-condition 1]	The authorizations and access levels of the users on existing outsourced dataset(s) has been displayed (without any change)

3.4.2.9 Edit authorisations of users on an outsourced dataset (UC-SEC-09)

UC-SEC-09	<i>Edit authorisations of users on existing outsourced dataset(s)</i>
Actors	Security manager
Description	The security manager wants to securely edit the permissions (read, update, delete) and access levels of the users on an existing outsourced dataset to make some changes
Pre-conditions	
[Pre-condition 1]	The [Pre-condition 1 to 3] and 'Step 1' to '6' described in 'UC-SEC-08' (3.4.2.8) are fulfilled : with the intention to edit authorizations to make an update in Step 3
Main scenario	Modify authorizations of a user which is not imported and which is not exported
Step 1	The security manager selects or unselects the read, update and delete authorizations displayed by the system.
Step 2	The security manager modifies the access level (e.g. a percentage of the accuracy or a degree of coarsening/generalization)
Step 3	The security manager confirms the modifications
Step 4	The system saves the modified authorizations and the modified access level
Step 5	The system modifies the user's appropriate materials for accessing the selected outsourced dataset, according to the modified authorizations and the modified access level
Variation 0 bis	Modify authorizations of an imported user or of an exported user
Step 6	The system notifies the other CLARUS proxy(ies) of the modified authorizations, the modified access level and modified appropriate materials for accessing the outsourced dataset
Step 7	The other CLARUS proxies overwrite authorizations, access level and appropriate materials for accessing the selected data
Variation 1	Add authorizations for a user which is not imported and which is not exported
Step 1	The security manager adds authorizations for a user
Step 2	The system lists the registered users that don't have authorizations defined on the outsourced dataset
Step 3	The security manager selects a user
Step 4	The system saves the default authorizations (read only) and default access level for the

	user
Step 5	The security manager selects or unselects the read, update and delete authorizations
Step 6	The security manager modifies the access level (e.g. a percentage of the accuracy or a degree of coarsening/generalization)
Step 7	The security manager confirms the modifications
Step 8	The system saves the new authorizations and the new access level
Step 9	The system creates appropriate materials for accessing the selected outsourced dataset, according to the new authorizations and the new access level
Variation 1 bis	Add authorizations for an imported user or for an exported user
Step 10	The system notifies the other CLARUS proxy(ies) of the new authorizations, the new access level and new appropriate materials for accessing the selected outsourced dataset
Step 11	The other CLARUS proxy(ies) save new authorizations, new access level and new appropriate materials for accessing the selected outsourced dataset
Variation 2	Remove authorizations of a user which is not imported and which is not exported
Step 1	The security manager selects a user to remove authorization
Step 2	The system prompts the security manager for confirmation
Step 3	The security manager confirms
Step 4	The system deletes the authorizations and access level for the selected user
Step 5	The system deletes materials for accessing the selected outsourced dataset
Step 6	The system lists the users which have authorizations on the outsourced dataset. The user for which authorizations have been removed does not appear anymore
Variation 2 bis	Remove authorizations of an imported user or of an exported user
Step 7	The system notifies the other CLARUS proxy(ies) that the authorizations, access level and appropriate materials for accessing the selected outsourced dataset have been removed
Step 8	The other CLARUS proxy(ies) delete authorizations, access level and appropriate materials for accessing the selected outsourced dataset
Post-conditions	
[Post-condition 1]	The authorizations and access levels of the users on existing outsourced dataset(s) have been modified on the system
[Post-condition 2]	The appropriate materials for accessing the outsourced dataset(s) managed by the system have been modified on the system
[Post-condition 3]	On the other CLARUS proxy(ies) , the authorizations and access levels of the users on existing outsourced dataset(s) managed by the system have been modified
[Post-condition 3]	On the other CLARUS proxy(ies) , the appropriate materials for accessing existing the outsourced dataset(s) managed by the system have been modified

3.5 Security policy management

3.5.1 Overview

This scenario describes how a security manager can define and configure the security policy for a CLARUS proxy.

Security policy is managed through the notion of dataspace. A dataspace is a virtual space that links the applications/services running on IT premises with the cloud storage services.

The security manager configures the dataspace selecting the protocols that are required to transfer data from the applications/services to the cloud and configuring the internal and external endpoints.

The security manager defines the security policy to apply to the dataspace selecting the data attributes to protect and defining the usages on the data.

Finally, the security manager refines the appropriate security techniques that are automatically selected by the system.

The system will automatically apply the associated security policy on the datasets created by the users. The system determines the dataspace of a dataset thanks to the endpoints used by the applications/services.

3.5.1.1 System

In this scenario the system is the CLARUS proxy running on trusted local premises.

3.5.1.2 UML diagram

The UML diagram below represents the corresponding use case (and actors).

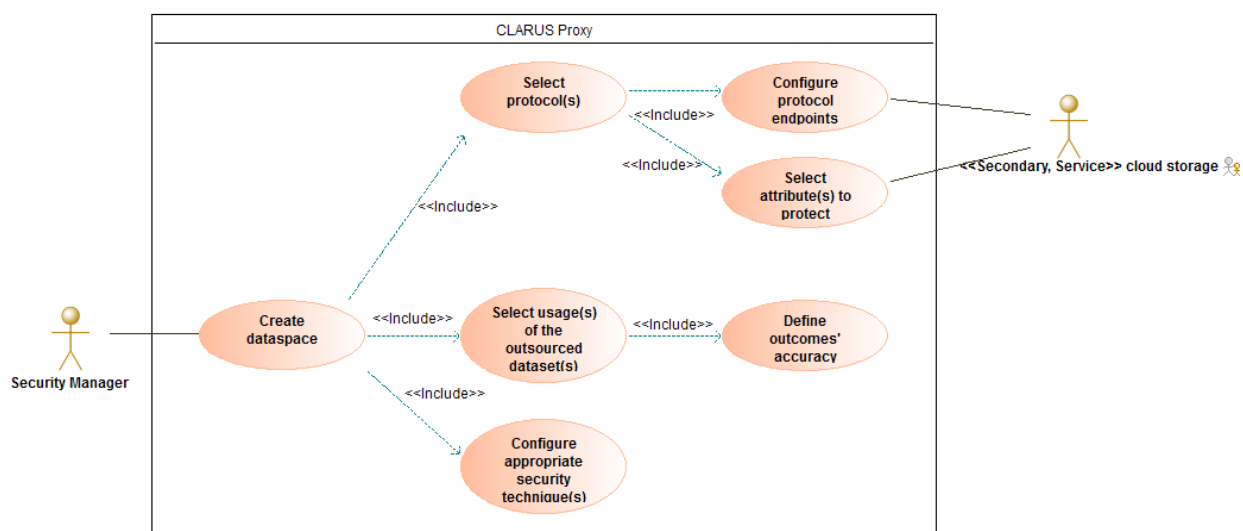


Figure 6 - Security Policy Management Use Case Diagram

3.5.2 Use cases

3.5.2.1 Create dataspace (UC-SEC-10)

UC-SEC-10	<i>Create dataspace</i>
Actors	Security manager
Description	The security manager wants to create a new dataspace, defining the protocol(s) for access, the usage of outsourced dataset(s) within the dataspace and configuring appropriate security techniques
Pre-conditions	
[Pre-condition 1]	The security manager navigates through the CLARUS dashboard and has opened the 'CLARUS dataspace manager'
Main scenario	<i>Create a dataspace</i>
Step 1	The security manager fills the name of the new dataspace
Step 2	The security manager selects and configures the protocol(s) (see UC-SEC-11 : 3.5.2.2)
Step 3	The security manager selects the usage(s) of the outsourced dataset(s) within the new dataspace (see UC-SEC-14 : 3.5.2.5)
Step 4	The security manager configures appropriate security technique(s) (see UC-SEC-16 : 3.5.2.7)
Step 5	The system creates the new dataspace
Post-conditions	
[Post-condition 1]	A new dataspace has been created and configured

3.5.2.2 Select protocol(s) (UC-SEC-11)

UC-SEC-11	<i>Select protocol(s)</i>
Actors	Security manager
Description	The security manager wants to select and configure the protocol(s) to access outsourced dataset(s) within an existing dataspace
Pre-conditions	
[Pre-condition 1]	The security manager has created a dataspace (see UC-SEC-10 : 3.5.2.1)
Main scenario	<i>Select protocol(s)</i>
Step 0	The system provides the security manager with a the list of protocols supported by the CLARUS proxy
Step 1	The security manager selects the appropriate protocol(s)
Step 2	The security manager configures endpoints for each selected protocol (see UC-SEC-12 : 3.5.2.3)
Step 3	The security manager selects attribute(s) to protect for each selected protocol (see UC-SEC-13 : 3.5.2.4)
Step 4	The system saves the selected protocol(s) with their configuration
Post-conditions	

<i>[Post-condition 1]</i>	The protocol(s) to access outsourced dataset(s) within a dataspace are selected and configured
---------------------------	--

3.5.2.3 Configure protocol endpoint(s) (UC-SEC-12)

UC-SEC-12	<i>Configure protocol endpoint(s)</i>
Actors	Security manager, cloud storage service(s) (secondary)
Description	The security manager wants to configure a selected protocol to access outsourced dataset(s) within a dataspace
Pre-conditions	
<i>[Pre-condition 1]</i>	The security manager has created a dataspace (see UC-SEC-10 : 3.5.2.1) and selected a protocol (see UC-SEC-11 : 3.5.2.2)
Main scenario	<i>Configure protocol endpoints</i>
Step 0	The system prompts the security manager to fill the parameters to build the input endpoint (used by IT applications/services) and the output endpoint(s) (used to access the cloud storage service(s))
Step 1	The security manager fills the parameters to build the input endpoint
Step 2	For each CSP(s) access configured in the CLARUS proxy (see UC-ADM-06 : 3.2.2.6), the security manager optionally fills the endpoint URL for the dataspace
Step 3	For each CSP, the system checks the endpoint URL is coherent and compatible with the configuration of access to the CSP
Step 4	For each CSP, the system verifies the cloud storage service is accessible with the endpoint URL
Step 5	For each CSP, the system prepares (or initializes) a backend for the dataspace (e.g. a bucket or a database)
Step 6	The system saves the configuration of the selected protocol
Post-conditions	
<i>[Post-condition 1]</i>	The selected protocol is configured to access outsourced dataset(s) within a dataspace

3.5.2.4 Select attribute(s) to protect (UC-SEC-13)

UC-SEC-13	<i>Select attribute(s) to protect</i>
Actors	Security manager, cloud storage service(s) (secondary)
Description	The security manager selects attributes to protect when a user accesses outsourced dataset(s) within the dataspace being created
Pre-conditions	
<i>[Pre-condition 1]</i>	The supported transfer protocols (e.g., FTP, SCP, HTTP, etc.) and their description are provided through: - the modules implemented by the CLARUS team, - the modules implemented by the IT development team
<i>[Pre-condition 2]</i>	The supported file types and description of their structure are provided through: the modules implemented by the CLARUS team,

	the modules implemented by the IT development team
<i>[Pre-condition 3]</i>	The structure of a database are provided through: - a generic module (implemented by the CLARUS team) that is able to retrieve the database structure from any database, - a module implemented by the IT development team that provides a static description of the database structure
<i>[Pre-condition 4]</i>	The security manager has created a dataspace (see UC-SEC-10 : 3.5.2.1) , selected and configured a protocol (see UC-SEC-11 and UC-SEC-12 : 3.5.2.2 and 3.5.2.3)
Main scenario	<i>Select protocol attribute(s) to protect</i>
Step 0	The system provides the security manager with the list of protocol's attributes that could bring data to protect
Step 1	The security manager selects the attribute(s) to protect
Step 2	The system saves the attribute(s) to protect
Variation 1	<i>Select file type(s) to protect</i>
Step 0	For a file transfer protocol, the security manager wants to protect some file(s)
Step 1	The system provides the security manager with the list of supported types of file
Step 2	The security manager selects the types of file to protect
Step 3	For each type of file to protect, the security manager indicates to protect the entire file
Step 4	The system saves the type(s) of file to protect entirely
Variation 1 bis	<i>Select attribute(s) of file type(s)</i>
Step 3	For each type of file to protect, the security manager wants to select attribute(s) to protect
Step 4	The system provides the security manager with the structure of the file, showing the attributes that could bring data to protect
Step 5	The security manager selects the attribute(s) to protect
Step 6	The system saves the attribute(s) to protect
Variation 2	<i>Select database table(s) and column(s) to protect</i>
Step 0	For a database protocol, the security manager wants to protect table(s) and column(s)
Step 1	The system provides the security manager with a list of database schema
Step 2	The security manager selects a database schema
Step 3	The security manager selects the table(s) and the column(s) to protect
Step 4	The system saves the name(s) of the table(s) and of the column(s) to protect
Variation 2 bis	<i>Select database table(s) and column(s) to protect from the cloud storage service(s)</i>
Step 1	The system retrieves the database schema from the cloud storage service(s) through the protocol's endpoint(s)
Step 2	The system provides the security manager with the database schema
Step 3	The security manager selects the table(s) and the column(s) to protect
Step 4	The system saves the name(s) of the table(s) and of the column(s) to protect
Post-conditions	
<i>[Post-condition 1]</i>	Protocol(s) attribute(s) to protected are selected
<i>[Post-condition 2]</i>	File type(s) to protected entirely are selected

[Post-condition 3]	File type(s) attribute(s) to protected are selected
[Post-condition 4]	Database table(s) and column(s) to protected are selected

3.5.2.5 Select usage(s) of the outsourced dataset(s) (UC-SEC-14)

UC-SEC-14	<i>Select usage(s) of the outsourced dataset(s)</i>
Actors	Security manager
Description	The security manager selects the usage(s) of the outsourced dataset(s) within a dataspace. The selected usage(s) shall cover one of the following scenarios: • User data storage (see 3.8) • Search in the cloud (see 3.9) • Update in the cloud (see 3.10) • Compute in the cloud (see 3.11)
Pre-conditions	
[Pre-condition 1]	The security manager has created a dataspace (see UC-SEC-10 : 3.5.2.1) , selected the protocol(s) and configured their endpoints (see UC-SEC-11 and UC-SEC-12 : 3.5.2.2 and 3.5.2.3), and selected the attribute(s) to protect (see UC-SEC-13 : 3.5.2.4)
Main scenario	<i>Select usage(s) of the outsourced dataset(s)</i>
Step 0	The system provides the security manager with the list of possible usages
Step 1	The security manager selects one (or more) usage(s)
Step 2	The system verifies the selected usage(s) are not incompatible with each other(s)
Step 3	For each selected usage(s), the security manager defines the outcomes' accuracy (see UC-SEC-15 : 3.5.2.4)
Step 3	The system saves the usage(s) of the outsourced dataset(s) within the dataspace
Post-conditions	
[Post-condition 1]	The usage(s) has been defined for the outsourced dataset(s) within a dataspace

3.5.2.6 Define outcomes' accuracy (UC-SEC-15)

UC-SEC-15	<i>Define outcomes' accuracy</i>
Actors	Security manager
Description	The security manager defines the accuracy of the outcomes for a selected usage of the outsourced dataset(s) within the dataspace being created
Pre-conditions	
[Pre-condition 1]	The security manager has created a dataspace (see UC-SEC-10 : 3.5.2.1) , selected the protocol(s) and configured their endpoints (see UC-SEC-11 and UC-SEC-12 : 3.5.2.2 and 3.5.2.3), and selected the attribute(s) to protect (see UC-SEC-13 : 3.5.2.4)
[Pre-condition 5]	The security manager has selected an usage (see UC-SEC-14 : 3.5.2.5)
Main scenario	<i>Define outcomes' accuracy</i>
Step 0	The system prompts the security manager to define the accuracy of the outcomes for

	the selected usage
Step 1	The security manager defines the accuracy of the outcomes (e.g. with a degree of coarsening/generalization or with a percentage of the accuracy)
Step 2	The system saves the outcomes' accuracy of the selected usage
Post-conditions	
[Post-condition 1]	The outcomes' accuracy has been defined for the selected usage

3.5.2.7 Configure appropriate security technique(s) (UC-SEC-16)

UC-SEC-16	<i>Configure appropriate technique rule(s)</i>
Actors	Security manager
Description	The security manager configures the security techniques that are appropriate according to the dataspace's configuration (protocol(s), attribute(s) to protect, usage(s), outcomes' accuracy(ies))
Pre-conditions	
[Pre-condition 1]	The security manager has created a dataspace (see UC-SEC-10 : 3.5.2.1) , selected the protocol(s) and configured their endpoints (see UC-SEC-11 and UC-SEC-12 : 3.5.2.2 and 3.5.2.3), and selected the attribute(s) to protect (see UC-SEC-13 : 3.5.2.4)
[Pre-condition 2]	The security manager has selected the usage of the outsourced dataset(s) (see UC-SEC-14 : 3.5.2.5) and outcome's accuracies (see UC-SEC-15 : 3.5.2.6)
Main scenario	<i>Configure appropriate technique rule(s)</i>
Step 0	The system computes the list of security techniques that are appropriate according to the dataspace's configuration
Step 1	If several combinations of appropriate security techniques are possible, the system prompts the security manager to choose one of the combinations
Step 2	The security manager choose one of the combinations of appropriate security techniques
Step 3	For each security technique of the selected combination, the system prompts the security manager to refine the configuration to apply within the dataspace being created
Step 4	For each security technique of the selected combination, the security manager refines the configuration filling the required parameters
Step 5	The system saves the combination of the security techniques with their configuration to apply within the dataspace being created
Post-conditions	
[Post-condition 1]	The combination of the security techniques with their configuration has been defined

3.6 Standard monitoring

3.6.1 Overview

This scenario refers to the actions for enabling standard monitoring of CLARUS proxy.

3.6.1.1 System

In this scenario the system encompasses:

- The CLARUS proxy running on trusted local premises,
- The CLARUS adopting CSP.

One CLARUS proxy is involved. Additional CLARUS proxies may be involved, and monitoring should correlate the information provided by distributed CLARUS proxies.

3.6.1.2 UML diagram

The UML diagram below represents the corresponding use case (and actors).

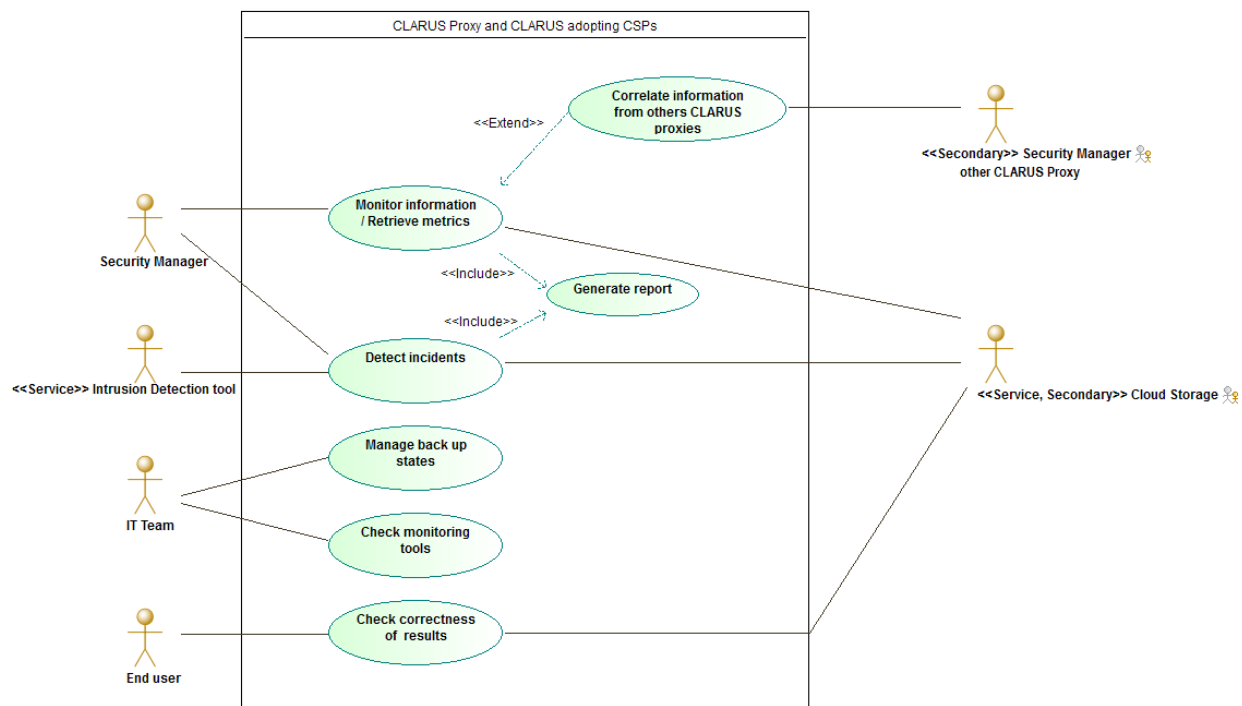


Figure 7 - Standard Monitoring Use Case Diagram

3.6.2 Use cases

3.6.2.1 Monitor information / Retrieve metrics (UC-MON-01)

UC-MON-01	Monitor information / Retrieve metrics
Actors	Security manager
Description	The Security Manager wants to supervise the standard use of CLARUS features in order to detect potential incidents

Pre-conditions	
[Pre-condition 1]	Monitoring tool installed in the system
[Pre-condition 2]	A risk analysis is already performed to define sensitive data to protect and security and business requirements. It also defines different risks and vulnerabilities of the proposed CLARUS solution.
[Pre-condition 3]	List of metrics and indicators to be monitored
Main scenario	<i>Periodical monitoring of the system status</i>
Step 1	The Security Manager connects to the monitoring interface
Step 2	The Security Manager selects the list of relevant reports (including business and security metrics) that he/she needs to monitor
Step 3	The monitoring tool provides the values of the requested metrics (as a set of graphs and tables) in a web-based report
Step 4	The Security Manager checks if there is any security incident and acts in consequence (See UC-MON-02 : 3.6.2.2)
Post-conditions	
[Post-condition 1]	The monitoring informations and relevant metrics retrieved are synthetised in a web-based reporting output.

3.6.2.2 Detect incidents (UC-MON-02)

UC-MON-02	<i>Detect incidents</i>
Actors	Security Manager
Description	The Security Manager supervising the standard use of CLARUS features detects incidents via the monitoring tools
Pre-conditions	
[Pre-condition 1]	List of security incidents to be detected already defined, as well as rules to detect those incidents from monitored information
[Pre-condition 2]	The monitoring tool continuously supervise the system at different strategic observation points (including the client device, the proxy, and the cloud provider)
Main scenario	<i>Security incident detection (manual response)</i>
Step 1	The monitoring tool detects a security incident
Step 2	The Security Manager is notified of the detected security incident (e.g., via the Web report)
Step 3	The Security Manager decides to apply a counter-measure to mitigate the detected security incident
Step 4	The Security Manager checks the impact of the security incident
Variation 1	<i>Security incident detection (automatic response)</i>
Step 2	The monitoring tool automatically executes the necessary counter-measures to mitigate the security incident
Step 3	The monitoring tool notifies the Security Manager about the security incident occurrence and the counter-measures taken

Step 4	The Security Manager checks the impact of the security incident
Post-conditions	
[Post-condition 1]	The detected incidents and taken counter-measures are synthetised in a web-based reporting output

3.6.2.3 Manage back up states (UC-MON-03)

UC-MON-03	<i>Manage back up states</i>
Actors	IT Team
Description	The IT Team handle some back-up states in order to prevent incidents as CLARUS failures, requiring to configure or restore a new CLARUS proxy
Pre-conditions	
[Pre-condition 1]	The IT Team provides a back-up state
Main scenario	<i>Restore a previous back-up state or configure the CLARUS proxy</i>
Step 1	The IT Team connects to the CLARUS proxy
Step 2	The IT Team restores/reconfigures the CLARUS proxy using a previous back-up state
Variation 1	<i>Make a periodical proxy configuration back-up</i>
Step 2	The IT Team copies the CLARUS proxy configuration to create a back-up state
Post-conditions	
[Post-condition 1]	A new CLARUS proxy is reconfigured or initial CLARUS proxy is restored after an incident has been detected

3.6.2.4 Check monitoring tools (UC-MON-04)

UC-MON-04	<i>Check monitoring tools</i>
Actors	IT Team
Description	The IT Team can check the functionality and performance of CLARUS monitoring tools in a dedicated integration environment
Pre-conditions	
[Pre-condition 1]	Monitoring tools already installed and configured in the system
Main scenario	<i>Functionality and performance checking of security tools</i>
Step 1	The IT Team connects to the monitoring tools management environment
Step 2	The IT Team checks the functionality of the monitoring tools
Step 3	The IT Team checks the performance of the monitoring tools
Step 4	The IT Team provides a report on the functionality and performance status of the monitoring tools
Post-conditions	
[Post-condition 1]	The functionality and performance of CLARUS monitoring tools has been tested in a

	dedicated integration environment
--	-----------------------------------

3.6.2.5 Check the correctness of results (UC-MON-05)

UC-MON-05	<i>Check the correctness of results</i>
Actors	End user, IT team
Description	The end-user checks that the data storage/search/update /computation has been correctly performed
Pre-conditions	
[Pre-condition 1]	The end-user is authorised to access the system
[Pre-condition 2]	The end-user has performed a data storage/search/update/computation
Main scenario	<i>The end-user verifies the correctness of results</i>
Step 1	The end-user verifies that the data storage/search/update/computation has been correctly performed
Variation 1	<i>An error detected</i>
Step 1	If an error in data storage/search/update/computation is detected, the end user notifies the IT Team to solve the issue
Post-conditions	
[Post-condition 1]	The end user has checked (and notified the IT Team if an error has been detected) the correctness of storage/search/update/computation results

3.6.2.6 Correlate information from other CLARUS proxies (UC-MON-06)

UC-MON-06	Correlate information from distributed CLARUS proxies
Actors	Security Manager , Security Manager from other CLARUS proxy
Description	The Security Manager performs information correlation between two (or more) distributed CLARUS proxies
Pre-conditions	
[Pre-condition 1]	The Security Manager is authorised to get information coming from distributed CLARUS proxies
[Pre-condition 2]	The information coming from the proxies needs to be in compatible format to allow direct correlation
Main scenario	<i>Correlation of information coming from distributed CLARUS proxies (e.g., proxies 1 and 2)</i>
Step 1	The Security Manager of proxy 1 requests the information from proxy 2
Step 2	The monitoring tool installed in proxy 1 performs the correlation
Step 3	The monitoring tool from proxy 1 provides a web-based report including the correlated information from proxies 1 and 2
Variation 1	<i>Correlation coming from n distributed proxies</i>

Step 1	The Security Manager of proxy 1 requests the information from proxies 2 to n
Step 2	The monitoring tool installed in proxy 1 performs the correlation
Step 3	The monitoring tool from proxy 1 provides a web-based report including the correlated information from proxies 2 to n
Post-conditions	
<i>[Post-condition 1]</i>	The correlated information from distributed proxies is synthetised in a web-based reporting output.

3.7 Attack monitoring

3.7.1 Overview

This scenario refers to the actions for enabling monitoring of CLARUS proxy in the context of an attack.

3.7.1.1 System

In this scenario the system encompasses:

- The CLARUS proxy running on trusted local premises,
- The CLARUS adopting CSP.

One CLARUS proxy is involved. Additional CLARUS proxies may be involved, and monitoring should correlate the information provided by distributed CLARUS proxies.

3.7.1.2 UML diagram

The UML diagram below represents the corresponding use case (and actors).

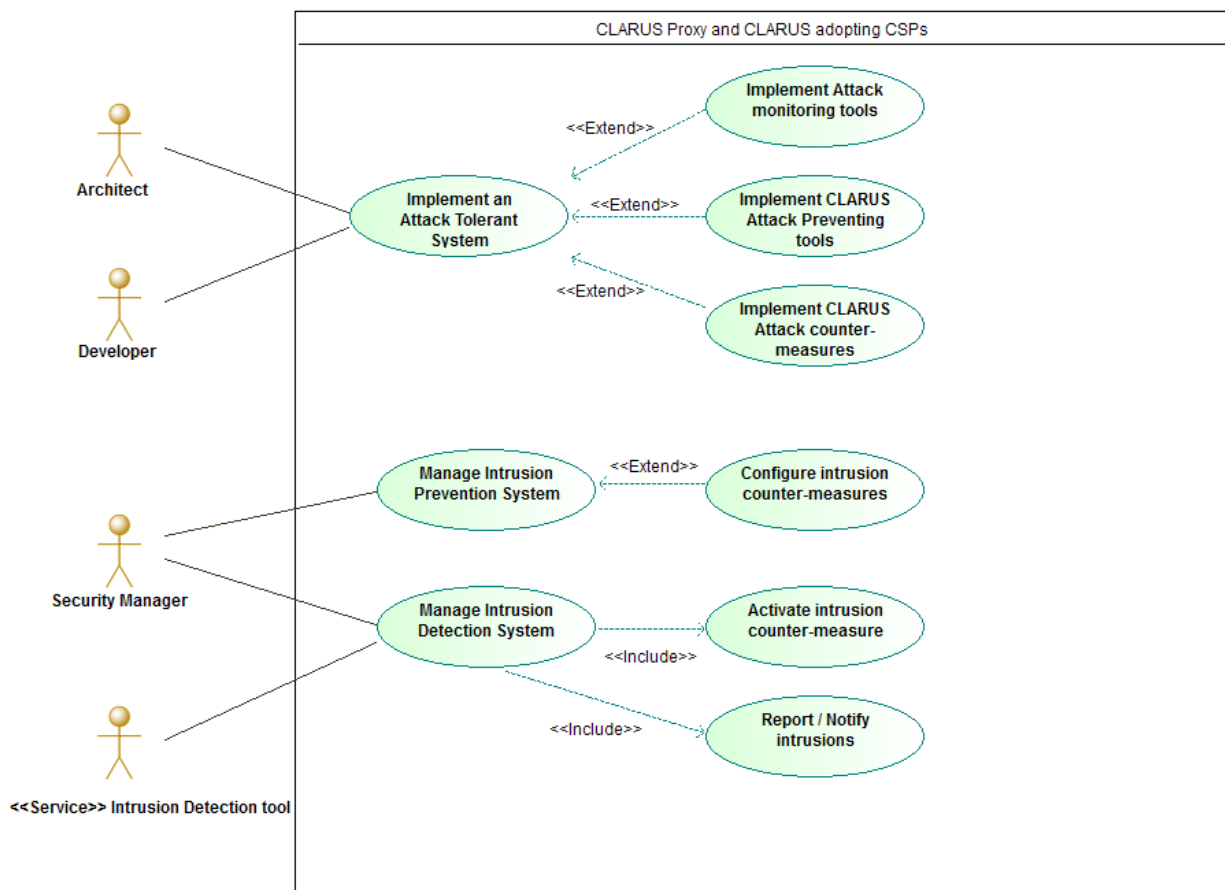


Figure 8 - Attack Monitoring Use Case Diagram

3.7.2 Use cases

3.7.2.1 Implement an Attack Tolerant System (UC-MON-07)

UC-MON-07	Implement an Attack Tolerant System
Actors	Architect ,Developer
Description	The architect and developer wants to develop intrusion management tools in an “Attack Tolerant System “ perspective
Pre-conditions	
[Pre-condition 1]	Implementation of Attack Monitoring tools
[Pre-condition 2]	Implementation of CLARUS Attack Preventing tools
[Pre-condition 3]	Implementation of CLARUS Attack counter-measures
Main scenario	Implementation of an Attack Tolerant System
Step 1	The Architect , considering the Attack Monitoring tools, the CLARUS Attack Preventing tools, and the CLARUS Attack counter-measures designs the Attack Tolerant System to guarantee the system operation even in the presence of an attack
Step 2	The Developer , considering the Attack Monitoring tools, the CLARUS Attack Preventing tools, and the CLARUS Attack counter-measures implements the Attack Tolerant System to guarantee the system operation even in the presence of an attack

Variation 1	<i>Mitigation mechanisms</i>
Step 1	The Architect/Developer implements mitigation mechanisms in case the system needs additional measures if an attack compromises the system
Post-conditions	
[Post-condition 1]	CLARUS system continues running and delivers the required services under the event of an attack.

3.7.2.2 Manage Intrusion Prevention System (UC-MON-08)

UC-MON-08	<i>Manage Intrusion Prevention System</i>
Actors	Security Manager
Description	The Security Manager handle the Intrusion Prevention monitoring tools in order to provide relevant countermeasures in case of intrusion detection
Pre-conditions	
[Pre-condition 1]	Intrusion prevention tools are installed, configured and running
[Pre-condition 2]	List of intrusions / attacks
[Pre-condition 3]	List of counter-measures
Main scenario	<i>Manage Intrusion Prevention System</i>
Step 1	The Security Manager connects to the Intrusion Prevention System
Step 2	The Security Manager analyses the list of intrusions/attacks and counter-measures to establish a correspondence between intrusions/attacks and counter-measures
Step 3	The Security Manager configures the system to execute the necessary counter-measures in presence of an intrusion/attack
Post-conditions	
[Post-condition 1]	The preventive measures, as the configuration of intrusion countermeasures, have been taken

3.7.2.3 Manage Intrusion Detection System (UC-MON-09)

UC-MON-09	<i>Manage Intrusion Detection System</i>
Actors	Security Manager
Description	The Security Manager handles the Intrusion Detection System, after an intrusion has been detected
Pre-conditions	
[Pre-condition 1]	Intrusion detection tools are installed, configured and running
[Pre-condition 2]	List of intrusions/attacks
[Pre-condition 3]	List of counter-measures
Main scenario	<i>Manage Attack Detection System (manual mode)</i>
Step 1	The Security Manager is notified on real-time via web-based reporting of the occurrence of an intrusion/attack in the system

Step 2	The Security Manager manually activates the corresponding counter-measures to mitigate the effects of the detected intrusion/attack
Step 3	A web-based report is created detailing the detected intrusion/attack and the counter-measures taken
Variation 1	<i>Manage Intrusion Detection System (automatic mode)</i>
Step 0	The Intrusion Detection Tool detects an attack
Step 1	The Intrusion Detection Tool automatically executes the necessary counter-measures to mitigate the effects of the detected intrusion/attack
Step 2	A web-based report is created detailing the detected intrusion/attack and the counter-measures taken
Post-conditions	
[Post-condition 1]	The dedicated countermeasures has been launched after an intrusion/attack has been detected. An associated notification has been generated as a web-based reporting output.

3.8 User data storage

3.8.1 Overview

This scenario considers one or more user(s) (authenticated by a CLARUS proxy) who wants to store and/or access data on the cloud through the CLARUS proxy.

In this scenario, two alternatives are considered:

- A single user (or multiple users seen as a single user by the CLARUS proxy) who wants to store his/her data on the cloud and modifies them at any time through CLARUS.
- Multiple users (identified by the CLARUS proxy) who store and share their data on the cloud and intend to modify them at any time.

All the data are accessed by their identifier, i.e. their “id” (e.g. a path, unique filename, database identifier, etc.)

3.8.1.1 System

In this scenario the system is the CLARUS proxy running on trusted local premises.

One or more CLARUS proxy(ies) is(are) involved. One or several CSP(s) may be involved, but only for the purpose of data storage. Consequently, there is no CLARUS software needed on the cloud.

3.8.1.2 UML diagram

The UML diagrams below represent the corresponding use case (and actors).

- In a first scenario a single user stores, modifies, accesses and deletes his/her data on the cloud.

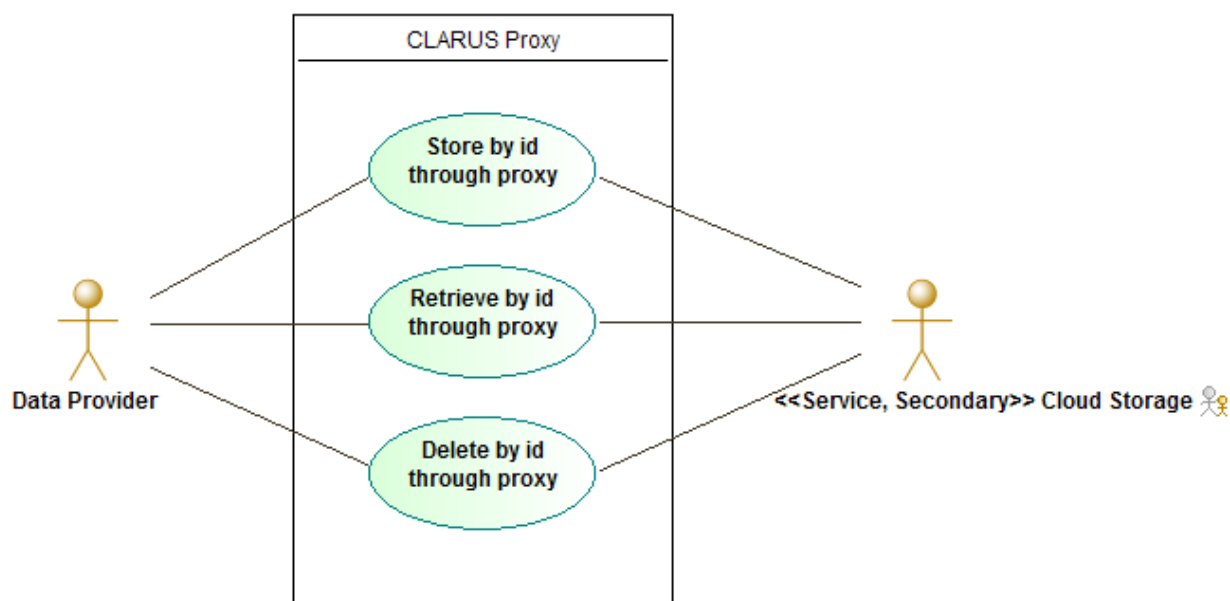


Figure 9 – Single-User Data Storage Use Case Diagram

- In a second scenario multiple users store and share their data on the cloud.

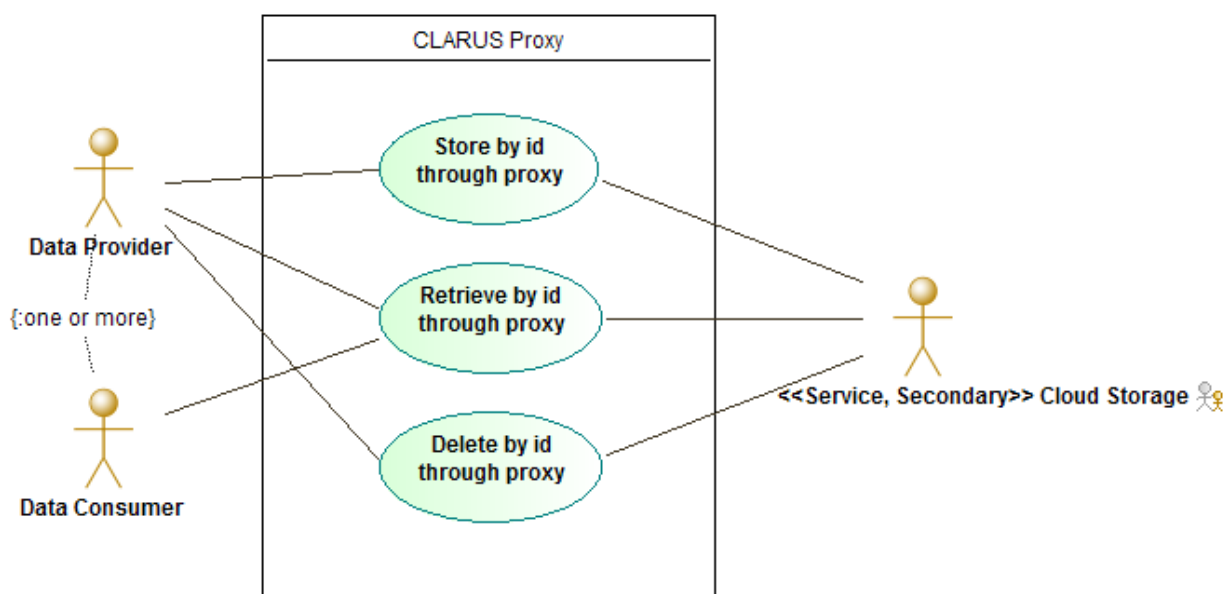


Figure 10 – Multi-User Data Storage Use Case Diagram

The UML diagram below represents store, retrieve and delete by id on protected data:

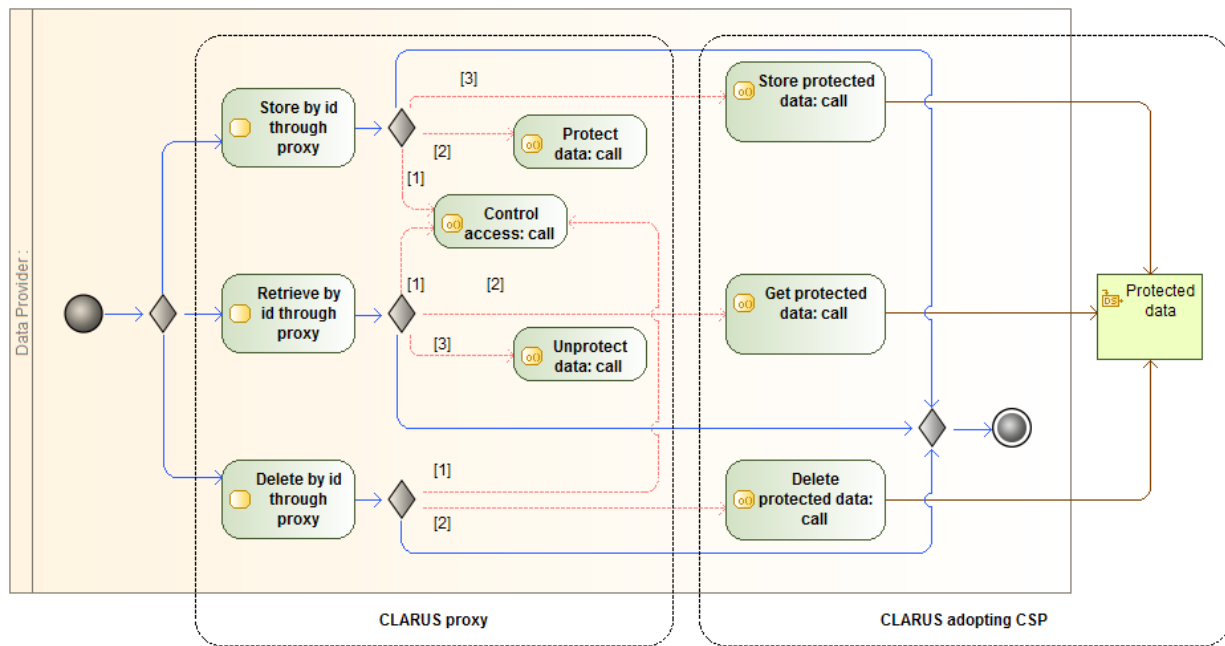


Figure 11 - User Data Storage Activity Diagram

3.8.2 Use cases

3.8.2.1 Store by id through proxy (UC-EUP-01)

UC-EUP-01	Store by id through proxy
Actors	Data provider, cloud storage service (secondary)
Description	The data provider wants to securely create new data or update existing data in the cloud through the CLARUS proxy, having the guarantee that the data will not be disclosed to the CSP(s)
Pre-conditions	
[Pre-condition 1]	The data to store is either structured (e.g. a table row in a database) or unstructured (e.g. a text file, an image, a binary file) in the cloud, using the storage service provided by the CSP
[Pre-condition 2]	The IT team has set up a user account in one (or many) cloud service provider(s)
[Pre-condition 3]	The security manager has set up the security policy
[Pre-condition 4]	The data provider is authorized to create new data and/or update existing data
Main scenario	<i>Create new data</i>
Step 1	The data provider , via a dedicated application, stores his/her data in the cloud through the CLARUS proxy
Step 2	The system identifies the dataspace in which the data to create will be stored
Step 3	The system controls the data provider is authorized to create new data (according to the access rights defined in his/her profile) in the dataspace
Step 4	The system protects the data according to the security policy defined for the dataspace
Step 5	The system stores the protected data on the cloud using the cloud storage service(s) provided by the CSP(s)

Step 6	The cloud storage service(s) confirms the system about the success of the storage operation
Step 7	The system confirms the data provider about the success of the storage operation
Variation 1	<i>Update existing data</i>
Step 1	The data provider , via a dedicated application, updates existing data in the cloud through the CLARUS proxy
Step 2	The system identifies the dataspace and the dataset in which the data will be updated
Step 3	The system controls the data provider is authorized to update existing data (according to the access rights defined in his/her profile and according to his/her authorizations on the dataset)
Step 4	The system protects the data according to the security policy defined for the dataspace
Step 5	The system stores the protected data on the cloud using the cloud storage service(s) provided by the CSP(s)
Step 6	The cloud storage service(s) confirms the system about the success of the storage operation
Step 7	The system confirms the data provider about the success of the storage operation
Post-conditions	
<i>[Post-condition 1]</i>	Protected data are stored in the cloud

3.8.2.2 Retrieve by id through proxy (UC-EUP-02)

UC-EUP-02	<i>Retrieve by id through proxy</i>
Actors	Data provider, data consumer, cloud storage service (secondary)
Description	The data provider (or the data consumer) wants to retrieve data from the cloud through the CLARUS proxy, having the guarantee that the data are not disclosed to the CSP(s)
Pre-conditions	
<i>[Pre-condition 1]</i>	Data have previously been protected and stored in the CSP(s) through the proxy
<i>[Pre-condition 2]</i>	The data provider (or the data consumer) is authorized to read existing data
Main scenario	
Step 1	The data provider (or the data consumer), via a dedicated application, retrieves data from the cloud, by means of their id
Step 2	The system identifies the dataspace and the dataset from which the data will be retrieved
Step 3	The system controls the data provider (or the data consumer) is authorized to retrieve existing data (according to the access rights defined in his/her profile and according to his/her authorizations on the dataset)
Step 4	The system retrieves the protected data from the cloud using the cloud storage service(s) provided by the CSP(s), by means of their id
Step 5	The system unprotects the data according to protection set initially
Step 6	The system provides the data provider (or the data consumer) with clear (unprotected)

	data
Post-conditions	
[Post-condition 1]	The data provider (or the data consumer) has retrieved clear (unprotected) data without disclosure to CSP(s)

3.8.2.3 Delete by id through proxy (UC-EUP-03)

UC-EUP-03	<i>Delete by id through proxy</i>
Actors	Data provider, cloud storage service (secondary)
Description	The data provider wants to delete data in the cloud through the CLARUS proxy, having the guarantee that the data are properly deleted by the CSP(s)
Pre-conditions	
[Pre-condition 1]	Data have previously been protected and stored in the CSP(s) through the proxy
[Pre-condition 2]	The data provider is authorized to delete existing data
Main scenario	
Step 1	The data provider , via a dedicated application, requests a deletion of data in the cloud, by means of their id
Step 2	The system identifies the dataspace and the dataset from which the data will be deleted
Step 3	The system controls the data provider is authorized to delete existing data (according to the access rights defined in his/her profile and according to his/her authorizations on the dataset)
Step 4	The system deletes the protected data from the cloud using the cloud storage service(s) provided by the CSP(s), by means of their id
Step 5	The cloud storage service(s) confirms the system about proper deletion of the protected data
Step 5	The system confirms the data provider about proper deletion of the protected data
Post-conditions	
[Post-condition 1]	The data provider has deleted data associated to an id: these data are not in the cloud any more

3.9 Search in the cloud

3.9.1 Overview

This scenario considers one or more user(s) (authenticated by the CLARUS proxy) who publish their data on the cloud. Any user can search data by attributes either through the CLARUS proxy, or directly accessing the cloud search service.

3.9.1.1 System

In the search scenario the system encompasses:

- The CLARUS proxy running on trusted local premises,
- The CLARUS adopting CSP(s).

Only one CLARUS proxy is involved. The CSP(s) is(are) involved to search data by attributes: consequently, CLARUS software might be needed on the cloud side, namely “CLARUS inside”.

3.9.1.2 UML diagrams

The UML diagrams below represent two different use cases (and their actors).

- In a first *standard* scenario data access is done by searching by attributes **through the CLARUS proxy** on protected data.

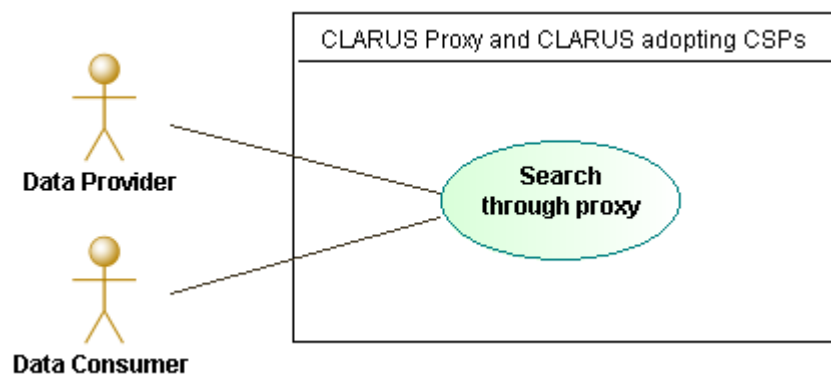


Figure 12 - Search Through Proxy Use Case Diagram

- In a second scenario data access is done by attributes directly in the cloud, without need of the CLARUS proxy.

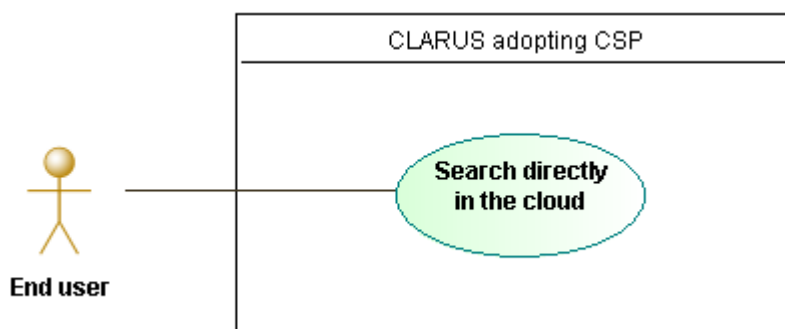


Figure 13 - Search Directly in the Cloud Use Case Diagram

The UML diagram below represents search on protected data either through the CLARUS proxy or directly in the cloud:

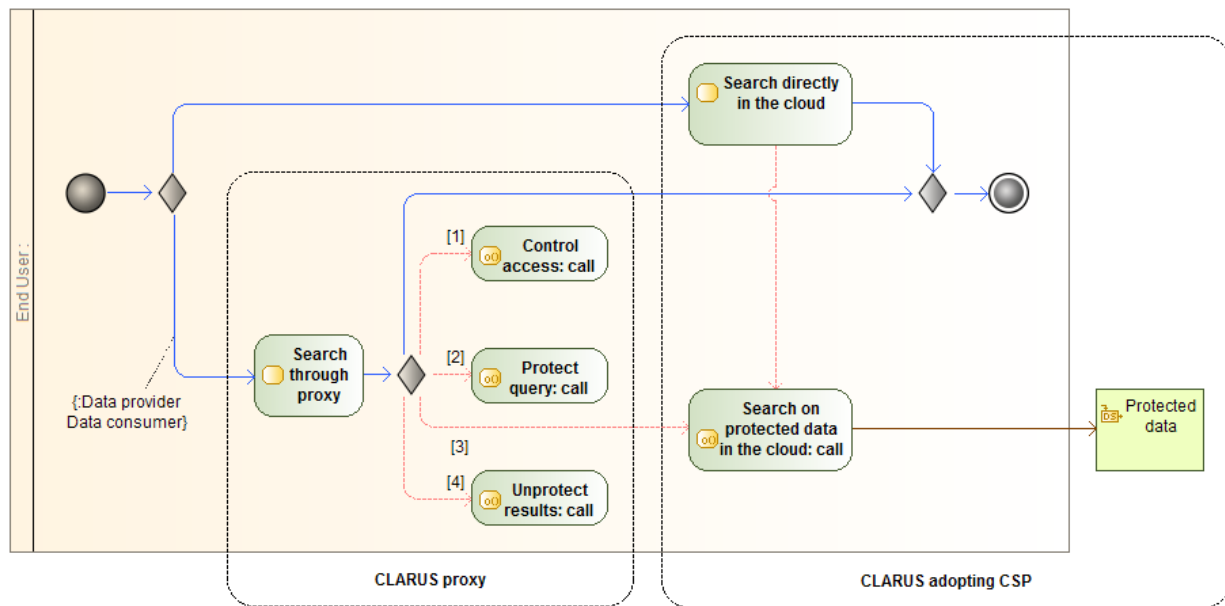


Figure 14 - Search Activity Diagram

3.9.2 Use cases

3.9.2.1 Search through proxy (UC-EUP-04)

UC-EUP-04	Search through proxy
Actors	Data provider, data consumer
Description	The data provider wants to search for data by attributes, having the guarantee that the content of his/her query will not be disclosed to the CSP(s) or any other unauthorized party
Pre-conditions	
[Pre-condition 1]	Data have previously been protected and stored in the CLARUS-enabled CSP(s)
[Pre-condition 2]	Data stored are mainly structured data, but can also be unstructured data (e.g. text)
[Pre-condition 3]	The Security manager has defined the users authorized to access the data
Main scenario	<i>Data provider (or data consumer) searches for data by attributes</i>
Step 1	The data provider (or the data consumer), via a dedicated application, searches for data in the cloud, by means of an attribute-based query
Step 2	The system identifies the dataspace and the dataset from which the data will be searched
Step 3	The system controls the data provider (or the data consumer) is authorized to search data (according to the access rights defined in his/her profile and according to his/her authorizations on the dataset)
Step 4	The proxy protects the query according to the protection set initially
Step 5	The proxy invokes the search service provided by the CLARUS-enabled CSP
Step 6	The search service provided by the CLARUS-enabled CSP retrieves and returns the protected data to the proxy
Step 7	The proxy unprotects the data according to the protection set initially

Step 8	The proxy provides the data provider (or the data consumer) with the clear (unprotected) data he requested
Post-conditions	
[Post-condition 1]	The data provider (or the data consumer), is provided with the clear (unprotected) search result
[Post-condition 2]	The clear (unprotected) query and the clear (unprotected) search result are not disclosed to the CSP or to any other unauthorized party

3.9.2.2 Search directly in the cloud (UC-EUP-05)

UC-EUP-05	<i>Search directly in the cloud</i>
Actors	End-user
Description	The end-user wants to search for data by attributes, directly in the cloud
Pre-conditions	
[Pre-condition 1]	Data have previously been protected and stored in the CSP(s) by the data provider
Main scenario	<i>End-user searches for data by attributes that are not protected</i>
Step 1	The end-user searches for data in the cloud, by means of a clear (unprotected) attribute-based query
Step 2	The search service provided by the CLARUS-enabled CSP provides the end-user with a correct search result. The correct search result contains the protected data
Variation 1	<i>End-user searches for data by attributes that are protected</i>
Step 2	The search service provided by the CLARUS-enabled CSP provides the end-user with an incorrect search result. The incorrect search result either is empty or contains invalid (but protected) data
Post-conditions	
[Post-condition 1]	In case of search by non protected attributes, the end-user is provided with protected data
[Post-condition 2]	In case of search by non protected attributes, the protected data are exploitable or not exploitable according to the protection set initially
[Post-condition 3]	In case of search by protected attributes, the end-user is provided with an empty or invalid search result

3.10 Update in the cloud

3.10.1 Overview

This scenario considers one or more user(s) (authenticated by the CLARUS proxy) who publish their data on the cloud and intend to modify it. Data to be modified are accessed by attribute(s), in this case the CSP searches for the data to modify.

3.10.1.1 System

In this scenario the system encompasses:

- The CLARUS proxy running on trusted local premises,
- The CLARUS adopting CSP(s).

Only one CLARUS proxy is involved. The CSP(s) is(are) involved to store the data and to search and modify data by attributes: consequently, CLARUS software might be needed on the cloud side, namely “CLARUS inside”.

3.10.1.2 UML diagrams

The UML diagram below represents the corresponding use case. In this scenario data access is done by attributes:

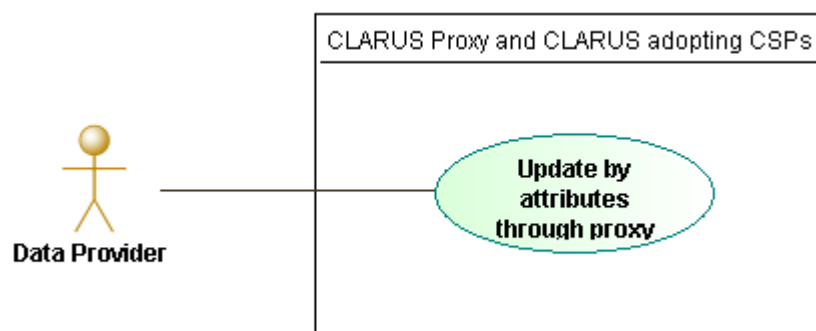


Figure 15 - Update by Attributes Use Case Diagram

The UML diagram below represents update on protected data:

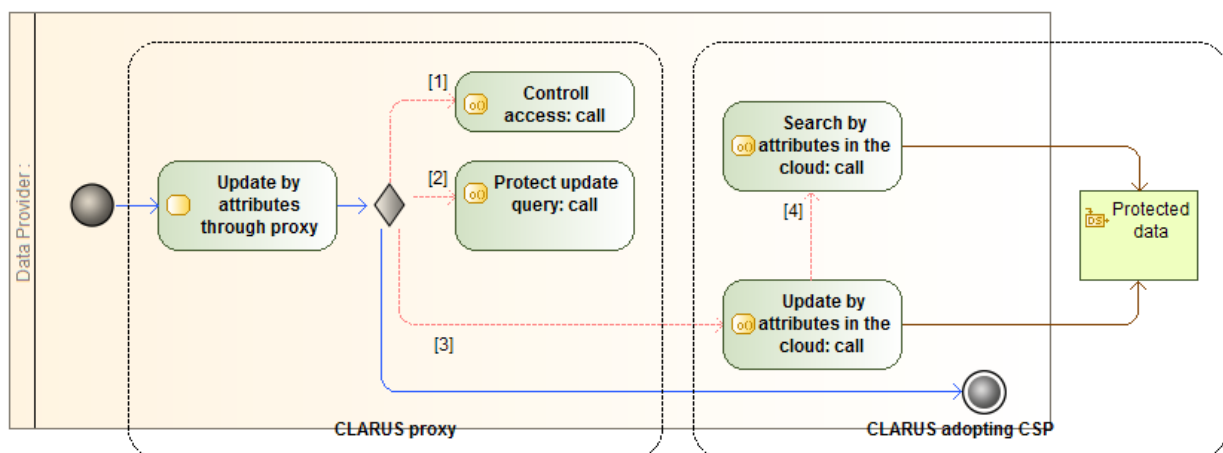


Figure 16 - Update by Attributes Activity Diagram

3.10.2 Use cases

3.10.2.1 Update by attributes through proxy (UC-EUP-06)

UC-EUP-06	<i>Update by attributes through proxy</i>
Actors	Data provider
Description	The data provider wants to update his/her data (searched by attributes), having the guarantee that the content of his/her query will not be disclosed to the CSP(s) or any other unauthorized party
Pre-conditions	
[Pre-condition 1]	Data have previously been protected and stored in the CLARUS-enabled CSP(s)
[Pre-condition 2]	The Security manager has defined the users authorized to access the data
Main scenario	<i>Data provider updates data by attributes</i>
Step 1	The data provider , via a dedicated application, updates data in the cloud, by means of an attribute-based query
Step 2	The system identifies the dataspace and the dataset in which the data will be updated
Step 3	The system controls the data provider is authorized to update data (according to the access rights defined in his/her profile and according to his/her authorizations on the dataset)
Step 4	The proxy protects the update query according to the protection set initially
Step 5	The proxy invokes the update service provided by the CLARUS-enabled CSP
Step 6	The update service provided by the CLARUS-enabled CSP updates the protected data and confirms the proxy about proper update of protected data
Step 7	The proxy confirms the data provider about proper update of his/her protected data
Post-conditions	
[Post-condition 1]	The data are updated
[Post-condition 2]	The clear (unprotected) update query and the clear (unprotected) data are not disclosed to the CSP or to any other unauthorized party

3.11 Computation in the cloud

3.11.1 Overview

This scenario considers one or more user(s) (authenticated by the CLARUS proxy) that publish their data on the cloud. The end-users can request for a computation either through CLARUS proxy services, or directly from the cloud.

3.11.1.1 System

In this scenario the system encompasses:

- The CLARUS proxy running on trusted local premises,
- The CLARUS adopting CSP(s).

Only one CLARUS proxy is involved. The CSP(s) is(are) involved to store the data and to execute the computation process: consequently, CLARUS software might be needed on the cloud side, namely “CLARUS inside”.

3.11.1.2 UML diagrams

The UML diagrams below represent two different use cases (with their actors). In both cases computation services are hosted in the cloud.

- In the first scenario, computation services are performed on protected data through the CLARUS proxy. The respective results being protected need to be unprotected on the CLARUS proxy side.

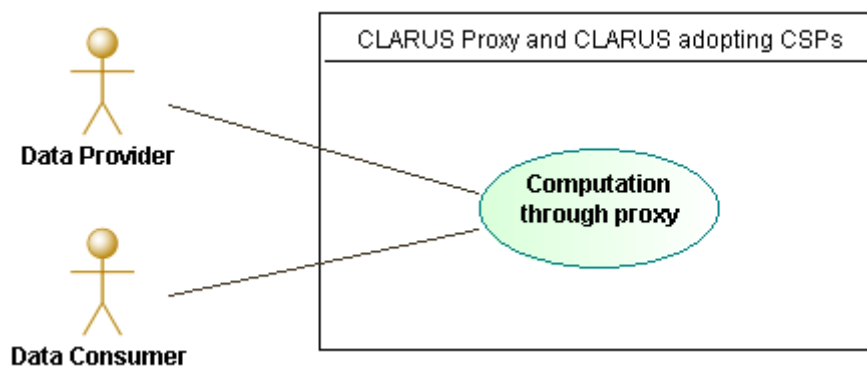


Figure 17 - Computation Through Proxy Use Case Diagram

- In a second scenario any user can request for a computation directly accessing the cloud process service, without need of the CLARUS proxy.

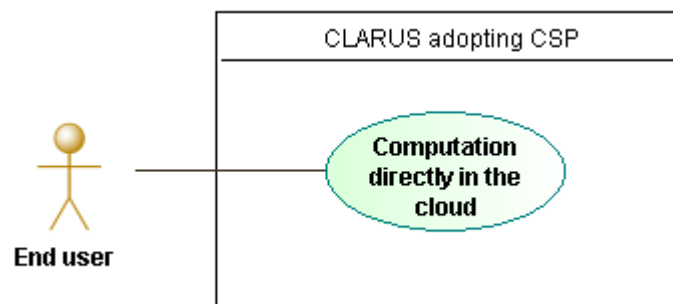


Figure 18 - Computation Directly in the Cloud Use Case Diagram

The UML diagram below represents computation on protected data either through the CLARUS proxy or directly in the cloud:

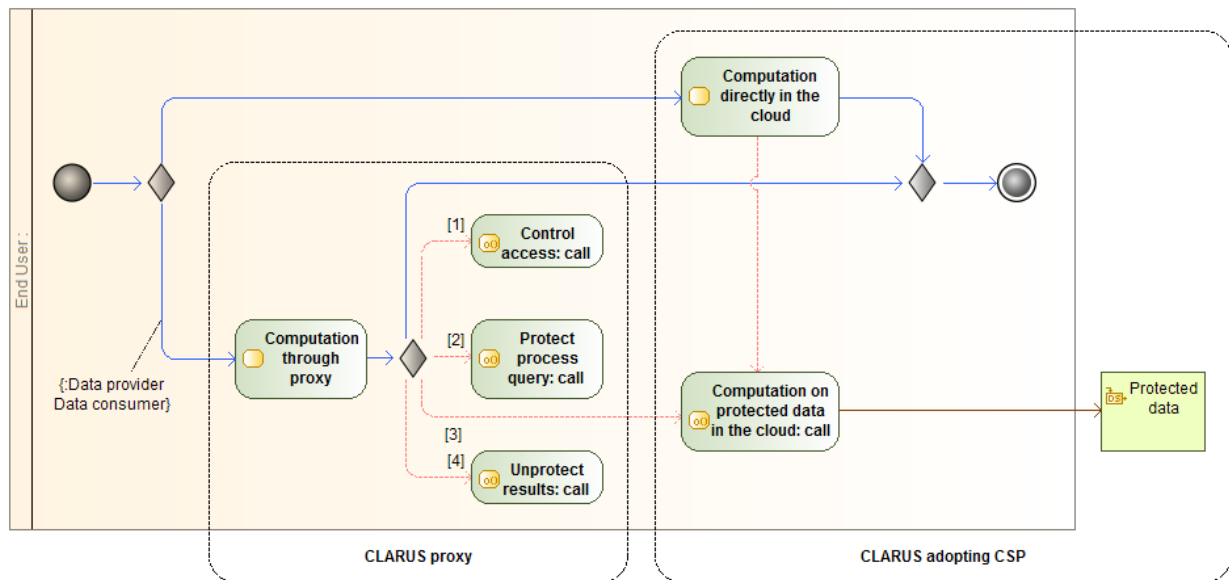


Figure 19 - Computation Activity Diagram

3.11.2 Use cases

3.11.2.1 Computation through proxy (UC-EUP-07)

UC-EUP-07	Computation through proxy
Actors	Data consumer, data provider
Description	The data provider wants to perform a computation over protected data in the cloud, having the guarantee that his/her query and the response will not be disclosed to the CSP or any other unauthorized party
Pre-conditions	
[Pre-condition 1]	Data have previously been protected and stored in the CLARUS-enabled CSP(s)
[Pre-condition 2]	The Security manager has defined the users authorized to access the process
Main scenario	<i>Data provider (or data consumer) performs a computation through the proxy</i>
Step 1	The data provider (or the data consumer), via a dedicated application, performs a computation in the cloud, over data previously protected
Step 2	The system identifies the dataspace and the dataset from which the computation service will access data
Step 3	The system controls that the data provider (or the data consumer) is authorized to perform computation (according to the access rights defined in his/her profile and according to his/her authorizations on the dataset)
Step 4	The proxy protects the process query according to protection set initially
Step 5	The proxy invokes the computation service provided by the CLARUS-enabled CSP
Step 6	The computation service provided by the CLARUS-enabled CSP performs the computation over protected data and returns the protected result to the proxy
Step 7	The proxy unprotects the result of the computation
Step 8	The proxy provides the data provider (or the data consumer) with the clear

	(unprotected) result of the computation
Post-conditions	
[Post-condition 1]	The data provider (or the data consumer), is provided with clear (unprotected) result of the computation
[Post-condition 2]	The clear (unprotected) process query and the clear (unprotected) computation result are not disclosed to the CSP or to any other unauthorized party

3.11.2.2 Computation directly in the cloud (UC-EUP-08)

UC-EUP-08	<i>Computation directly in the cloud</i>
Actors	End-user
Description	The end-user wants to perform a computation over protected data in the cloud
Pre-conditions	
[Pre-condition 1]	Data have previously been protected and stored in the CSP(s) by the data provider
Main scenario	<i>End-user invokes a computation service that accepts clear (unprotected) parameters</i>
Step 1	The end-user performs a computation directly in the cloud, over data previously protected, passing clear (unprotected) parameters
Step 2	The cloud computation service provided by the CLARUS-enabled CSP performs the computation over protected data and provides the end-user with a correct computation result
Variation 1	<i>End-user invokes a computation service that expects protected parameters</i>
Step 1	The end-user performs a computation directly in the cloud, over data previously protected, passing clear (unprotected) parameters
Step 2	The cloud computation service provided by the CLARUS-enabled CSP performs the computation over protected data and provides the end-user with an incorrect computation result
Post-conditions	
[Post-condition 1]	If the computation service accepts clear (unprotected) parameters, the end-user is provided with a protected result
[Post-condition 2]	If the computation service accepts clear (unprotected) parameters, the protected result is exploitable or not exploitable according to the protection set initially
[Post-condition 3]	If the computation service expects protected parameters, the end-user is provided with an invalid computation result

4 Requirements

4.1 Overview

The aim of this section is to describe all the identified requirements of the different functions to be performed by CLARUS software. This gathering of requirements is inherited from users requirements previously developed in the use cases section.

These different requirements will provide a baseline for the subsequent steps of :

- Design and implementation of the CLARUS software solution (WP4/WP5 - architecture design and CLARUS implementation),
- Validation and verification (WP6 – evaluation, testing and validation),
- Proposal for a V2.0 release of the same deliverable, which may contain areas for enhancements, as optional features which would be proposed in the final and definitive version of the CLARUS solution.

As previously mentioned (Section 1.1), defining these requirements will aim to specify :

- Which functionalities should be supported in order to have a complete description of what CLARUS software is supposed to do and thus try to cover all the facets of its potential use (i.e functional requirements). We will base in particular on the different perspectives outlined in the use cases definition section (Section 0).
- Which are the expected outcome in terms of integrity, reliability, quality of service, security needs, legal constraints...

The review of the requirements is therefore organized in two main sections, namely: functional and non functional requirements. Each contains subsections, related to the issues identified.

4.1.1 Naming Scheme

In a standardisation purpose, the requirements will be assigned with an unique identifier defined by the following nomenclature :

<Requirement ID>= REQ-<Category code>-<Progressive number>
--

Figure 20 - Requirement Naming Scheme

Such a nomenclature will ensure the subsequent requirement traceability.

4.1.2 Priorities

“Requirements are organized and classified in a priority hierarchy in order to facilitate later adjustments to any limitations in their implementation that may be due to time and budget constraints” [A1].

Using the following classification of four categories allows us to propose a priority level organization in descending order :

- Must have [MH]: the project will be seriously impacted if this requirement is not met.

- Should have [SH]: a requirement that, if it is not met, seriously impacts the project, but it can be delivered on a different timescale than the must-haves.
- Could have [CH]: optional requirements which can improve the project outcome and the value of its results, but are not essential to the main delivery (not too many Could-have requirements should be dropped in the design).
- Would like [WL]: truly optional features which can be delivered if there is sufficient effort available within a task, or if they can be found elsewhere and integrated easily.

The priorities assigned to each requirement reflect the current state of our analysis of the CLARUS solution, and they might be subject to modification in the next version of the requirement specification according to constraints, risks or technical limitations that may emerge during the project development.

4.1.3 Description

The template table below will be used for requirements description :

Req. No.	Description	Priorities	Comment
<Requirement ID> REQ-<Category code>-<Nb>		<MH SH CH WL>	Ref : <Use Case Id > or <comment>

4.1.4 Categories code

We refer here to two different conditions:

1. If we focus on the description of the features required, defining CLARUS system behaviour and general functionality we consider the category of 'functional requirements'.
2. If we focus on term of constraints, targets or control mechanisms, we refer to the 'non functional requirements' category.

The different sub-classes of requirements for each of these two main categories are :

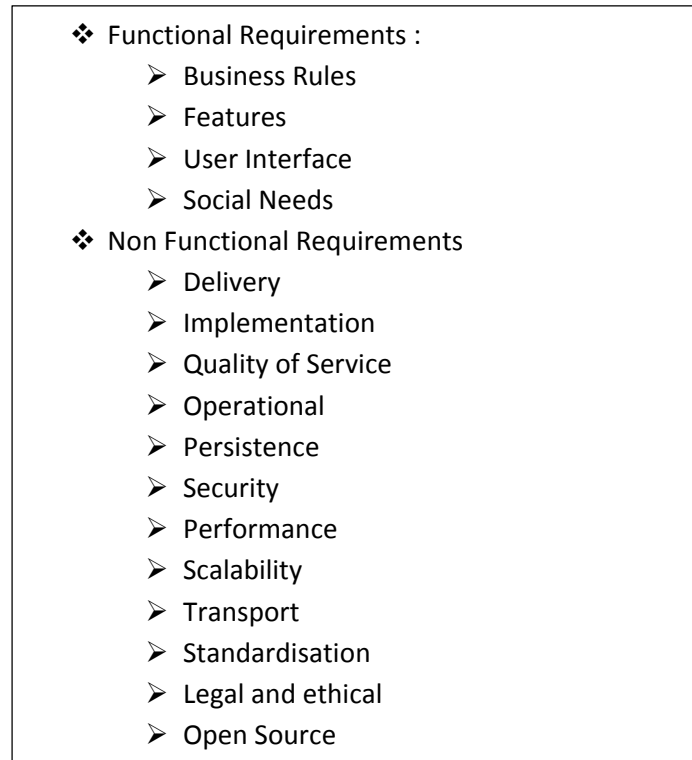


Figure 21 - Classification of Requirements

4.2 Functional Requirements

Functional requirements are the functions and features that the user needs for the system to work.

In this part, we address 4 main topics (Section 4.1.4), namely : business rules, features, user interface and social needs.

4.2.1 Business Rules

This set of requirements is linked to the category code : <F_BRU>.

We set out here the business requirements to enable the implementation and compliance with a business rule. These requirements correspond to high level needs or wants of a business organization. They should describe what CLARUS system or solution should do. It is by their fulfilment that the organization achieves its objectives.

Req. No.	Description	Priorities	Comment
REQ-F_BRU-1.1	The CLARUS solution must help cloud customers increase their control over their (critical and sensitive) outsourced data.	MH	Ref: UC-EUP-01

REQ-F_BRU-1.2	CLARUS must offer security and privacy enhanced mechanisms, monitorable by the security manager (or assimilated) of the customer's organization.	MH	
REQ-F_BRU-1.3	The CLARUS solution must integrate seamlessly some client API on premises, as well as some Applications and Services offered by cloud PaaS, without interfering with the normal operations of these features.	MH	Ref: UC-EUP-01
REQ-F_BRU-1.4	At any moment in the data securisation workflow, CLARUS utilities must not apply some unprotection mechanism that may lead to disclosure or leakage of sensitive or confidential data to the CSP(s).	MH	Ref: UC-EUP-01, UC-EUP-02, UC-EUP-04
REQ-F_BRU-1.5.1	The CLARUS solution implemented must offer intrusion prevention and detection systems.	MH	Ref: UC-MON-02, UC-MON-07, UC-MON-08, UC-MON-09
REQ-F_BRU-1.5.2	The intrusion detection system offered by CLARUS must provide dedicated countermeasures to manage any episode of intrusion and vulnerability on the client device, the CLARUS proxy, or the cloud storage platform	SH	Ref: UC-MON-02, UC-MON-07, UC-MON-08, UC-MON-09
REQ-F_BRU-1.6	Any intrusion detection should be treated in an 'Attack Tolerant System' perspective allowing applications to continue to operate.	SH	Ref: UC-MON-07
REQ-F_BRU-1.7	The implementation of CLARUS must allow some of its users to specify the level of security they want to apply.	MH	Ref: UC-EUP-01
REQ-F_BRU-1.8	The architecture/design of CLARUS solution must allow the communication between several CLARUS proxies able to interoperate with each others.	MH	Ref: UC-SEC-01, UC-SEC-03, UC-MON-06
REQ-F_BRU-1.9	The CLARUS solution must be generic enough (protocols, data formats used) and implement open standards to ensure	MH	Ref: UC-EUP-01, UC-EUP-02, UC-EUP-03, UC-SEC-11

	interoperability and collaboration between several Cloud Service Providers.		
REQ-F_BRU-1.10	CLARUS must not induce a loss of efficiency or data utility for both cloud costumers and CSPs.	MH	
REQ-NF_BRU-1.11	CLARUS must not compromise the benefits offered by cloud computing (e.g. cost savings, ubiquitous access, etc.).	MH	
REQ-NF_BRU-1.12	The CLARUS solution should be able to partially preserve the analytical utility of the user data for the CSP so that it could still (partially) mine user data	SH	

4.2.2 Features

This set of requirements is linked to the category code : <F_FEA>.

4.2.2.1 Secure data storage

Req. No.	Description	Priorities	Comment
REQ-F_FEA-1.1	Only authorized users (data owner / data provider) by the CLARUS system can run some data storage operations to a CSP.	MH	Ref: UC-EUP-01
REQ-F_FEA-1.2	The CLARUS proxy must ensure protection of the dataset that the end-user wants to store in the cloud.	MH	Ref: UC-EUP-01
REQ-F_FEA-1.3	The data protection during the storage through the CLARUS proxy must not interfere with the interaction of the user with the application / services (transparency).	MH	Ref : UC-EUP-01
REQ-F_FEA-1.4	Any storage operation of protected data must notify the confirmation of success / unsuccess of the storage operations.	MH	Ref : UC-EUP-01, UC-MON-05
REQ-F_FEA-1.5	The CLARUS proxy must allow the outsourcing of data in different CSP(s) by	SH	Ref : UC-EUP-01, UC-EUP-02

	distribution/replication methodology.		
REQ-F_FEA-1.6	Only authorized users (as data owner / data provider) by the CLARUS system can run some data retrieving operations on protected data outsourced.	MH	Ref : UC-EUP-02
REQ-F_FEA-1.7	Some authorized users (as data consumer) different than the owner of the data may perform retrieving operations on outsourced protected data.	MH	Ref : UC-EUP-02
REQ-F_FEA-1.8	The CLARUS proxy must be able to retrieve the protected data previously stored in the cloud by means of an identifier.	MH	Ref : UC-EUP-02
REQ-F_FEA-1.9	During a retrieving operation, the CLARUS system must define the mechanism, according to the protection of data set initially, to obtain the clear data.	MH	Ref : UC-EUP-02
REQ-F_FEA-1.10	During a retrieving operation, the unprotection mechanisms are applied in a way that only authorized users and CLARUS proxy have access to the unprotected (clear) data outsourced. Especially, the clear unprotected data must not be disclosed in any moment to the CSP or other unauthorized party	MH	Ref : UC-EUP-02
REQ-F_FEA-1.11	Only authorized users (as data owner / data provider) by the CLARUS system can run some delete operations on protected data outsourced.	MH	Ref : UC-EUP-03
REQ-F_FEA-1.12	The CLARUS proxy must delete the protected data previously stored in the cloud by means of an identifier.	MH	Ref : UC-EUP-03
REQ-F_FEA-1.13	Any delete operation of protected data must notify the confirmation of success / unsuccess of the deletion operation.	MH	Ref : UC-EUP-03, UC-MON-05
REQ-F_FEA-1.14	In case of several CSPs implicated in a data storage application field, the CLARUS proxy	MH	Ref : UC-EUP-03

	must allow the deletion of data stored in different CSP(s), by means of their data ids.		
--	---	--	--

4.2.2.2 Secure data search

Req. No.	Description	Priorities	Comment
REQ-F_FEA-2.1	Only authorized users (data provider) by the system can run some search operations on outsourced protected data.	MH	Ref: UC-EUP-04
REQ-F_FEA-2.2	Some authorized users (as data consumers) different than the owner of the data may perform search operations on outsourced protected data.	MH	Ref: UC-EUP-04
REQ-F_FEA-2.3	The CLARUS solution must manage some attributes based search on protected data previously stored in the cloud, in such a way that the system is able to process the “confidential” queries based on data in their protected format.	MH	Ref: UC-EUP-04
REQ-F_FEA-2.4	The CLARUS solution must manage some attributes based search on protected data previously stored in the cloud, in such a way that the system is able, from “confidential queries”, to provide the end user with the associated result in its unprotected format.	MH	Ref: UC-EUP-04
REQ-F_FEA-2.5	During a search operation, no party except authorized users and CLARUS proxy, is aware of the content of the search query.	MH	Ref: UC-EUP-04
REQ-F_FEA-2.6	During a search operation, the system must define the unprotection mechanism to apply according to the protection of data set initially.	MH	Ref: UC-EUP-04
REQ-F_FEA-2.7	During a search operation, the unprotection mechanisms are applied in a way that only authorized users and CLARUS proxy have	MH	Ref: UC-EUP-04

	access to the unprotected (clear) data outsourced, corresponding to the query result. Especially, the clear unprotected data must not be disclosed in any moment to the CSP or other unauthorized party.		
REQ-F_FEA-2.8	In the case of public outsourced data in the cloud, i.e. after using some CLARUS protection methods on sensitive or confidential data have been applied (as coarsening or anonymization), users different than the authorized users of CLARUS solution may perform search operations on outsourced CLARUS managed data, directly from the cloud without the need of the CLARUS proxy.	SH	Ref: UC-EUP-05
REQ-F_FEA-2.9	In the case of public outsourced data in the cloud, i.e. after using some CLARUS protection methods on sensitive or confidential data have been applied (as coarsening or anonymization), the CSP service may provide the coarsened or anonymised associated result, the end user has requested with the search query.	SH	Ref: UC-EUP-05, UC-MON-05

4.2.2.3 Secure data update

Req. No.	Description	Priorities	Comment
REQ-F_FEA-3.1	Only users authorized by the system with proper access rights, can run some update operations on outsourced protected data.	MH	Ref: UC-EUP-06
REQ-F_FEA-3.2	The CLARUS solution must manage update accessed by attributes.	MH	Ref: UC-EUP-06
REQ-F_FEA-3.3	The CLARUS solution must manage some attributes based updates on protected data previously stored in the cloud, in such a way that the system is able to process attributes based searches queries, on data in their	SH	Ref: UC-EUP-06

	protected format.		
REQ-F_FEA-3.4	During a search operation resulting from an update request, no party except authorized users and CLARUS proxy, is aware of the content of the “confidential” search query.	MH	Ref: UC-EUP-06
REQ-F_FEA-3.5	The CLARUS solution must manage some attributes based updates on protected data previously stored in the cloud, in such a way that the system is able, from “confidential search queries”, to update the corresponding data with the new protected value.	SH	Ref: UC-EUP-06
REQ-F_FEA-3.6	Any update query on protected data must notify the confirmation of success / unsuccess of the update operation.	MH	Ref : UC-EUP-06, UC-MON-05

4.2.2.4 Secure data computation

Req. No.	Description	Priorities	Comment
REQ-F_FEA-4.1	Only authorized users (data provider) by the system can run compute operations on outsourced protected data.	MH	Ref: UC-EUP-07
REQ-F_FEA-4.2	Some authorized users (as data consumers) different than the owner of the data may perform compute operations on outsourced protected data.	MH	Ref: UC-EUP-07
REQ-F_FEA-4.3	The CLARUS solution must enable computation services hosted in the cloud to perform compute operations on data previously stored in the cloud, in their protected format.	SH	Ref: UC-EUP-07
REQ-F_FEA-4.4	The CLARUS solution must manage the results of computation services on protected data previously stored in the cloud, in such a way that the system is able to provide the	MH	Ref: UC-EUP-07

	end user with the associated result in its unprotected format.		
REQ-F_FEA-4.5	During a compute operation, the system must define the unprotection mechanism to apply according to the protection of data set initially.	MH	Ref: UC-EUP-07
REQ-F_FEA-4.6	During a compute operation, the unprotection mechanisms are applied in a way that only authorized users and CLARUS proxy have access to the unprotected (clear) computation result over the outsourced data. Especially, the clear unprotected result must not be disclosed in any moment to the CSP or other unauthorized party.	MH	Ref: UC-EUP-07
REQ-F_FEA-4.7	In the case of public outsourced data in the cloud, i.e. after using some CLARUS protection methods on sensitive or confidential data have been applied (e.g. coarsening or anonymization), users other than the authorized users of the CLARUS solution may directly request computation services hosted in the cloud on outsourced CLARUS managed data, without the need of CLARUS proxy.	SH	Ref: UC-EUP-08
REQ-F_FEA-4.8	In the case of public outsourced data in the cloud, i.e. after using some CLARUS protection methods on sensitive or confidential data have been applied (e.g. coarsening or anonymization), the CSP computation service may provide the end user with the coarsened or anonymised associated computation result he/she has requested.	SH	Ref: UC-EUP-08, UC-MON-05

4.2.2.5 Manage access control

- Configure user repository

Req. No.	Description	Priorities	Comment
REQ-F_FEA-5.0	The CLARUS proxy must provide the IT team with the ability to configure the user repository from which the CLARUS proxy will retrieve the list of user identifiers.	MH	Ref : UC-ADM-03, UC-ADM-04
REQ-F_FEA-5.0.1	The CLARUS proxy must be able to accept a simple list of user identifiers as a “user repository” (e.g. text file, csv, etc.).	MH	
REQ-F_FEA-5.0.2	The CLARUS proxy must be able to accept a remote database as a “user repository”.	MH	
REQ-F_FEA-5.0.3	The CLARUS proxy must be able to accept a company directory as a “user repository”.	MH	

- Register users

Req. No.	Description	Priorities	Comment
REQ-F_FEA-5.1	The CLARUS proxy must provide the security manager with the ability to register new users among the users found in the configured user repository.	MH	Ref : UC-SEC-02
REQ-F_FEA-5.1.1	The CLARUS proxy must provide the security manager with the ability to search for users to be registered (by id and/or attributes) in the user repository.	MH	Ref : UC-SEC-02
REQ-F_FEA-5.1.2	The CLARUS proxy should be able to access the identity service provided by the IT infrastructure, if any.	SH	Ref : UC-SEC-02
REQ-F_FEA-5.2	The CLARUS proxy must save the users that the security manager has registered (persistence).	MH	Ref : UC-SEC-02
REQ-F_FEA-5.3	The CLARUS proxy could provide the security manager with the possibility to select among two options for the newly registered users “default permission”:	CH	Ref : UC-SEC-02

	- Creation of outsourced datasets allowed. - Creation of outsourced datasets denied.		
REQ-F_FEA-5.4	Upon registration of a new user, the CLARUS proxy must generate the appropriate materials required for creating outsourced datasets.	MH	Ref : UC-SEC-02
REQ-F_FEA-5.5	The CLARUS proxy must offer the security manager a search facility to list all the CLARUS registered users.	MH	Ref : UC-SEC-02

- Import users from another proxy

Req. No.	Description	Priorities	Comment
REQ-F_FEA-5.10	The CLARUS proxy must provide the IT team and/or the security manager with the ability to define other CLARUS proxies to interoperate with.	MH	Ref : UC-SEC-03
REQ-F_FEA-5.10.1	The CLARUS proxy should make it possible for the IT team and/or the security manager to define if the users from another given CLARUS proxy are: - Allowed to create/delete dataset(s) (e.g. from the same organization). - Denied to create/delete dataset(s) (e.g. from another organization).	SH	Ref : UC-SEC-03
REQ-F_FEA-5.11	The CLARUS proxy must provide the security manager with the ability to register new users from the user repository of the other CLARUS proxies (user import).	MH	Ref : UC-SEC-03
REQ-F_FEA-5.12	In case a user having the right to create datasets is imported from another proxy (e.g. a user from a proxy in the same organization), the CLARUS proxy must import the appropriate materials required for creating outsourced datasets.	MH	Ref : UC-SEC-03

- Activate users

Req. No.	Description	Priorities	Comment
REQ-F_FEA-5.15	The CLARUS proxy must provide the security manager with the ability to activate users, i.e. allow them to access the outsourced datasets managed by the system.	MH	Ref : UC-SEC-05
REQ-F_FEA-5.16	The CLARUS proxy must provide the security manager with a search facility to list the activated CLARUS users (i.e. registered users with an activation flag set to YES).	MH	Ref : UC-SEC-05
REQ-F_FEA-5.17	Upon activation of a user imported from another proxy, the CLARUS proxy must notify its counterpart that the user is allowed to access the outsourced datasets managed by the system.	MH	Ref : UC-SEC-05
REQ-F_FEA-5.18	Upon activation of a user exported to another proxy, the CLARUS proxy must notify its counterpart that the user is allowed to access the outsourced datasets managed by the system.	MH	Ref : UC-SEC-05
REQ-F_FEA-5.19	Upon notification by another proxy of a user activation, the CLARUS proxy must activate the corresponding user and allow it to access the outsourced datasets managed by the system.	MH	Ref : UC-SEC-05

- Deactivate users

Req. No.	Description	Priorities	Comment
REQ-F_FEA-5.20	The CLARUS proxy must provide the security manager with the ability to deactivate users, i.e. deny them the access to the outsourced datasets managed by the system.	MH	Ref : UC-SEC-04
REQ-F_FEA-5.21	The CLARUS proxy must provide the security	MH	Ref : UC-SEC-04

	manager with a search facility to list the deactivated CLARUS users (i.e. registered users with an activation flag set to NO).		
REQ-F_FEA-5.22	Upon deactivation of a user imported from another proxy, the CLARUS proxy must notify its counterpart that the user is not allowed any more to access the outsourced datasets managed by the system.	MH	Ref : UC-SEC-04
REQ-F_FEA-5.23	Upon deactivation of a user exported to another proxy, the CLARUS proxy must notify its counterpart that the user is not allowed any more to access the outsourced datasets managed by the system.	MH	Ref : UC-SEC-04
REQ-F_FEA-5.24	Upon notification by another proxy of a user deactivation, the CLARUS proxy must deactivate the corresponding user and deny it the access to the outsourced datasets managed by the system.	MH	Ref : UC-SEC-04

- Unregister users

Req. No.	Description	Priorities	Comment
REQ-F_FEA-5.25	The CLARUS proxy must provide the security manager with the ability to unregister users .	MH	Ref : UC-SEC-06
REQ-F_FEA-5.25.1	When a user is unregistered by the security manager, the CLARUS proxy must delete his/her profile.	MH	Ref : UC-SEC-06
REQ-F_FEA-5.25.2	When a user is unregistered by the security manager, the CLARUS proxy must delete all the materials thanks to which he/she could access the outsourced datasets managed by the system.	MH	Ref : UC-SEC-06
REQ-F_FEA-5.26	In case no other users have authorization on the outsourced dataset(s) created by the unregistered user, the CLARUS proxy must request the CSP to delete it/them.	MH	Ref : UC-SEC-06

REQ-F_FEA-5.27	In case other users have authorizations on the outsourced dataset(s) created by the unregistered user, the CLARUS proxy must provide the security manager with the ability to chose whether the outsourced datasets will be deleted or not.	MH	Ref : UC-SEC-06
REQ-F_FEA-5.28	In case the user being unregistered was imported from another proxy, the CLARUS proxy must notify the other proxy (unregistration notification).	MH	Ref : UC-SEC-06
REQ-F_FEA-5.29	In case the user being unregistered has been exported to another proxy, the CLARUS proxy must notify the other proxy (unregistration notification).	MH	Ref : UC-SEC-06
REQ-F_FEA-5.30	When the CLARUS proxy receives an unregistration notification from another proxy, the CLARUS proxy must unregister the corresponding user.	MH	Ref : UC-SEC-06
REQ-F_FEA-5.30.1	When the CLARUS proxy receives an unregistration notification from another proxy, the CLARUS proxy must delete his/her profile.	MH	Ref : UC-SEC-06
REQ-F_FEA-5.30.2	When the CLARUS proxy receives an unregistration notification from another proxy, the CLARUS proxy must delete his/her authorizations on the outsourced datasets managed by the system.	MH	Ref : UC-SEC-06

- Edit user profile

Req. No.	Description	Priorities	Comment
REQ-F_FEA-5.35	The CLARUS proxy must provide the security manager with the ability to edit a user profile.	MH	Ref : UC-SEC-07
REQ-F_FEA-5.36	The CLARUS proxy must provide the security manager with the ability to select the permission of a given user, among two	MH	Ref : UC-SEC-07

	options: - Creation of new outsourced datasets allowed. - Creation of new outsourced datasets denied.		
REQ-F_FEA-5.37	The CLARUS proxy must not allow the security manager to update the profile of a user who was imported from another proxy.	MH	Ref : UC-SEC-07
REQ-F_FEA-5.37.1	In case the user was imported from another proxy, the CLARUS proxy must provide the security manager with the ability to synchronise the profile with the origin proxy (pull from origin).	MH	Ref : UC-SEC-07
REQ-F_FEA-5.37.2	In case the CLARUS proxy receives a user profile synchronisation request from another proxy, the CLARUS proxy must return the corresponding user profile.	MH	Ref : UC-SEC-07
REQ-F_FEA-5.37.3	In case the CLARUS proxy receives a user profile synchronisation request from another proxy, the CLARUS proxy must return the corresponding materials for creation of outsourced datasets.	MH	Ref : UC-SEC-07
REQ-F_FEA-5.38	In case the user being updated has been exported to another proxy, the CLARUS proxy must notify the other proxy (profile update notification).	MH	Ref : UC-SEC-07
REQ-F_FEA-5.38.1	In case the CLARUS proxy receives a profile update notification from another proxy, the CLARUS proxy must update the corresponding user profile with the notified values.	MH	Ref : UC-SEC-07
REQ-F_FEA-5.38.2	In case the CLARUS proxy receives a profile update notification from another proxy, the CLARUS proxy must update the corresponding materials for creation of outsourced datasets.	MH	Ref : UC-SEC-07

- View user authorizations

Req. No.	Description	Priorities	Comment
REQ-F_FEA-5.40	The CLARUS proxy must provide the security manager with the possibility to list the outsourced datasets stored through it.	MH	Ref : UC-SEC-08
REQ-F_FEA-5.41	The CLARUS proxy must provide the security manager with the possibility to view, for a given outsourced dataset: - The owner of this outsourced dataset. - The list of users who have authorizations on this outsourced dataset.	MH	Ref : UC-SEC-08
REQ-F_FEA-5.42	The CLARUS proxy must provide the security manager with the possibility to view, for a given user and a given outsourced dataset: - The authorizations the selected user has on this outsourced dataset: read, update, delete, - The access level the selected user has on this outsourced dataset.	MH	Ref : UC-SEC-08

- Edit user authorizations

Req. No.	Description	Priorities	Comment
REQ-F_FEA-5.45	The CLARUS proxy must provide the security manager with the possibility to edit, in a secure way, the permissions (read, update, delete) and access levels of the users on an existing outsourced dataset.	MH	Ref : UC-SEC-09
REQ-F_FEA-5.45.1	The CLARUS proxy must provide the security manager with the possibility to update, for a given user and a given outsourced dataset: - The read permission of this user on this outsourced dataset (yes/no) - The update permission of this user on this outsourced dataset (yes/no) - The delete permission of this user on	MH	Ref : UC-SEC-09

	this outsourced dataset (yes/no).		
REQ-F_FEA-5.45.2	The CLARUS proxy must provide the security manager with the possibility to update, for a given user and a given outsourced dataset, the access levels the user has on this outsourced dataset.	MH	Ref : UC-SEC-09
REQ-F_FEA-5.46.1	When the permissions of a user on an existing outsourced dataset have been modified, the CLARUS proxy must modify the user materials for accessing this outsourced dataset so that they match the modified authorizations (e.g. read, update, delete).	MH	Ref : UC-SEC-09
REQ-F_FEA-5.46.2	When the access level of a user on an existing outsourced dataset has been modified, the CLARUS proxy must modify the user materials for accessing this outsourced dataset so that they match the modified access level.	MH	Ref : UC-SEC-09
REQ-F_FEA-5.47	In case the authorizations being updated correspond to a user that was imported from another proxy, the CLARUS proxy must notify the other proxy (authorizations update notification)	MH	Ref : UC-SEC-09
REQ-F_FEA-5.48	In case the authorizations being updated correspond to a user that has been exported to another proxy, the CLARUS proxy must notify the other proxy (authorizations update notification).	MH	Ref : UC-SEC-09
REQ-F_FEA-5.48.1	In case the CLARUS proxy receives an authorization update notification from another proxy (for a given user and a given outsourced dataset), the CLARUS proxy must update the permissions of the corresponding user on the corresponding outsourced dataset accordingly.	MH	Ref : UC-SEC-09
REQ-F_FEA-5.48.2	In case the CLARUS proxy receives an authorization update notification from another proxy (for a given user and a given outsourced dataset), the CLARUS proxy must	MH	Ref : UC-SEC-09

	update the access level of the corresponding user on the corresponding outsourced dataset accordingly.		
REQ-F_FEA-5.48.3	In case the CLARUS proxy receives an authorization update notification from another proxy (for a given user and a given outsourced dataset), the CLARUS proxy must update the corresponding materials for accessing the outsourced datasets.	MH	Ref : UC-SEC-09
REQ-F_FEA-5.49	The CLARUS proxy must provide the security manager with the possibility to list the registered users who do not have any authorization defined on a given outsourced dataset.	MH	Ref : UC-SEC-09
REQ-F_FEA-5.50	The CLARUS proxy must provide the security manager with the ability to add, for a given outsourced dataset, a new user in the list of users who have authorizations on this outsourced dataset, defining : - Read, update, delete permissions - Access level.	MH	Ref : UC-SEC-09

4.2.2.6 Manage security policy

- Create dataspace

Req. No.	Description	Priorities	Comment
REQ-F_FEA-6.1	The CLARUS proxy must provide the security manager with the possibility to create a dataspace in the context of CLARUS.	MH	Ref : UC-SEC-10

- Define dataspace usages and outcome accuracy

Req. No.	Description	Priorities	Comment
REQ-F_FEA-6.5	The CLARUS proxy must provide the security manager with the ability to specify one or	MH	Ref : UC-SEC-14

	several data usage(s) for a given dataspace.		
REQ-F_FEA-6.5.1	The data usages proposed by the CLARUS proxy to the security manager configuring a dataspace should be: - Create - Retrieve by id - Update by id - Search by attributes - Update by attributes - Compute.	SH	Ref : UC-SEC-14 These possible usages are deduced from the use cases and scenarios
REQ-F_FEA-6.6	The CLARUS proxy must provide the security manager with the ability to specify the outcome accuracy for a given dataspace and a given data usage (e.g. a percentage of accuracy).	MH	Ref : UC-SEC-14, UC-SEC-15

- Define protocols and data attributes to protect

Req. No.	Description	Priorities	Comment
REQ-F_FEA-6.10	The CLARUS proxy must provide the security manager with the ability to specify, for a given dataspace, one or several protocols to access the data (e.g. Amazon S3, SQL, Web Service).	MH	Ref : UC-SEC-11
REQ-F_FEA-6.10.1	The CLARUS proxy must provide the security manager with the ability, for a given protocol, to associate a dataspace with an endpoint in the cloud (e.g. either a directory in the file system or a database).	MH	Ref : UC-SEC-12 The endpoint of the cloud storage service running in the cloud
REQ-F_FEA-6.10.2	The CLARUS proxy must provide the security manager with the ability to describe an endpoint according to the protocol, e.g.: - File access protocol (SCP, SFTP, etc.): the file URI scheme, - SQL protocol: the JDBC url (host, port, database), - Web service protocol: the WSDL url - Amazon S3 protocol: the access url (REST) - etc.	MH	Ref : UC-SEC-12

REQ-F_FEA-6.10.3	The CLARUS proxy should provide a facility for the security manager to test the connection to the protocol endpoint specified (result: connection ok / connection failed).	SH	Ref : UC-SEC-12
REQ-F_FEA-6.10.4	The CLARUS proxy must provide the security manager with the possibility to use additional modules for defining protocols not natively supported by CLARUS.	MH	Ref : UC-ADM-08
REQ-F_FEA-6.11	The CLARUS proxy should control that the protocol endpoint(s) associated to the dataspace belong to the CSP(s) configured during CLARUS installation.	SH	Ref : UC-SEC-12
REQ-F_FEA-6.12	The CLARUS proxy must control that there are more than one protocol endpoint specified when the security technique applied to protect the dataspace is split/merge.	MH	Ref : UC-SEC-12
REQ-F_FEA-6.13	The CLARUS proxy must provide the security manager with the possibility to define, for a given dataspace, the supported types of data (e.g. the types of files, the database schema).	MH	Ref : UC-SEC-13
REQ-F_FEA-6.14	The CLARUS proxy must provide the security manager with the possibility to select the data attributes to be protected, with regard to: - The selected dataspace, - The selected protocol, - The selected type of data.	MH	Ref : UC-SEC-13
REQ-F_FEA-6.14.1	The CLARUS proxy could be able to visualize the structure of a database schema (given the connection url and credentials) and display it to the security manager for him/her to select data attributes to be protected.	CH	Ref : UC-SEC-13

- Select appropriate security technique

Req. No.	Description	Priorities	Comment
REQ-F_FEA-6.15	The CLARUS proxy must provide the security	MH	Ref : UC-SEC-16

	manager with the ability to configure, for a given dataspace, the security rules that are the most appropriate.		
REQ-F_FEA-6.15.1	The CLARUS proxy must present to the security manager only those options that are compatible with: - The data usages, - The accuracy of the outcome, - The type of data.	MH	Ref : UC-SEC-15, CLARUS therefore assists the security manager in configuring the security rules that are appropriate for the data usages and the outcome accuracy, i.e. presenting a reduced set of options.

4.2.2.7 Manage monitoring

Req. No.	Description	Priorities	Comment
REQ-F_FEA-7.1	The CLARUS proxy must provide the Security Manager and the IT team with the ability to monitor CLARUS usage and retrieve relevant metrics via additional relevant Monitoring and Attack management tools.	MH	Ref : UC-MON-01
REQ-F_FEA-7.2	The Monitoring tools and Attack management, additional to CLARUS, must contain an Intrusion Prevention System based upon a list of security incidents allowing definition and of dedicated counter-measures.	MH	Ref : UC-MON-08
REQ-F_FEA-7.3	The Monitoring tools and Attack management, additional to CLARUS, must contain an Intrusion Detection System based upon a list of metrics and indicators monitored allowing definition of rules to detect incidents.	MH	Ref : UC-MON-01, UC-MON-02, UC-MON-09
REQ-F_FEA-7.4	The Monitoring tools and Attack management, additional to CLARUS, must provide the authorized users with a web-based report which synthesizes all the monitoring information and relevant metrics retrieved, as well as intrusion detected and counter-measures associated in case of	MH	Ref : UC-MON-01, UC-MON-02, UC-MON-09

	system attack.		
REQ-F_FEA-7.5	The Monitoring tools and Attack management, additional to CLARUS, must provide the Security Manager with the ability to get information and generates a synthesizing web-based report coming from distributed CLARUS proxies.	MH	Ref : UC-MON-06
REQ-F_FEA-7.6	The Monitoring tools and Attack management, additional to CLARUS, must provide the Security Manager with the ability to be notified in real-time of an intrusion in the system.	MH	Ref : UC-MON-02, UC-MON-09
REQ-F_FEA-7.7	The Monitoring tools and Attack management, additional to CLARUS proxy(ies), must provide the authorized users with the ability to manually activate dedicated counter-measure(s) depending the attack type detected.	MH	Ref : UC-MON-02, UC-MON-09
REQ-F_FEA-7.8	The Monitoring tools and Attack management, additional to CLARUS proxy(ies), should provide the ability to trigger automated counter-measure(s) process by an attack detection.	SH	Ref : UC-MON-02, UC-MON-09
REQ-F_FEA-7.9	Functionality and performance of the Monitoring tools and Attack management, additional to CLARUS proxy(ies), should be tested in a dedicated integration environment.	SH	Ref : UC-MON-04
REQ-F_FEA-7.10	The Monitoring tools and Attack management, additional to CLARUS proxy(ies), must include the means to create and restore back-up states of one or several CLARUS proxy(ies) in case of failure of the latter.	MH	Ref : UC-MON-03, UC-MON-06, UC-ADM07

4.2.2.8 Manage installation/configuration

Req. No.	Description	Priorities	Comment
----------	-------------	------------	---------

REQ-F_FEA-8.1	It must be possible for an IT administrator to configure : - the parameters of the IT authentication service (if any) that CLARUS will use to authenticate the users (e.g. company SSO). - an embedded authentication mechanism (if any) that CLARUS will use to authenticate the users.	MH	Ref : UC-ADM-03, UC-ADM-05
REQ-NF_FEA-8.2	It must be possible for an IT administrator to configure the cloud storage endpoints (URLs) and credentials of one (or more) cloud storage service(s) towards which CLARUS will direct the end-users.	MH	Ref : UC-ADM-03, UC-ADM-06
REQ-F_FEA-8.3	It must be possible for an IT administrator to add a module developed using the CLARUS development API.	MH	Ref : UC-IMPL-01
REQ-F_FEA-8.4	It must be possible for an IT administrator to configure the failover mode (activated/deactivated).	MH	Ref : UC-ADM-07
REQ-F_FEA-8.5	It must be possible for an IT administrator to configure the parameters specifying that a CLARUS proxy is part for the same failover cluster as other CLARUS proxies.	MH	Ref : UC-ADM-07

4.2.3 User Interface

This set of requirements is linked to the category code : <F_USI>.

4.2.3.1 General

Req. No.	Description	Priorities	Comment
REQ-F_USI-1.1	The CLARUS solution must integrate some rich client applications running on premises, or cloud applications to perform the management of data on the cloud (storage/retrieve/update or compute).	MH	Ref: UC-EUP-01, UC-EUP-02, UC-EUP-03, UC-EUP-04, UC-EUP-05, UC-EUP-06

4.2.3.2 User management dashboard

Req. No.	Description	Priorities	Comment
REQ-F_USI-2.1	There must be a user interface for the security manager to connect to the CLARUS proxy.	MH	
REQ-F_USI-2.2	The user interface of the CLARUS proxy must contain a dashboard for the security manager to manage the CLARUS users.	MH	Ref: UC-SEC-02
REQ-F_USI-2.2.1	The user management interface of the CLARUS proxy should display a Register user screen upon selection of a dedicated button/menu item.	SH	Ref: UC-SEC-02
REQ-F_USI-2.2.2	The Register User screen of the user management interface should display an item to search for user by attributes (e.g. login, mail address, name).	SH	Ref: UC-SEC-02
REQ-F_USI-2.2.3	The Register User screen of the user management interface should display a list of users matching search criteria with their registration status (yes/no).	SH	Ref: UC-SEC-02, UC-SEC-06
REQ-F_USI-2.2.4	The Register User screen of the user management interface should provide an item to select a user to be registered/unregistered in the list of users.	SH	Ref: UC-SEC-02, UC-SEC-06
REQ-F_USI-2.2.5	The Register User screen of the user management interface should provide a button to register/unregister a selected user.	SH	Ref: UC-SEC-02, UC-SEC-06
REQ-F_USI-2.3	The user management interface of the CLARUS proxy should display a Import user screen upon selection of a dedicated button/menu item.	SH	Ref: UC-SEC-03
REQ-F_USI-2.3.1	The Import User screen of the user management interface should display the list of registered CLARUS proxies.	SH	Ref: UC-SEC-03

REQ-F_USI-2.3.2	The Import User screen of the user management interface should display an item to search for a user in a given proxy by attributes (e.g. login).	SH	Ref: UC-SEC-03
REQ-F_USI-2.3.3	The Import User screen of the user management interface should display informations about a given user from another proxy (e.g. already imported yes/no).	SH	Ref: UC-SEC-03
REQ-F_USI-2.3.4	The Import User screen of the user management interface should provide a button to import a selected user.	SH	Ref: UC-SEC-03
REQ-F_USI-2.4	The user management interface of the CLARUS proxy should display a Activate user screen upon selection of a dedicated button/menu item.	SH	Ref: UC-SEC-03
REQ-F_USI-2.4.1	The Activate User screen of the user management interface should display the list of registered users with their activation status.	SH	Ref: UC-SEC-04, UC-SEC-05
REQ-F_USI-2.4.2	The Activate User screen of the user management interface should provide an item to select a user to be activated/deactivated in the list of registered users.	SH	Ref: UC-SEC-04, UC-SEC-05
REQ-F_USI-2.4.3	The Activate User screen of the user management interface should provide a button to activate/deactivate a selected user.	SH	Ref: UC-SEC-04, UC-SEC-05
REQ-F_USI-2.5	The user management interface of the CLARUS proxy should display an Edit user profile screen upon selection of a dedicated button/menu item.	SH	Ref: UC-SEC-07
REQ-F_USI-2.5.1	The Edit User profile screen of the user management interface should display the list of registered users.	SH	Ref: UC-SEC-07

REQ-F_USI-2.5.2	The Edit User profile screen of the user management interface should provide an item to select a user to be edited in the list of registered users.	SH	Ref: UC-SEC-07
REQ-F_USI-2.5.3	The Edit User profile screen of the user management interface should display the user profile of a given user.	SH	Ref: UC-SEC-07
REQ-F_USI-2.6	The user management interface of the CLARUS proxy should display a Edit user authorizations screen upon selection of a dedicated button/menu item.	SH	Ref: UC-SEC-09
REQ-F_USI-2.6.1	The Edit user authorizations screen of the user management interface should display the list of outsourced datasets stored through the system (dataspaces).	SH	Ref: UC-SEC-09
REQ-F_USI-2.6.2	The Edit user authorizations screen of the user management interface should provide an item to select an outsourced dataset.	SH	Ref: UC-SEC-09
REQ-F_USI-2.6.3	The Edit user authorizations screen of the user management interface should display the list of users which have authorizations on a selected outsourced dataset.	SH	Ref: UC-SEC-09
REQ-F_USI-2.6.4	The Edit user authorizations screen of the user management interface should provide an item to select a user which have authorizations on a given outsourced dataset.	SH	Ref: UC-SEC-09
REQ-F_USI-2.6.5	The Edit user authorizations screen of the user management interface should display an item to view/modify the authorizations a selected user has on a given dataset (read, update, delete).	SH	Ref: UC-SEC-09
REQ-F_USI-2.6.6	The Edit user authorizations screen of the user management interface should display an item to view/modify the access level the selected user has on a given outsourced dataset.	SH	Ref: UC-SEC-09

4.2.3.3 Security policy dashboard

Req. No.	Description	Priorities	Comment
REQ-F_USI-3.1	The user interface of the CLARUS proxy must contain a dashboard for the security manager to manage the security policy rules.	MH	
REQ-F_USI-3.2	The security policy management interface of the CLARUS proxy should display a Create Dataspace screen upon selection of a dedicated button/menu item.	SH	
REQ-F_USI-3.3	The Create Dataspace screen of the security policy management interface should display an input to fill in the name of the dataspace to be created.	SH	
REQ-F_USI-3.4	The Create Dataspace screen of the security policy management interface should provide an item to select among the list of available protocols.	SH	e.g. File access protocol (SCP, SFTP, etc.), SQL protocol, Web service protocol, Amazon S3 protocol, REST, etc.
REQ-F_USI-3.5	The Create Dataspace screen of the security policy management interface should provide an input to configure the endpoint for each selected protocol	SH	e.g. a file URI scheme, a JDBC url (host, port, database), a WSDL url, etc.
REQ-F_USI-3.6.1	The Create Dataspace screen of the security policy management interface should display the list of attributes that could be protected	SH	i.e. protocol attributes, attributes of file types, database attributes and columns to protect given a database schema Ref: UC-SEC-13
REQ-F_USI-3.6.2	The Create Dataspace screen of the security policy management interface should provide an item to select the attributes to be	SH	Ref: UC-SEC-13

	protected		
REQ-F_USI-3.6.3	The Create Dataspace screen of the security policy management interface should provide a button to save the attributes to be protected	SH	Ref: UC-SEC-13
REQ-F_USI-3.7.1	The Create Dataspace screen of the security policy management interface should display the list of possible data usages	SH	see. REQ-F_FEA-6.5.1
REQ-F_USI-3.7.2	The Create Dataspace screen of the security policy management interface should provide an item to select, among the usage list, the usage(s) of the outsourced datasets within the dataspace	SH	Ref: UC-SEC-14
REQ-F_USI-3.8	The Create Dataspace screen of the security policy management interface should prompt for the accuracy of the outcome of the selected usage (e.g. percentage of accuracy)	SH	Ref: UC-SEC-15
REQ-F_USI-3.9.1	The Create Dataspace screen of the security policy management interface should display the list of security techniques that are the most appropriate according to the dataspace configuration that has been set	SH	Ref: UC-SEC-16
REQ-F_USI-3.9.2	The Create Dataspace screen of the security policy management interface should provide an item to select, among the list of possible options, the security technique to be applied	SH	Ref: UC-SEC-16
REQ-F_USI-3.10	The Create Dataspace screen of the security policy management interface should provide a button to create the dataspace with all the configuration parameters selected	SH	
REQ-F_USI-3.11	The security policy management interface of the CLARUS proxy should display a Configure Dataspace screen upon selection of a dedicated button/menu item	SH	
REQ-F_USI-3.12	The Configure Dataspace screen of the security policy management interface should	SH	

	provide an item to chose the dataspace to be configured among the list of existing dataspace		
REQ-F_USI-3.13	The Configure Dataspace screen of the security policy management interface should propose the same configuration items as the Create Dataspace screen	SH	i.e. select protocols, select attributes to protect, select usages, select security techniques, etc.

4.2.3.4 Monitoring and attack management interface

Req. No.	Description	Priorities	Comment
REQ-F_USI-4.1	The CLARUS interface must include a Monitoring and Attack Management Dashboard (MAMD) that will be accessible by the Security Manager and the IT Team.	MH	Ref: UC-MON-01
REQ-F_USI-4.2	The MAMD must include a list of metrics that can be measured by the monitoring tools and options to show (description, current value), select, and unselect specific metrics.	MH	Ref: UC-MON-01, UC-MON-07
REQ-F_USI-4.3	The MAMD must include a list of counter-measures and the following options: show description, select, unselect, activate, and deactivate.	MH	Ref: UC-MON-07, UC-MON-08, UC-MON-09
REQ-F_USI-4.4	The MAMD must include a list of possible attacks, an option to show the description of each attack and the suggested counter-measures to mitigate it, as well as the possibility of configuring (i.e., adding and removing) counter-measures to be automatically applied for a specific attack.	MH	Ref : UC-MON-08, UC-MON-09
REQ-F_USI-4.5	The MAMD must include the possibility of selecting and retrieving the value of a list of metrics: - From an individual CLARUS proxy, - Correlated from a list of CLARUS	MH	Ref: UC-MON-01, UC-MON-06

	proxies.		
REQ-F_USI-4.6	The MAMD must include the possibility of activating one or several counter-measures.	MH	Ref : UC-MON-08, UC-MON-09
REQ-F_USI-4.7	The MAMD should include the possibility of configuring the receiver of the attack alarms and reports (in case there are more than one Security Manager or that any member from the IT Team should be informed).	SH	Ref : UC-MON-08, UC-MON-09
REQ-F_USI-4.8	The MAMD should have the possibility of scheduling automatic report forwarding.	SH	Ref : UC-MON-02, UC-MON-09
REQ-F_USI-4.9	The MAMD must include the possibility of selecting and downloading existing reports.	MH	Ref : UC-MON-02, UC-MON-09
REQ-F_USI-4.10	The MAMD must include the means to create and restore back-up states.	MH	Ref : UC-MON-03

4.2.4 Social needs

This set of requirements is linked to the category code : <F_SOC>.

Req. No.	Description	Priorities	Comment
REQ-F_SOC-1.1	The CLARUS solution, adopted by a customer who outsources his/her data to a CSP, must improve the trust of the customers in data storage with a cloud computing solution.	MH	
REQ-F_SOC-1.2	The architecture of CLARUS solution must be designed in a service-oriented architecture, with friendly interfaces to run the security mechanisms.	SH	Ref: UC-EUP-01
REQ-F_SOC-1.3	The usability of CLARUS interfaces could be improved especially as the user interfaces are graphical.	CH	
REQ-F_SOC-1.4	The usability of CLARUS interfaces could be improved especially as the interfaces are delivered with user manuals.	CH	

REQ-F_SOC-1.5	The usability of CLARUS interfaces could be improved especially as the interfaces include clear and non ambiguous instructions as well as informative error messages.	CH	
REQ-F_SOC-1.6	The usability of CLARUS interfaces could be improved especially as the interfaces are supplemented by help facilities.	WL	
REQ-F_SOC-1.7	The instructions displayed to end users by CLARUS interfaces must be submitted in the same language as used for CLARUS agreement document [A2].	MH	
REQ-F_SOC-1.8	The usability of CLARUS interfaces could be improved especially as the interfaces are supplemented with multi-lingual features, providing to the end users the choice to display the instructions in a selected language.	WL	

4.3 Non Functional Requirements

We call non-functional requirements the characteristics of the system that are important to the users even though they may not realize it. As the name suggests these requirements do not deal with a functionality of the system, but rather place constraints on how the system should operate.

This set of requirements allows to express some performance characteristics of CLARUS system, sometimes measurable.

We will address in this part the following main topics already set, namely :

- Delivery
- Implementation
- Quality of Service
- Operational
- Persistence
- Security
- Performance
- ScalabilityTransportStandardisation
- Legal and ethical
- Open Source

4.3.1 Delivery

This set of requirements is linked to the category code : <NF_DLV>.

- Delivery requirements

Req. No.	Description	Priorities	Comment
REQ-NF_DLV-1.1	An installation package should be available for an IT administrator to install easily the CLARUS proxy on a given operating system (e.g. Unix/Linux).	SH	Ref : UC-ADM-01
REQ-NF_DLV-1.2	CLARUS solution assumes a collaborative agreement by the CLARUS adopting CSPs, which must be willing to comply the need to install a CLARUS library in the cloud platform.	MH	Ref : UC-ADM-09
REQ-NF_DLV-1.3	It must be possible to upgrade CLARUS to a newer version, containing additional/updated modules.	MH	Ref : UC-ADM-01, UC-ADM-02, UC-ADM-10
REQ-NF_DLV-1.4	The CLARUS solution must be delivered with an installation guide (or CSP installation guide).	MH	Ref : UC-ADM-01, UC-ADM-09
REQ-NF_DLV-1.5	The CLARUS solution must be delivered with a configuration guide.	MH	Ref : UC-ADM-03
REQ-NF_DLV-1.6	The CLARUS solution must be delivered with a development API.	MH	Ref : UC-IMPL-01
REQ-NF_DLV-1.7	The development API for CLARUS modules must be delivered with a development guide.	MH	Ref : UC-IMPL-01
REQ-NF_DLV-1.8	The CLARUS solution must be delivered with a monitoring and intrusion management solution.	MH	Ref : REQ-F_FEA-7.1, UC-MON-01

- Configuration Management

Req. No.	Description	Priorities	Comment
REQ-NF_DLV-1.9	A configuration management methodology could be applied on the CLARUS software changes to handle the different releases of the modules/packages developed.	CH	Ref : UC-ADM-02
REQ-NF_DLV-1.10	A configuration management methodology	CH	Ref : UC-ADM-01, UC-ADM-03

	could be applied on the documentation delivered with the CLARUS solution (installation guide, configuration guide, etc.) to handle the different releases of each document.		
REQ-NF_DLV-1.11	A configuration management methodology could be applied on software changes of the CLARUS development API to handle the different releases of the CLARUS development API.	CH	Ref : UC-IMPL-01
REQ-NF_DLV-1.12	A configuration management methodology could be applied on the documentation delivered with the CLARUS development API (development manual) to handle the different releases of the document.	CH	Ref : UC-IMPL-01
REQ-NF_DLV-1.13	A configuration management methodology could be applied on software changes of the CLARUS Monitoring and Attack management tools, to handle the different releases.	CH	Ref : UC-MON-01
REQ-NF_DLV-1.14	A configuration management methodology could be applied on the documentation delivered with the CLARUS Monitoring and Attack management tools to handle the different releases of the associated documents.	CH	Ref :C-MON-01

4.3.2 Implementation

This set of requirements is linked to the category code : <NF_IMP>.

Req. No.	Description	Priorities	Comment
REQ-NF_IMP-1.1	The CLARUS solution implementation must allow multi OS (e.g. Windows, Linux...) compatibility.	MH	
REQ-NF_IMP-1.2	Storage service must be configured to allow access only from the end user premises.	MH	Ref : UC-EUP-01
REQ-NF_IMP-1.3	For storage application case, the end user of	MH	Ref : UC-EUP-01, UC-EUP-02, UC-

	CLARUS solution must have a user account with sufficient privileges to access the CSP data storage space.		EUP-03
REQ-NF_IMP-1.4	The CLARUS system must be able to identify the dataset to which a data belongs by means of an identifier.	MH	Ref : UC-EUP-01
REQ-NF_IMP-1.5	The CLARUS system must support different data types as structured data (e.g. relational database) as well as unstructured (e.g. binary file, image, text file).	MH	Ref : UC-EUP-01, UC-EUP-02, UC-EUP-03
REQ-NF_IMP-1.6	CLARUS solution assumes a collaborative agreement by the CLARUS adopting CSPs, which must be willing to comply the need to adapt the implementation of cloud services if required (in order to integrate CLARUS features required to enhance security and trust in storage or some computational cloud services).	MH	Ref : UC-IMPL-02

4.3.3 Quality of Service

This set of requirements is linked to the category code : <NF_QOS>.

Req. No.	Description	Priorities	Comment
REQ-NF_QOS-1.1	Any end-user applications interfaced with CLARUS features, must maintain their ability to perform the required usual functions as a guaranteed reliability.	MH	
REQ-NF_QOS-1.2	CLARUS securing tool should be an “Attack Tolerant System” for the cloud, so that it continues to operate and deliver the required services under the event of an attack.	SH	Ref : UC-MON-07
REQ-NF_QOS-1.3.1	CLARUS must not impair typical cloud functionality (including the functionality provided by high-level services such as data storage, management, retrieval, transformation, etc.)	MH	CLARUS must not imply loss of functionality, i.e. make it hard or impossible to perform certain operations on data

REQ-NF_QOS-1.3.2	The benefits of an ubiquitous access of data outsourced offered by a high availability service of a CSP, should be maintained after adopting the CLARUS solution.	SH	
REQ-NF_QOS-1.4	CLARUS must implement the appropriate techniques to ensure the reliability and availability of the service (e.g. provide a fail-over mode).	MH	To be validated through metrics e.g. failure rate, mean time between failures (example: shall have an availability of 95%)
REQ-NF_QOS-1.5	The CLARUS solution should be able to prevent the complete loss of services availability during attacks.	SH	I.e. graceful degradation of services

4.3.4 Operational

This set of requirements is linked to the category code : <NF_OPE>.

Req. No.	Description	Priorities	Comment
REQ-NF_OPE-1.1	Each application case of data outsourcing to the cloud (and data modification) must be complemented by data distribution/replication between distincts CSPs or in distincts location/data centers of an unique CSP, in a Disaster Recovery approach.	MH	Ref : UC-EUP-01, UC-EUP-03
REQ-NF_OPE-1.2	In case of failure of the main CSP, data integrity and availability must be ensured by the “Disaster Recovery” center designated by the CSP.	MH	Ref : UC-EUP-01, UC-EUP-02, UC-EUP-03
REQ-NF_OPE-1.3	The solution must be robust against local CLARUS failures.	MH	

4.3.5 Persistence

This set of requirements is linked to the category code : <NF_PER>.

The persistence set of requirements details operational and performance criteria relating to the storage of information back-up of database and files system and other persistent storage mechanisms.

Req. No.	Description	Priorities	Comment
REQ-NF_PER-1.1	The business continuity linked to the persistence and the recovering of the data outsourced, after protection by the CLARUS solution, will be ensured by the CSP in a Disaster Recovery plan.	SH	Ref : REQ-NF_OPE-1.1, REQ-NF_OPE-1.1
REQ-NF_PER-1.2	All the configuration components of a CLARUS proxy must be subject to a back-up in case of outage of the CLARUS proxy (requiring a new CLARUS proxy reconfiguration or the initial CLARUS proxy restoration).	MH	Ref : REQ-F_FEA-7.10, UC-MON-03
REQ-NF_PER-1.3	The metadata stored by CLARUS must be stored as a backup.	MH	Either on premises or in the cloud (protected)
REQ-NF_PER-1.4	The backup of the metadata stored by CLARUS must be protected against the CSP and any other unauthorised party.	MH	

4.3.6 Security

This set of requirements is linked to the category code : <NF_SEC>.

The security set of requirements details both the information security (data access) and physical security (servers, applications and critical hardware access).

Req. No.	Description	Priorities	Comment
REQ-NF_SEC-1.1	The access permissions to CLARUS system may only be changed by the CLARUS security manager, or the like, of the organization concerned.		
REQ-NF_SEC-1.2	The CLARUS solution must ensure that data is protected from unauthorised access.	MH	
REQ-NF_SEC-1.3	The CLARUS solution should be able to isolate attacks and contain them.	SH	
REQ-NF_SEC-1.4	The CLARUS solution must be able to regenerate after an attack, and ensure the	MH	i.e. get back to a previous state

	integrity of the system.		
REQ-NF_SEC-1.5	The CLARUS solution must offer a quality of data protection satisfying for the organization concerned, by the means of dedicated security mechanisms, on each type of data where privacy or confidentiality is a major concern.	MH	CLARUS must not imply a lower degree of security, e.g. in the use of function-preserving encryption methods.

4.3.7 Performance

This set of requirements is linked to the category code : <NF_PERF>.

The performance set of requirements can be linked with parameters such as response time and such measurable aspects of the system that govern overall speed and responsiveness.

Req. No.	Description	Priorities	Comment
REQ-NF_PERF-1.1	The securisation mechanisms introduced and embodied by CLARUS must not significantly degrade the response time compared to the normal behaviour of data processing.	MH	
REQ-NF_PERF-1.2	CLARUS should improve the performance of requests (e.g. search, operations) compared to naïve implementations of security-enabling solutions already available in some commercial products.	SH	Naïve implementations would be e.g. download the whole encrypted/ anonymised dataset from the CSP to the CLARUS proxy; perform the search operations on the proxy over clear data; update and re-encrypt / anonymise the results and re-upload them to the CSP
REQ-NF_PERF-1.3	CLARUS should improve the performance of requests (e.g. search, operations) compared to the same approach directly on the end user's computer.	SH	Without considering the case of not using the cloud at all (i.e. to run the whole application on the end user computer).

4.3.8 Scalability

This set of requirements is linked to the category code : <NF_SCA>.

Scalability is the ability of the system to handle an increased workload without significantly increasing the transaction processing time.

Req. No.	Description	Priorities	Comment
REQ-NF_SCA-1.1	The CLARUS solution must implement the appropriate techniques (distributed based system) to ensure the scalability of the system.	MH	e.g. replication, parallelization

4.3.9 Transport

This set of requirements is linked to the category code : <NF_TRSP>.

The transport set of requirements defines constraints affecting the transmission of information between nodes.

Req. No.	Description	Priorities	Comment
REQ-NF_TRSP-1.1	The CLARUS solution must integrate and support secured communication protocols for the data management and transfers to the cloud storage environment.	MH	Ref : UC-SEC-11 e.g. HTTPS, SCP, SFTP
EQ-NF_TRSP-1.2	The CLARUS solution must support the secured protocols used by cloud services with which CLARUS interacts for data processing.	MH	Ref : UC-SEC-11
REQ-NF_TRSP-1.3	The CLARUS solution must seamlessly integrate the network infrastructure, in order to not induce any significant latency in the requests processing.	SH	

4.3.10 Standardisation

This set of requirements is linked to the category code : <NF_STD>.

As previously mentioned, as part of WP2, this current document is part of a suite of related deliverable, in particular D2.5 [R2]. This latter, outlines standardisation requirements and provides a roadmap aiming to integrate the existing standards used, function of the different CLARUS fields of application. Consequently, we voluntarily approach this topic in the form of a generic requirement. Refer to D2.5 for the CLARUS compliance approach and the standardisation effort which must be associated.

Req. No.	Description	Priorities	Comment
REQ-NF_STD-1.1	The CLARUS solution must support most of the standards used in the different business domains on which CLARUS is deployed.	MH	

4.3.11 Legal and ethical

This set of requirements is linked to the category code : <NF_LEG>.

The goal of this section is to furnish a first framework of requirements for the implementation of a legally compliant CLARUS solution. These first set of requirements will be more extended and developed in the D2.4 deliverable **¡Error! No se encuentra el origen de la referencia.**, where constraints will be moreover examined from CLARUS application cases in domains (i.e Geospatial publications and e-Health) where data privacy and confidentiality are high concerns.

Req. No.	Description	Priorities	Comment
REQ-NF_LEG-1.1	Any access authentication controls to CLARUS proxy(s) (based on authorisations delivered to an end user on existing outsourced dataset, see UC-EUP-06), processing personal data to do so in a secure manner, respecting Article 17 Directive 95/46/EC. The data minimization principle and purpose limitation principle (as contained in Article 6 Directive 95/46/EC) should also be respected.	MH	Directive 95/46/EC : [R3] Ref : UC-SEC-09
REQ-NF_LEG-1.2	Any personal data related to the authenticated users, must be securely protected throughout the data lifecycle. As per Article 6 Directive 95/46/EC, any data that is not necessary for the personal data processing should be deleted in compliance with the limited retention and data minimisation principles.	MH	Directive 95/46/EC : [R3] Ref : UC-SEC-09
REQ-NF_LEG-1.3	Any personal data related to the authenticated users, must be securely protected throughout the data lifecycle : • Encryption should be employed	MH	Directive 95/46/EC : [R3] Ref : UC-SEC-09

	throughout with the default state of data being unreadable if there is a data leak. • In respect of the data subject's rights, any request to delete personal data should be respected. • Any processing of personal data must be transparent so that the end user is aware of the nature of the processing and can exercise his/her data subject rights in an unencumbered fashion.		
REQ-NF_LEG-1.4	Any communications sent between the authenticated users and the CLARUS adopting CSP to be encrypted in order to prevent any unauthorized access and ensure the security of the data being shared. Systems should be designed to ensure that even where personal data are transmitted, any data elements which are not necessary to fulfil the purpose of the transmission are filtered out or removed.	MH	Directive 95/46/EC : [R3] Ref : UC-SEC-09
REQ-NF_LEG-1.5	The cloud storage of any personal (sensitive) data, processed by CLARUS solution must be compliant in respect of article 17 Directive 95/46/EC, namely that such storage must be secure.	MH	Directive 95/46/EC : [R3] Ref : UC-EUP-01
REQ-NF_LEG-1.6	CLARUS system must be designed so as to allow access to the transferred personal data only to the extent necessary for the role being performed and the individual performing it. Such access rights should be easily revocable.	MH	Directive 95/46/EC : [R3] Ref : UC-EUP-01, UC-SEC-04, UC-SEC-06, UC-SEC-09
REQ-NF_LEG-1.7	Ensure that additional legal and ethical requirements, set out in deliverable D2.4, are embedded in CLARUS solution.	MH	D2.4 deliverable ¡Error! No se encuentra el origen de la referencia.

4.3.12 Open Source

This set of requirements is linked to the category code : <NF_OPS>.

Req. No.	Description	Priorities	Comment
----------	-------------	------------	---------

REQ-NF_OPS-1.1	The CLARUS softwares implemented should be an open source solution.	SH	
----------------	---	----	--

5 Conclusion

In this document, the CLARUS functional and non-functional requirements are described, in relation with the main use cases of the expected solution.

Summarizing the security expectations arising from the cloud-based scenarios described in the D2.1 deliverable was the starting point of the analysis. These application cases in the eHealth and the geospatial domains first help to get a clear vision of the different actors involved in the context of CLARUS.

These actors can be divided in two distinct categories: on the one hand the actors involved in the infrastructure management, and on the other the end-users of the solution.

- The first set of actors is involved in the installation and configuration of the CLARUS software, either on premises (IT team) or in the cloud (CLARUS-adopting CSP), in the development of additional modules (developers) or in the implementation of the attack-tolerant system (architects/developers).
- As for end-users they could further be divided in two groups, according to whether :
 - o they use the CLARUS interface to manage access control and security policy – that is the security manager,
 - o they use transparently the cloud functionalities supported and secured by the solution, thanks to the security policy configured – that is the data providers/consumers.

The data provider/consumer perspective is of utmost interest, because from it we can deduce the cloud functionalities to be supported and their expected outcomes. Those are the most straightforward inputs for functional requirements. Four scenarios are proposed that reflect different use cases: User data storage, Search in the cloud, Update in the cloud and Computation in the cloud. These scenarios are a combination of basic features which are perfectly suitable for becoming one of the building blocks of the CLARUS security policy, that is “data usage” – create, retrieve by id, update by id, search by attributes, update by attributes, compute.

The security manager perspective is precisely important for defining further the way to configure a security policy with CLARUS. In this prospect, it made sense to introduce the notion of “dataspace” – a dataspace being a virtual space that links the applications/services running on IT premises with the cloud storage services. This notion is very important to understand how CLARUS could protect the outsourced datasets according to their usage. The task of the security manager is to specify the usage(s) and outcomes' accuracy for a given dataspace, along with the the supported protocols, the appropriate endpoints of the cloud storage services, the supported types of data and the attributes to be protected. Given all these parameters, the security manager has to select the security rules that are the most appropriate. The CLARUS proxy, presenting a reduced set of options compatible with the data usage and accuracy, assists the security manager in this task. The security manager would also be responsible for the management of access control (authorizations of users on existing outsourced datasets in a dataspace) and the monitoring (detection of intrusions and prevention of attacks thanks to monitoring tools).

The perspectives from other actors mainly relate to the installation of CLARUS and to the development of additional modules. The IT team/administrator perspective provides a set of functional and non-functional (delivery) requirements regarding the installation of the CLARUS proxy on premises, while the

CSP perspective shows how to integrate CLARUS in the cloud platform (thanks to a dedicated API) and provide the end-users with security-enhanced and trust-enhanced cloud services. Lastly, the Developer perspective provides requirements regarding the development of modules, allowing the CLARUS proxy to support features not natively supported (e.g. a new protocol used to transfer data).

The CLARUS requirements listed in this document have been derived from these different perspectives and their corresponding scenarios either directly (functional requirements) or indirectly (non-functional requirements). The requirements here listed allow to cover all facets of the CLARUS solution proposed, and will be refined, as well as new requirements will be defined at a later stage of this WP, namely in a D2.3 document.

Appendix A. Mapping CLARUS use cases with Application cases

A document describing the application cases in the e-Health and geospatial domains and matching them with the CLARUS use cases here defined is joined in the additional document :

“CLARUS D2.2 Annex I - Mapping CLARUS use cases with application cases” [A5].

**** End of the Document ****



A framework for user centred privacy and security in the cloud

Requirements Specification – Annex I: Mapping CLARUS use cases with Application cases

Type (distribution level)	Public
Contractual date of Delivery	30-06-2015
Actual date of delivery	30-07-2015
Deliverable number	D2.2-AnnexI
Deliverable name	Requirements Specification – Annex I: Mapping CLARUS use cases with application cases
Version	V1.0
Number of pages	19
WP/Task related to the deliverable	Task 2.2
WP/Task responsible	AKKA
Author(s)	AKKA Team
Partner(s) Contributing	
Document ID	CLARUS-D2.2-RequirementsSpecification-AnnexI-v1.0

Abstract

This document describes the business cases in the e-Health and geospatial domains, as identified during the definition of application cases, and matches them cases with the CLARUS use cases defined in the requirement dossier.

Disclaimer

CLARUS (G.A. 644024) is a Research and Innovation Actions project funded by the EU Framework Programme for Research and Innovation Horizon 2020. This document contains information on CLARUS core activities, findings and outcomes. Any reference to content in this document should clearly indicate the authors, source, organization and publication date. The content of this publication is the sole responsibility of the CLARUS consortium and cannot be considered to reflect the views of the European Commission.

Table of Contents

1	INTRODUCTION.....	6
1.1	SCOPE OF THE DOCUMENT.....	6
1.2	APPLICABLE AND REFERENCE DOCUMENTS	6
1.3	REVISION HISTORY	6
1.4	NOTATIONS, ABBREVIATIONS AND ACRONYMS	6
2	BUSINESS CASES.....	7
2.1	E-HEALTH	7
2.1.1	<i>Securing Passive Medical Health Records storage in the cloud</i>	<i>7</i>
2.1.2	<i>Securing Passive Medical Health Records access and retrieval from the cloud</i>	<i>7</i>
2.1.3	<i>Securing Passive Medical Health Records for the Advanced Query</i>	<i>7</i>
2.1.4	<i>Securing Passive Medical Health Records for Statistics Computation query</i>	<i>8</i>
2.2	GEO-PUBLICATION.....	8
2.2.1	<i>Geodata Storage in the cloud.....</i>	<i>8</i>
2.2.2	<i>Geodata Publication in the cloud</i>	<i>8</i>
2.2.2.1	Web mapping (metadata and map creation)	8
2.2.2.2	Geo-processing (Kriging computation)	8
2.2.3	<i>Collaboration on Geodata in the cloud</i>	<i>9</i>
2.3	GENERIC BUSINESS CASES	9
3	MAPPING OF DEMONSTRATION CASES WITH USE CASES	10
3.1	SINGLE-USER STORAGE	10
3.1.1	<i>Store by id through proxy.....</i>	<i>10</i>
3.1.1.1	E-Health storage	10
3.1.1.2	Geodata storage	11
3.1.2	<i>Retrieve by id through proxy.....</i>	<i>12</i>
3.1.2.1	E-Health Access and Retrieval	12
3.1.2.2	Geodata storage	12
3.2	SEARCH IN THE CLOUD	13

3.2.1	<i>Search through proxy</i>	13
3.2.1.1	E-Health advanced query	13
3.2.2	<i>Search directly in the cloud</i>	14
3.2.2.1	Web mapping (Geo-publication)	14
3.2.2.2	E-Health advanced query	15
3.3	UPDATE IN THE CLOUD.....	15
3.3.1	<i>Update data by attributes through the CLARUS proxy</i>	16
3.3.1.1	Geodata storage	16
3.3.1.2	Geodata collaboration	17
3.4	COMPUTATION IN THE CLOUD	17
3.4.1	<i>Compute through proxy</i>	18
3.4.1.1	E-Health statistics computation query	18
3.4.1.2	Geo-processing (Kriging computation)	19
3.4.2	<i>Compute directly in the cloud</i>	19

1 Introduction

1.1 Scope of the document

The aim of this document is to:

- describe the business cases in the e-Health and geospatial domains, as identified during the definition of application cases
- match these business application cases with the CLARUS use cases defined in the requirements specification

1.2 Applicable and reference documents

This document refers to the following documents:

- CLARUS Grant Agreement
- CLARUS-D2.1-Definition of Application Cases-V1.0
- CLARUS-D2.2-Requirements Specification-V1.0

1.3 Revision History

Version	Date	Author	Description
0.1	01/06/2015	AKKA	Initial version
1.0	30/06/2015	AKKA	V1.0 Version

1.4 Notations, Abbreviations and Acronyms

<i>API:</i>	Application Programming Interfaces
<i>CSP :</i>	Cloud Service Provider
<i>GA:</i>	Grant Agreement
<i>GIS:</i>	Geographical Information System
<i>HIS :</i>	Hospital Information System
<i>ICD :</i>	International Classification of Diseases
<i>INSPIRE :</i>	Infrastructure for Spatial Information in Europe
<i>LOINC :</i>	Logical Observation Identifiers Names and Codes
<i>OGC :</i>	Open Geospatial Consortium
<i>WFS :</i>	Web Feature Service
<i>WMS :</i>	Web Map Service
<i>WPS :</i>	Web Processing Service

2 Business Cases

In this section we summarize the cloud-based scenarios where data confidentiality is an issue described in the D2.1 deliverable. These business cases demonstrate the appropriateness and applicability of CLARUS solutions in two different contexts, respectively the domain of geospatial data publication and of e-Health.

2.1 E-Health

2.1.1 Securing Passive Medical Health Records storage in the cloud

In the context of an HIS, the storage facilities of the cloud are of utmost interest in order to fill the need to “passivate” active Medical Records for the purpose of archiving or lower frequency consultation. The expectations in terms of security for this business case are obviously high, as medical records represent highly personal data to be protected against the cloud service provider or any other party – e.g. results reports, clinical Notes, Diagnostics (ICD-9), Lab Results (LOINC), in-patient medication, etc.

Passivation of medical records is done by a “medical record passivation responsible”, using the “Clinical Workstation interface” of the Hospital Information System (HIS) – i.e. a rich-client application running on premises, while the medical data are stored in a database in the cloud.

2.1.2 Securing Passive Medical Health Records access and retrieval from the cloud

Once stored, it is often necessary for the medical doctors to retrieve the passive medical health records according to the patient ID, for medical practice, medical research, legal purposes, etc. Here again, the security needs are very high, and access and retrieval of medical records stored in the cloud must be properly protected, fully justifying the use of the CLARUS innovative set of security-enabling techniques.

Retrieval of passive medical records is done through the Clinical Workstation interface running on premises, while the medical data are stored in a database in the cloud.

2.1.3 Securing Passive Medical Health Records for the Advanced Query

Another typical need in an HIS is to retrieve health data information using multiple fields as search criteria, i.e. defining different conditions to be met by multiple attributes – e.g. specific values or ranges of values, duration, etc. The Healthcare professionals performing this Advanced Query are able to view the resulting dataset, and also to download it. The security needs here relate mainly to data anonymisation.

This business case is particularly useful for research purpose (population assessment, scientific research, etc.). Here again the research is done through the Clinical Workstation interface running on premises, while the medical data are stored in a database in the cloud.

2.1.4 Securing Passive Medical Health Records for Statistics Computation query

This business scenario relates to the case where a healthcare professional wants to make statistics computation on the medical data stored in the cloud. These computations (e.g. calculation of means, standard deviations, and frequencies) are performed *in the cloud*, i.e. without requiring to download any data on premises. The Healthcare professionals are able to view the results of the computation and also to download them. In this case, the healthcare professionals want the statistics computation to be performed on anonymised data (security expectation).

2.2 Geo-Publication

2.2.1 Geodata Storage in the cloud

Cloud storage facilities are of great interest to geospatial professionals as the volume of geodata to be stored becomes more and more important over time. Geospatial professionals and content providers already need to move all or part of their data to the cloud, but at the same time part of these data (i.e. geographical coordinates and/or scientific data such as measurements) need to be secured in some specific cases, e.g. Rare Earth Elements or Minerals Resources.

Storage of geodata is often done thanks to rich-client application providing support for spatial databases and GIS file formats. These applications run on premises while the geodata are stored on the cloud.

2.2.2 Geodata Publication in the cloud

2.2.2.1 Web mapping (metadata and map creation)

Following the INSPIRE directive, public authorities responsible for geodata (e.g. national geological surveys) need to publish their data on the internet and make them accessible through research, consultation and downloading thanks to metadata and services. In some cases (borehole data for instance) they also need to restrict the access to some of their data for public security, national defence or business reasons.

Publication of geodata is done thanks to various services, notably geo web services (e.g. OGC WMS and WFS). Dataset files, prepared on premises, are first uploaded to the cloud, then map layers are published as WMS/WFS services. Access to these maps is done through a dedicated web application running on the cloud, compatible with OGC services.

2.2.2.2 Geo-processing (Kriging computation)

In the geospatial field, companies and/or institutions may sell geostatistics services, such as the Kriging interpolation (a well-known interpolation technique in geology). In such a business case, security needs are strong as the service provider often wants to keep secret the sample locations and measurements on which the computation is based, and as the service consumer may want the specific location and its estimated prediction to remain confidential too. The confidential coordinates of the sample locations, and their associated measurements, therefore need to be protected (e.g. via encryption or any other alternative).

The geo-processing (here Kriging) service is published as an OGC WPS service and accessed by the customer via a map application hosted in the cloud.

2.2.3 Collaboration on Geodata in the cloud

Providing support to situations of emergency is one of the key roles of a Geographical Information System. It is crucial that on the immediate aftermath of a disaster, geohazard experts in the field can collect and share information about the spatial extent and the consequences of these events (for the coordination of emergency efforts). Security needs are strong here because without any guarantee in terms of security and confidentiality, companies may be reluctant to share their data, even in case of emergencies.

The collaboration on geodata is done through mobile applications using OGC WFS-T service for transactional updates of map features – in this case geospatial servers supporting the update through WFS-T are running in the cloud.

2.3 Generic Business Cases

One aim of the CLARUS framework is to be *generic* enough to apply to other domains than the strict fields of e-health and geo-publication.

Actually the application cases described allow to cover a broader range of business domains concerned by both privacy protection and data confidentiality. For instance the Finance domain, concerned by privacy protection and personal data, could be related to the cases described in the e-Health section. Other business domains, more concerned by data confidentiality (e.g. for public security or commercial purposes), could be related to the cases described in the geo-publication section.

In addition, the two domains and their corresponding security expectations could be cross-correlated. A good example is given by the security concerns in the geospatial field which are about to extend beyond the perimeter of data confidentiality. It is indeed interesting to note that, as the use of geo-data becomes widespread and as this kind of data shows more and more bundling powers (i.e. through location one is able to link data together), this will surely drive Geospatial Information to the field of Privacy protection, along with e-Health.

3 Mapping of Demonstration Cases with Use Cases

This section shows the possible mapping of the various demonstration cases proposed in the application case definition with the CLARUS Use Cases.

3.1 Single-user storage

eHealth / GeoData Demonstration cases		CLARUS use case Single-user storage	
		Store by id through proxy	Retrieve by id through proxy
		UC-EUP-01	UC-EUP-02
eHealth	Storage	✓	
	Access and Retrieval		✓ (multi-user)
	Advanced query		
	Statistics computation query		
GeoData	Geodata storage	✓	✓
	Web mapping		
	Geo-processing		
	Geodata collaboration		

3.1.1 Store by id through proxy

3.1.1.1 E-Health storage

- Description**

The “medical record passivation responsible” (i.e. an expert in collecting medical data in the Hospital Information System) wants to securely store health data in the cloud, using CLARUS.

- Pre-conditions**

The HIS CLARUS security manager has set up the security policy (e.g. encryption or alternatives, split-merge between CSPs) for this dataset, and a user account in one (or many) cloud service provider(s) has been set up.

- Main scenario**

Step 0	The medical record passivation responsible, via the clinical workstation interface, stores passive health data in the cloud through the CLARUS proxy
Step 1	The CLARUS proxy controls the access rights of the medical record passivation responsible
Step 2	The CLARUS proxy protects the data according to the security policy defined (e.g. storing fake medical records having the same structure)
Step 3	The CLARUS proxy stores the protected data on the cloud using the cloud storage service provided by the CSP
Step 4	The cloud storage service confirms the CLARUS proxy about the success of the storage operation
Step 5	The CLARUS proxy confirms the medical record passivation responsible about the success of the storage operation

3.1.1.2 Geodata storage

- **Description**

A GIS professional wants to store and protect the geological dataset he/she is responsible for (e.g. mineral resource shapefiles), in order to use securely the cloud storing facilities.

- **Pre-conditions**

The organization CLARUS security manager has set up the security policy (e.g. encryption or alternatives, split-merge between CSPs) for this dataset, and a user account in one (or many) cloud service provider(s) has been set up.

- **Main scenario**

Step 0	The GIS professional, via a dedicated application (e.g. QGIS), stores shapefiles in the cloud through the CLARUS proxy
Step 1	The CLARUS proxy controls the access rights of the GIS professional
Step 2	The CLARUS proxy protects the data according to the security policy defined (e.g. encryption or any other alternative)
Step 3	The CLARUS proxy stores the protected data on the cloud using the cloud storage service provided by the CSP
Step 4	The cloud storage service confirms the CLARUS proxy about the success of the storage operation
Step 5	The CLARUS proxy confirms the GIS professional about the success of the storage operation

3.1.2 Retrieve by id through proxy

3.1.2.1 E-Health Access and Retrieval

- **Description**

Healthcare professionals (e.g. medical doctors) can access the passive medical records of certain patients by their patient ID.

In this case, the healthcare professional (data consumer) wants to retrieve, through the CLARUS proxy, the passive medical records securely stored in the cloud by the medical record passivation responsible. The healthcare professional retrieving the data wants to have the guarantee that the results of his/her query is not disclosed to the cloud service provider or any other unauthorized party.

- **Pre-conditions**

- Medical records have previously been protected and stored in the CSP(s)
- The CLARUS security manager for the HIS has given the proper rights to the healthcare professional

- **Main scenario**

Step 0	The healthcare professional, via the Clinical Workstation Interface, searches for the protected medical record in the cloud (e.g. encrypted clinical note), by means of the patient id
Step 1	The CLARUS proxy controls the access rights of the healthcare professional for the corresponding data
Step 2	The CLARUS proxy retrieves the protected medical record (e.g. anonymised clinical note) from the cloud using the cloud storage service provided by the CSP, by means of the patient id
Step 3	The CLARUS proxy unprotects the protected medical record according to the protection set initially
Step 4	The CLARUS proxy provides the healthcare professional with the clear (unprotected) medical record

3.1.2.2 Geodata storage

- **Description**

A GIS professional wants to retrieve part of the geological dataset he/she has stored in the cloud (e.g. mineral resource shapefiles).

- **Pre-conditions**

The geodata files have been stored and protected in the cloud.

- **Main scenario**

Step 0	The GIS professional, via a rich-client application, searches for his/her protected vector geodata (e.g. encrypted shapefile) in the cloud, by means of their id (e.g. filename)
Step 1	The CLARUS proxy controls the access rights of the GIS professional for the corresponding data (control ownership)
Step 2	The CLARUS proxy retrieves the protected data from the cloud using the cloud storage service provided by the CSP, by means of their id
Step 3	The CLARUS proxy unprotects the vector geodata according to the protection set initially (e.g. encryption)
Step 4	The CLARUS proxy provides the GIS professional with his/her clear (unprotected) data

3.2 Search in the cloud

eHealth / GeoData Demonstration cases		CLARUS use case Search in the cloud	
		Search through proxy	Search directly in the cloud
		UC-EUP-07	UC-EUP-08
eHealth	Storage		
	Access and Retrieval		
	Advanced query	✓	✓
	Statistics computation query		
GeoData	Geodata storage		
	Web mapping	✓	✓
	Geo-processing		
	Geodata collaboration	✓	

3.2.1 Search through proxy

3.2.1.1 E-Health advanced query

- **Description**

A medical doctor, in order to retrieve specific information for clinician purpose, wants to perform an advanced search on protected data, based on multiple criteria (e.g. within a specific range of values, a duration, with certain medication records, etc.).

- **Pre-conditions**

- Medical records have previously been protected (e.g. anonymised) and stored in the CLARUS-enabled CSP(s), using CLARUS.
- the HIS CLARUS security manager has authorized the medical doctor to access the data

- **Main scenario**

Step 0	The medical doctor, via the clinical workstation interface, searches for specific data in the cloud, by means of an attribute-based query (e.g. list of patient IDs that meet given diagnostic conditions and follow-up duration)
Step 1	The CLARUS proxy controls the access rights of the medical doctor for the corresponding dataset
Step 2	The CLARUS proxy retrieves the protected data from the cloud using the cloud storage service provided by the CLARUS-enabled CSP
Step 3	The CLARUS proxy unprotects the data according to protection set initially
Step 4	The CLARUS proxy provides the medical doctor with the unprotected data he requested

3.2.2 Search directly in the cloud

3.2.2.1 Web mapping (Geo-publication)

- **Description**

For national statistics purpose, a geologist wants to search for groundwater borehole locations on a map published by the national geological survey, directly in the cloud.

- **Pre-conditions**

The borehole data have been published, i.e. stored and protected using CLARUS, under certain restrictions related to the map resolution (scale).

- **Main scenario**

Step 0	The geologist, via the map application published in the cloud, searches for borehole locations (i.e. without the CLARUS proxy), by means of an attribute-based query (e.g.
--------	--

	boreholes within a bounding box)
Step 1	The map service provides the geologist with coarsened borehole locations (e.g. locations are degraded from the 1/100,000 scale)

3.2.2.2 E-Health advanced query

- Description**

For research purposes, a medical doctor wants to perform directly in the cloud an advanced search over public anonymised data, based on multiple criteria (e.g. within a specific range of values, a duration, with certain medication records, etc.).

- Pre-conditions**

Medical records have previously been protected (anonymised) and stored in the CLARUS-enabled CSP(s), using CLARUS.

- Main scenario**

Step 0	The medical doctor, via the clinical workstation interface, searches for data directly in the cloud (i.e. without the CLARUS proxy), by means of an attribute-based query (e.g. lab results within a specific range of values)
Step 1	The cloud service provides the medical doctor with the anonymised data he requested for.

3.3 Update in the cloud

eHealth / GeoData Demonstration cases		CLARUS use case Update in the cloud
		Update by attributes through proxy
		UC-EUP-09
eHealth	Storage	
	Access and Retrieval	
	Advanced query	

	<i>Statistics computation query</i>	
GeoData	<i>Geodata storage</i>	(✓)
	<i>Web mapping</i>	
	<i>Geo-processing</i>	
	<i>Geodata collaboration</i>	✓

3.3.1 Update data by attributes through the CLARUS proxy

This CLARUS use case can be related to the Geodata collaboration use case.

3.3.1.1 Geodata storage

- **Description**

A GIS professional (data provider) wants to update geodata stored in a geodatabase thanks to a rich-client application (e.g. QGIS). The update query is based on geographical attributes. The GIS professional wants the content of his/her query not to be disclosed to the cloud service provider or any other unauthorized party.

- **Pre-conditions**

Geodata have previously been protected and stored in the CSP(s) by the organization GIS team (data provider). A security manager has provided the geomatics professional with proper access rights.

- **Main scenario**

Step 0	The GIS professional, via a rich-client application updates data in the cloud geoDB by means of an attribute-based SQL query
Step 1	The CLARUS proxy controls the access rights of the GIS professional for the corresponding data
Step 2	The CLARUS-enabled CSP updates the protected (e.g. encrypted) data in the cloud geoDB using the CLARUS-enabled SQL update service
Step 3	The CLARUS-enabled SQL update service confirms the CLARUS proxy about proper update of his/her protected data
Step 4	The CLARUS proxy confirms the GIS professional about proper update of his/her protected data

3.3.1.2 Geodata collaboration

- Description**

In the immediate aftermath of a disaster, a geohazard expert in the field wants to update several features of a map (via an attribute-based query from a mobile device) in order to report the field situation. For security reasons, it is important that the content of his/her query is not disclosed to the cloud service provider or any other unauthorized party.

- Pre-conditions**

Map features have previously been protected and stored in the CSP(s) by the disaster-response team (data provider). A security manager has provided the geohazard expert with proper access rights.

- Main scenario**

Step 0	The geohazard expert, via a mobile application updates several features in a map by means of an attribute-based WFS transaction (web service)
Step 1	The embedded CLARUS proxy controls the access rights of the geohazard expert for the corresponding data
Step 2	The CLARUS-enabled CSP updates the protected (e.g. encrypted) data in the cloud using the CLARUS-enabled WFS-T service
Step 3	The CLARUS-enabled WFS-T service confirms the CLARUS proxy about proper update of his/her protected data
Step 4	The CLARUS proxy confirms the geohazard expert about proper update of his/her protected data

3.4 Computation in the cloud

eHealth / GeoData Demonstration cases		CLARUS use case Computation in the cloud	
		Compute through proxy	Compute directly in the cloud
		UC-EUP-10	UC-EUP-11
eHealth	Storage		
	Access and Retrieval		

	<i>Advanced query</i>		
	<i>Statistics computation query</i>	✓	
GeoData	<i>Geodata storage</i>		
	<i>Web mapping</i>		✓
	<i>Geo-processing</i>	✓	
	<i>Geodata collaboration</i>		

3.4.1 Compute through proxy

3.4.1.1 E-Health statistics computation query

- **Description**

A medical doctor, in order to improve the hospital health service, wants to make statistics computation on structured data - e.g. Diagnostics (ICD-9), Lab Results (LOINC), in-patient medication records – that are stored in the cloud.

- **Pre-conditions**

Health data have previously been anonymised and stored in the CSP by a “medical record passivation responsible” (data provider) using CLARUS.

- **Main scenario**

Step 0	the medical doctor, via the Clinical Workstation interface, requests the execution in the cloud of a statistical calculation on the health data stored in the cloud (e.g. frequency of a given infection way category)
Step 1	the CLARUS-enabled CSP performs the statistics computation over the protected (e.g. anonymised) health data (cloud computation service)
Step 2	the CLARUS-enabled CSP stores the results of the computation (in the cloud)
Step 3	the medical doctor, via the Clinical Workstation interface, requests the download of the corresponding results from the cloud
Step 4	the CLARUS proxy controls the access rights of the medical doctor for the corresponding result
Step 5	the medical doctor retrieves the result of his/her statistics calculation

3.4.1.2 Geo-processing (Kriging computation)

- **Description**

A customer, in order to get the possible value of a geo-referenced scientific value (e.g. mineral concentration at a given place) wants to use a Kriging geo-statistics computation service.

- **Pre-conditions**

The confidential coordinates of the sample locations have previously been protected (e.g. via encryption or any other alternative) by the geodata provider, using CLARUS.

- **Main scenario**

Step 0	the kriging service customer (data consumer), via a map application hosted in the cloud, selects the point where he/she wants to get a prediction value
Step 1	the CLARUS proxy controls the access rights of the data consumer (execution rights)
Step 2	the CLARUS-enabled CSP performs the Kriging prediction calculation for the protected (e.g. encrypted) query location, the prediction value calculated being itself protected
Step 3	the CLARUS proxy unprotects the value of the Kriging prediction
Step 4	The Kriging service customer is provided with the clear result of the prediction

3.4.2 Compute directly in the cloud

This feature can be matched with the Web Mapping (GeoPublication) business case, through the process of Map Reprojection – a computation capability supported by map servers (e.g. GeoServer, MapServer, etc.). In this specific case we can talk about “Reprojection directly in the cloud”, i.e. on maps protected through CLARUS (e.g. coarsened).